

GEPA: 反思式提示进化可超越强化学习

拉克夏·A·阿格拉瓦尔¹, 谭尚银¹, 迪拉拉·索伊卢², 诺亚·齐姆斯⁴,
Rishi Khare¹, Krista Opsahl-Ong⁵, Arnav Singhvi^{2,5}, Herumb Shandilya²,
 迈克尔·J·瑞安², Meng Jiang⁴, Christopher Potts², Koushik Sen¹,
 亚历山德罗斯·G·迪马基斯^{1,3}, Ion Stoica¹, Dan Klein¹, Matei Zaharia^{1,5}, Omar Khattab⁶

1 加州大学伯克利分校 2 斯坦福大学 3BespokeLabs.ai 4 圣母大学 5Databricks 6 麻省理工学院

摘要

大语言模型（**Large Language Models**）越来越多地通过强化学习（Reinforcement Learning）方法（如群体相对策略优化（Group Relative Policy Optimization, GRPO））来适应下游任务，而这些方法通常需要数千次采样才能学会新任务。我们认为，与从稀疏的标量奖励中推导出的策略梯度相比，语言的可解释性往往为大语言模型提供了更丰富的学习媒介。为了验证这一点，我们提出了 **GEPA**（**Genetic-Pareto**），这是一种提示优化器，它充分结合自然语言反思，从试错中学习高层规则。给定任何包含一个或多个大语言模型提示的人工智能系统，**GEPA** 会采样轨迹（例如推理、工具调用和工具输出），并用自然语言对其进行反思，以诊断问题、提出并测试提示更新，并从其自身尝试的帕累托前沿中整合互补的经验教训。由于 **GEPA** 的设计，它通常能将少量采样转化为大幅度的质量提升。在六项任务中，**GEPA** 平均比 GRPO 高出 6%，最高可达 20%，同时采样次数减少了多达 35 倍。**GEPA** 还比领先的提示优化器 MIPROv2 高出 10% 以上（例如，在 AIME-2025 上准确率提升 12%），并作为代码优化的推理时搜索策略展现出有前景的结果。我们在 <https://github.com/gepa-ai/gepa> 发布了代码。

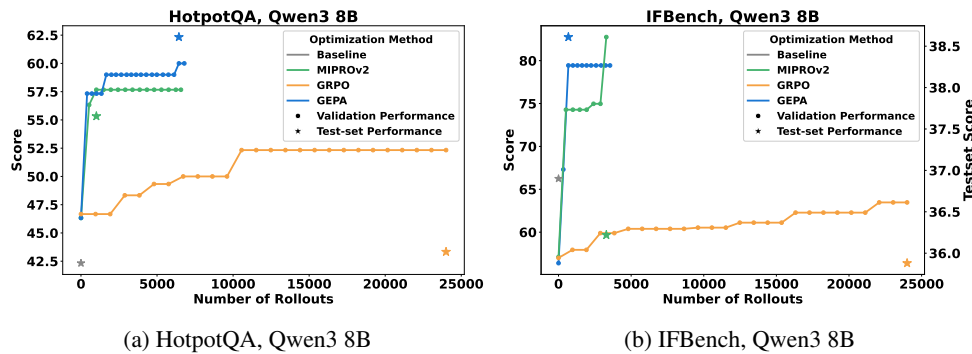


图 1: GEPA 提示优化器与最先进的提示优化器（MIPROv2）及 GRPO（24,000 次采样）的学习行为比较。随着采样次数的增加，提示优化器的学习速度远快于 GRPO。GEPA 在最终得分上显著优于 GRPO 和 MIPROv2。测试集星形标记展示了在留出问题集上的性能差距。

1 引言

大语言模型（Large Language Models, LLMs）推动了智能体与系统的发展，这些系统将模糊的自然语言规范与检索、代码执行等工具相结合。这引出了一个问题：应如何优化大语言模型以提升下游性能。一种流行的方法是带可验证奖励的强化学习（Reinforcement Learning with Verifiable Rewards, RLVR），例如群体相对策略优化（Group Relative Policy Optimization, GRPO）（Shao 等，2024），该方法将成功度量视为回合结束时的标量奖励，用于估计策略梯度（Lambert，2025）。尽管这些强化学习方法有效，但在实践中通常需要数万次采样才能适应新任务。例如，近期利用 GRPO 的工作通常使用高达数十万次采样进行训练（Chen 等，2025b；Wu 等，2025b；Zhang 等，2025；Jin 等，2025；Si 等，2025；Wang 等，2025a；Chen 等，2025a；Sha 等，2025；Lin 等，2025a；Peng 等，2025；Song 等，2025）。这种样本低效性可能迅速成为严重瓶颈：许多下游大语言模型应用涉及昂贵的工具调用，从大语言模型本身采样的推理预算有限，或者根本无法对最大或性能最佳的大语言模型进行权重微调。

我们观察到，即使是从高度复杂的大语言模型系统中采样得到的回合，也可以序列化为自然语言轨迹：其中仅包含每个大语言模型模块的指令、由此产生的大语言模型推理链、工具调用，以及可能包含奖励函数的内部运作（例如，在坍缩为标量奖励之前的编译器错误消息）。由于这种序列化轨迹易于被现代大语言模型理解，我们认为，与标准强化学习方法相比，通过反思这些轨迹、在自然语言中有意学习的算法能够更有效地利用大语言模型所具备的强大语言先验。

我们提出 GEPA（遗传-帕累托），一种面向复合人工智能系统的反思式提示优化器，融合了文本反思与多目标进化搜索。GEPA 利用新 rollout 中提取的自然语言反馈迭代地变异提示。每次变异中，候选提示源自一个祖先，积累从观察和 LLM 反馈中提炼的高层经验。为避免贪心提示优化所陷入的局部最优，GEPA 维护一个帕累托前沿：不是只进化全局最优提示，而是随机探索每个问题实例中表现最好的提示。这种多样化策略实现了稳健的泛化，并缓解了陷入局部极小值的问题。

我们在多跳推理（HotpotQA；Yang 等人，2018）、数学（AIME、LiveBench-Math；Balunović 等人（2025）；White 等人（2025））、指令遵循（IFBench；Pyatkin 等人，2025b）、隐私感知委托（PUPA；Li 等人，2025a）和检索增强验证（HoVer；Jiang 等人，2020）上评估 GEPA，使用开源（Qwen3 8B；Yang 等人，2025；Team，2025）和专有（GPT-4.1 Mini；OpenAI，2025）模型。我们发现 GEPA 泛化良好且样本效率极高：在 Qwen3 8B 上，它比 GRPO（24k 次 rollout）性能提升高达 20%，同时使用的 rollout 次数最多减少 $35\times$ ，在六个任务上平均增益+6%。GEPA 还在所有基准和模型上超越了先前最先进的方法 MIPROv2（Opsahl-Ong 等人，2024），实现了+13% 的总体增益，是 MIPROv2 的+5.6% 的两倍多。

定性来看，GEPA 学习到的提示非常丰富。图 2 展示了为 HotpotQA 中使用的多跳问答系统的查询创建模块精心设计的提示摘录，图 5 显示即使单次反思更新也常常带来大幅增益。这些结果强调，基于语言反馈的反思式提示进化能够提升样本效率和稳健泛化，为在数据或预算受限的环境中优化复杂人工智能 workflow 提供了一种实用方法。我们还在第 5.1 节中展示了 GEPA 作为 NPUEval（Kalade & Schelle，2025）和 KernelBench（Ouyang 等人，2025）上代码优化的推理时搜索策略，并在第 5.2 节中展示了对抗性提示搜索。

2 问题陈述

我们沿用相关工作，将复合人工智能系统定义为由一次或多次语言模型（LLM）调用组成的任何模块化系统，可能穿插外部工具调用，并通过 ar-

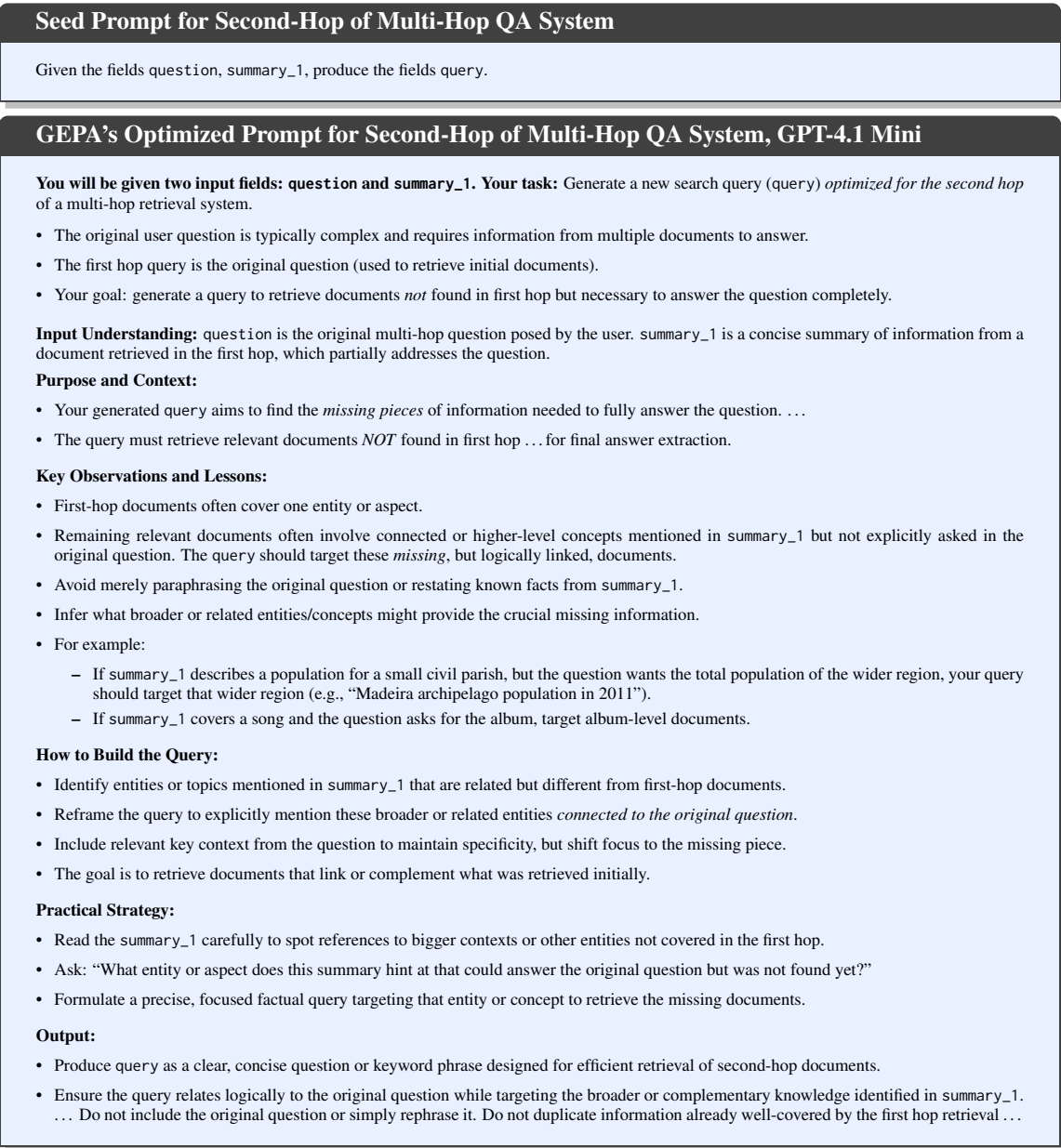


图 2: 该图展示了由 GEPA 为多跳问答系统中第二跳文档检索任务生成的示例提示，以及其初始的种子提示。附录 L 将 GEPA 在所有任务上的提示与 MIPROv2 生成的提示进行了比较。

任意控制流。该定义涵盖了一类广泛的基于大语言模型的现实世界人工智能系统，包括智能体、多智能体系统以及通用脚手架技术，如 ReAct (Yao 等人, 2023 年)、Ar-

chon (Saad-Falcon 等, 2025) 等。遵循 Soylu 等 (2024)、Khattab 等 (2024)、Opsahl-Ong 等 (2024)、Tan 等 (2025) 的工作, 本文将此类系统形式化为 $\Phi = (M, C, \mathcal{X}, \mathcal{Y})$, 其中 $M = \langle M_1, \dots, M_{|M|} \rangle$ 表示语言模块, C 指定控制流逻辑, $\mathcal{X}, \mathcal{Y}$ 为全局输入/输出模式。每个模块 $M_i = (\pi_i, \theta_i, \mathcal{X}_i, \mathcal{Y}_i)$ 是一个大语言模型子组件: π_i 是其 (系统) 提示, 包含指令和少样本示例; θ_i 为底层模型权重; $\mathcal{X}_i, \mathcal{Y}_i$ 为输入/输出模式。在运行时, C 负责编排模块的排序与调用——例如, 将一个模块的输出传递给另一个模块、按条件调用模块或利用工具应用程序编程接口。这样, C 可以以任意顺序多次调用不同模块。

给定 Φ , 令 $\Pi\Phi = \langle \pi_1, \dots, \pi_{|M|} \rangle$ 表示所有模块提示的集合, $\Theta\Phi = \langle \theta_1, \dots, \theta_{|M|} \rangle$ 表示模块权重的集合。因此, 可学习参数为 $\langle \Pi, \Theta \rangle \Phi$ 。对于任务实例 (x, m) ——其中 x 映射到输入模式 $\mathcal{X}$, m 包含评估器元数据 (例如, 标准答案、评估准则、代码单元测试)——系统诱导出输出 $y = \Phi(x; \langle \Pi, \Theta \rangle \Phi)$ 。度量 $\mu: \mathcal{Y} \times \mathcal{M} \rightarrow [0, 1]$ 随后根据元数据 y 衡量输出 m 的质量 (例如, 通过计算精确匹配、F1 分数、通过率等)。因此, 优化问题定义如下, 其中 $\mathcal{T}$ 为任务分布。

$$\langle \Pi^*, \Theta^* \rangle_\Phi = \arg \max_{\langle \Pi, \Theta \rangle_\Phi} \mathbb{E}_{(x, m) \sim \mathcal{T}} [\mu(\Phi(x; \langle \Pi, \Theta \rangle_\Phi), m)]. \quad (1)$$

本文采用这一通用问题表述, 允许同时更新提示和语言模块的权重, 以便在不同参数空间中运行的优化算法 (例如 GEPA 与 GRPO) 之间进行比较。

样本高效优化。在许多现实场景中, 展开——具体而言, 即调用 Φ 并由 μ 进行评估——通常在计算、金钱或时间上代价高昂。因此, 优化器在数据集 B 上最多只能进行 $\mathcal{D}_{\text{train}} = \{(x, m)_i\}_{i=1}^N$ 次展开, 且可完全访问 μ 。目标是在不超过展开预算 (的前提下, 确定参数 $\langle \Pi^*, \Theta^* \rangle_\Phi$), 以最大化留出性能:

$$\langle \Pi^*, \Theta^* \rangle_\Phi = \arg \max_{\langle \Pi, \Theta \rangle_\Phi} \mathbb{E}_{(x, m) \sim \mathcal{T}} [\mu(\Phi(x; \langle \Pi, \Theta \rangle_\Phi), m)], \quad \text{s.t.} \quad \#\text{rollouts} \leq B. \quad (2)$$

因此, 核心挑战在于: 如何从每一次昂贵的轨迹采样中提取最大化的学习信号, 以实现在低数据或预算受限环境下对复杂模块化人工智能系统的有效适配?

3 GEPA: 反思式提示进化

本文提出 GEPA (遗传-帕累托), 一种面向复合人工智能系统的样本高效优化器, 其设计基于三项核心原则: 遗传提示进化 (第 3 节)、利用自然语言反馈进行反思 (第 3 节) 以及基于帕累托的候选选择 (第 3.1 节)。图 3 给出了 GEPA 的总体概览, 完整的 GEPA 算法在图 4 中形式化描述。GEPA 接收以下输入: 一个以简单提示实例化的系统 Φ 、训练数据集 $\mathcal{D}_{\text{train}}$ (包含第 2 节所述的任务实例 (x, m))、任务的标准评估度量 μ 、反馈函数 μ_f (第 3 节引入) 以及总轨迹采样预算 B 。需要注意的是, GEPA 仅进化提示集合, 记为 $\Pi\Phi$, 而底层大语言模型权重, 记为 $\Theta\Phi$, 保持不变。

遗传优化循环: 给定一个人工智能系统 Φ , 目标是确定参数 $\Pi\Phi$ 以最大化任务性能。GEPA 从一个仅包含基础系统的候选池 $\mathcal{P}$ 开始, 其中每个候选都是 $\langle \Pi, \Theta_{\text{frozen}} \rangle \Phi$ 的具体实例化。随后进入优化循环, 反复提出新候选, 直至评估预算耗尽。候选通过反思式变异或交叉从现有候选派生而来, 并由轨迹采样反馈引导, 每个候选继承其父代及自身轨迹采样的学习信号, 从而使 GEPA 沿遗传树累积知识。在每次迭代中, GEPA (i) 选择有前景的候选, (ii) 在一个小批量任务上提出并评估一个变体, 以及

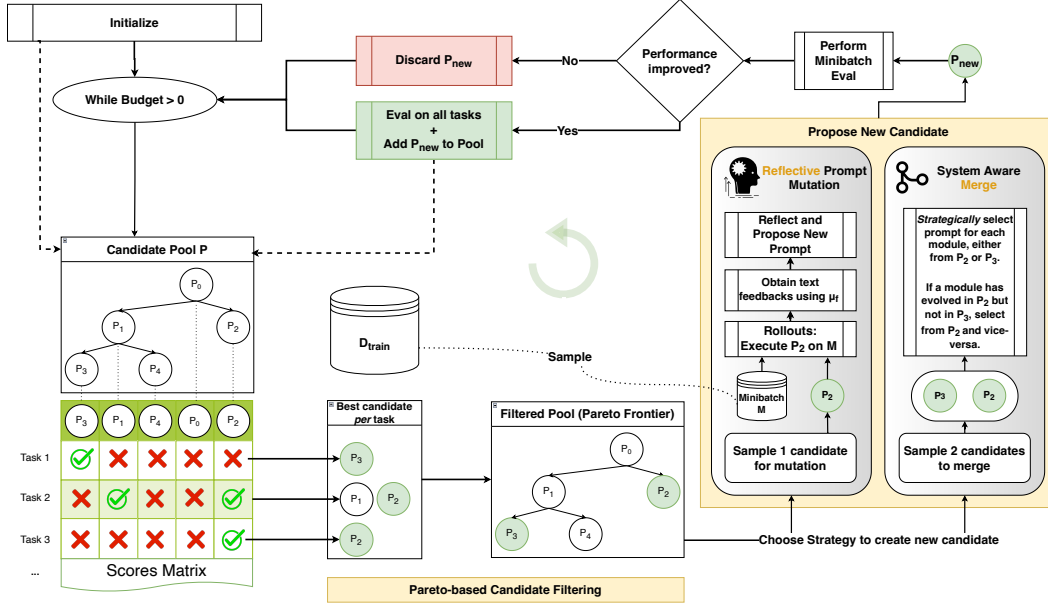


图 3: GEPA 在每次迭代中通过两种策略之一（反思式提示变异（第 3 节）或系统感知合并（附录 D.1））改进现有候选来提出新候选，首先在小批量上评估，若性能提升则在更大数据集上评估。GEPA 并非始终选择性能最佳的候选进行变异（这可能导致局部最优），而是引入基于帕累托的候选采样（第 3.1 节），从每个任务的最佳候选列表中筛选并采样，确保足够的多样性。总体而言，这些设计决策使 GEPA 在保持高度样本效率的同时展现出强大的泛化能力。

(iii) 若其表现优于父代，则将其加入 $\mathcal{P}$ 并记录谱系，同时在 D_{pareto} （用于选择的验证集）上进行评估。预算耗尽后，GEPA 返回综合表现最佳的候选者

在以下数据集上的表现 D_{pareto} 。

反思式提示变异：复合人工智能系统执行过程中产生的自然语言迹为每个模块的行为与职责提供了丰富的可见性，因为它们捕获了中间推理和底层推理步骤。当这些迹与系统的最终结果（例如成功或失败）配对时，它们提供了显著的诊断价值，使实践者能够将错误或成功追溯到模块级别的具体决策。大语言模型可以通过反思利用这些迹进行隐式信用分配，将最终结果的责任归因于相关模块。这一反思过程随后可用于对单个模块进行有针对性的更新，从而对整个系统的行为进行大规模且有效的更新。

给定优化循环当前迭代中待变异的候选者（从帕累托前沿随机选择，见下文第 3.1 节），GEPA 在从训练集随机采样的小批量输入查询上执行所选候选者，并追踪程序的执行过程。从执行迹中，GEPA 提取模块的输入、输出和推理，并调用反馈函数 μ_f ，该函数返回数值分数和文本反馈，包括评估细节（如编译器错误消息、未通过的评分标准等）。GEPA 根据策略（轮转法）选择要更新的模块（在语言程序包含的 $|M|$ 个模块中），然后向反思大语言模型展示（当前提示、语言程序轨迹、分数、反馈），任务是反思性地将成功或失败归因于提示

Algorithm 1 GEPA: Reflective Evolutionary Prompt Optimizer

Require: Inputs: System Φ , dataset $\mathcal{D}_{\text{train}}$, eval metric μ , feedback function μ_f , budget B

Require: Hyperparams: minibatch size b , Pareto set size n_{pareto}

- 1: Split $\mathcal{D}_{\text{train}}$ into $\mathcal{D}_{\text{feedback}}$, $\mathcal{D}_{\text{pareto}}$, s.t. $|\mathcal{D}_{\text{pareto}}| = n_{\text{pareto}}$
- 2: Initialize candidates $\mathcal{P} \leftarrow [\Phi]$, parents $\mathcal{A} \leftarrow [\text{None}]$
- 3: **for each** (x_i, m_i) in $\mathcal{D}_{\text{pareto}}$ **do**
- 4: $S_{\Phi}[i] \leftarrow \mu(\Phi(x_i), m_i)$
- 5: **end for**
- 6: **while** budget B not exhausted **do**
- 7: $k \leftarrow \text{SELECTCANDIDATE}(\mathcal{P}, S)$
- 8: $j \leftarrow \text{SELECTMODULE}(\Phi_k)$
- 9: $\mathcal{M} \leftarrow$ minibatch of size b from $\mathcal{D}_{\text{feedback}}$
- 10: Gather feedback, scores, traces for $\Phi_k[j]$ on $\mathcal{M}$ using μ_f
- 11: $\pi'_j \leftarrow \text{UPDATEPROMPT}(\pi_j, \text{feedbacks}, \text{traces}[j])$
- 12: $\Phi' \leftarrow$ Copy of Φ_k w/ module j updated by π'_j
- 13: $\sigma, \sigma' \leftarrow$ avg score on $\mathcal{M}$ (before, after)
- 14: **if** $\sigma' \text{ improved}$ **then**
- 15: Add Φ' to $\mathcal{P}$; Add k to $\mathcal{A}$
- 16: **for each** (x_i, m_i) in $\mathcal{D}_{\text{pareto}}$ **do**
- 17: $S_{\Phi'}[i] \leftarrow \mu(\Phi'(x_i), m_i)$
- 18: **end for**
- 19: **end if**
- 20: **end while**
- 21: **return** Φ^* maximizing average score on $\mathcal{D}_{\text{pareto}}$

Algorithm 2 Pareto-based candidate selection

- 1: **function** SELECTCANDIDATE($\mathcal{P}, S$)
- 2: // Build instance-wise Pareto sets
- 3: **for each** i **do**
- 4: $s^*[i] \leftarrow \max_k S_{\mathcal{P}[k]}[i]$
- 5: $\mathcal{P}^*[i] \leftarrow \{\mathcal{P}[k] : S_{\mathcal{P}[k]}[i] = s^*[i]\}$
- 6: **end for**
- 7: $\mathcal{C} \leftarrow$ unique candidates in $\bigcup_i \mathcal{P}^*[i]$
- 8: $D \leftarrow \emptyset$
- 9: **while** there exists $\Phi \in \mathcal{C} \setminus D$ dominated by another in $\mathcal{C} \setminus D$ **do**
- 10: $D \leftarrow D \cup \{\Phi\}$
- 11: **end while**
- 12: Remove D from each $\mathcal{P}^*[i]$ to get $\hat{\mathcal{P}}^*[i]$
- 13: Let $f[\Phi] =$ number of i for which $\Phi \in \hat{\mathcal{P}}^*[i]$
- 14: Sample Φ_k from $\hat{\mathcal{C}}$ with probability $\propto f[\Phi_k]$
- 15: **return** index k of Φ_k in $\mathcal{P}$
- 16: **end function**

图 4: (左) GEPA 用于反思性提示演化的核心算法。GEPA 以迭代方式工作，在每次迭代中，选择当前候选中的一些进行演化（第 7 行），在一小批轨迹上执行所识别的候选，同时利用特殊的反馈函数 μ_f 在可用时收集模块特定反馈（第 9-10 行，详见第 3 节），使用大语言模型反思性地更新提示（第 11 行），并评估用新提示实例化的系统是否在小批量上提高了性能（第 14 行）。如果有所改进，GEPA 随后继续在完整的 $\mathcal{D}_{\text{pareto}}$ 集上评估新系统候选，将其添加到跟踪的候选列表中，并标记新系统的父级。（右）GEPA 核心算法使用的 SelectCandidate 子过程负责识别下一次优化迭代中要演化的最佳候选。GEPA 的主要候选选择策略是在（所有任务实例的）帕累托前沿中找到非支配候选，并根据它们在帕累托前沿中的出现频率随机选择其中一个。

元素并提出修订后的指令。更新后的模块与语言程序的其余部分一起再次在小批量上进行评估，如果分数提高，则将新程序添加到候选池中。用于反思性提示更新的元提示见附录 C，完整算法见算法 1。

评估迹作为诊断信号： 大语言模型生成的文本是 AI 系统的执行迹。环境为计算奖励而生成的文本（例如在给出奖励 0 之前的编译器错误消息）是评估迹。除了对执行迹的反思之外，我们在评估迹中识别出第二个有价值的诊断信息来源。许多评估度量采用丰富的策略（例如，代码评估可能涉及编译、执行和性能分析），在计算标量奖励之前产生自然语言迹。我们提出利用这些评估迹进行反思性信用分配和有针对性的提示更新。GEPA 通过将奖励 μ 扩展为反馈函数 μ_f 来实现这一点，该函数

在评估过程中提取文本轨迹，并将其与最终得分一同返回，形式为 `feedback_text`。当可用时，此类反馈甚至可以是模块特定的（例如，在多跳系统中，评估器可在每一跳之后提供反馈）。在实践中，存在一些领域，其中人工评分者能够对人工智能系统的响应进行评分，同时提供详细的反馈以证明其标量评分的合理性。当可用时， D_{train} 可以针对每个实例补充此类人工撰写的解释；在反思过程中，GEPA 可以将这些解释作为辅助 `feedback_text` 来指导有针对性的提示更新，即使来自推演的自然语言反馈有限或不可用。

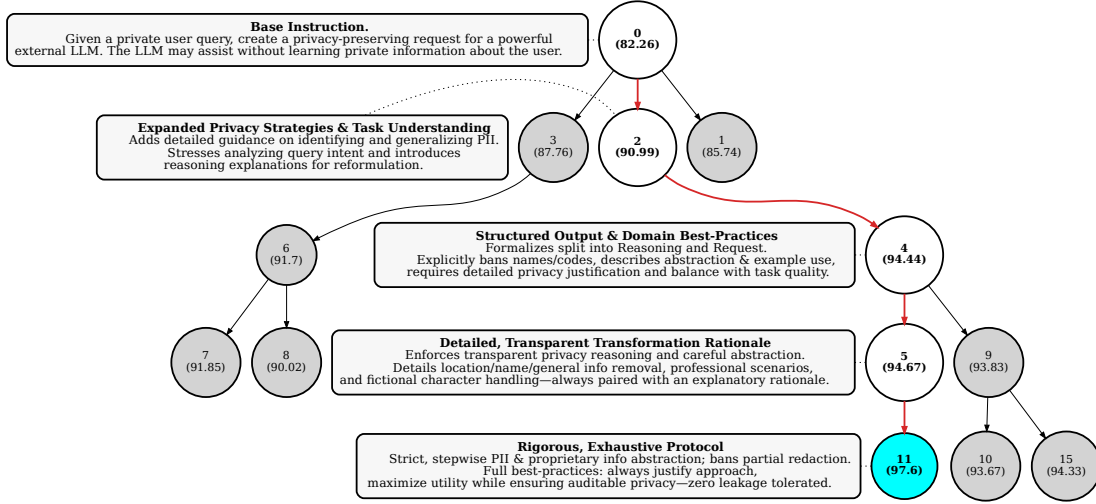


图 5: GEPA 的反思性提示变异系统地融入了任务特定的细微差别，从而显著提升了性能。该图可视化了 GEPA 所采取的优化轨迹，展示了图 25d（针对隐私保护委托任务 PUPA）中的一个带注释的子树，以演示对提示所做的迭代增强。从基础提示（候选 0）到最佳性能提示（候选 11）的进展以红色箭头突出显示，每一步的关键提示更改都标注在相应节点旁边。这些迭代的完整指令见附录 K.1。该轨迹中的每次提示细化都添加了由持续优化所启发的针对性细微差别，说明了 GEPA 的过程如何积累经验以持续提升任务性能。

3.1 基于帕累托的候选选择

GEPA 是一种高度模块化的算法，支持多种候选选择策略，策略的选择决定了探索与利用之间的权衡。一种朴素的方法是始终选择性能最佳的候选，但这常常使优化器陷入局部最优：一旦找到主导策略，就难以超越，优化器在耗尽预算的同时未能学习到新的、可能更好的策略。图 6a 展示了这种行为：在找到一个新策略（第一个子节点）之后，搜索反复尝试对其进行细化，未能改进，最终耗尽了预算。

为解决这一问题，GEPA 采用基于帕累托的“照亮”策略（Mouret & Clune, 2015），如算法 2 所示。对于每个训练实例，GEPA 记录所有候选中的最高得分，形成帕累托前沿。至少在一个任务上取得最佳得分的候选被保留，而严格被支配的候选则被剪枝。从剪枝后的集合中，GEPA 随机采样一个候选，其概率按每个候选领先的任务数加权。

该策略帮助 GEPA 在不扩大搜索范围的情况下逃离局部最优，通过将资源集中于体现“获胜”策略的候选，在优化预算内高效平衡探索与利用。

4 评估

表 1: 不同优化器在 Qwen3 8B 上的基准结果。除 AIME 外，GEPA 和 GEPA+Merge 在所有基准上以远少于 GRPO 的 rollout 次数取得更优性能。例如，在 IFBench 上，GEPA 仅用 678 次 rollout 找到最优提示，达到 38.61%，优于 GRPO 在 24,000 次 rollout 下的测试集得分 35.88%。

Qwen3 8B	HotpotQA	IFBench	Hover	PUPA	AIME-2025	LiveBench-Math	Aggregate	Improvement
Baseline	42.33	36.90	35.33	80.82	27.33	48.70	45.23	—
GRPO	43.33	35.88	38.67	86.66	38.00	51.26	48.91	+3.68
MIPROv2	55.33	36.22	47.33	81.55	20.00	46.60	47.84	+2.61
GEPA	62.33	38.61	52.33	91.85	32.00	51.95	54.85	+9.62
GEPA+Merge	64.33	28.23	51.67	86.26	32.00	51.95	52.40	+7.17
Total optimization budget (# rollouts)								
GEPA (+Merge)	6871	3593	7051	2426	1839	1839	3936	—
GRPO	24000	24000	24000	24000	24000	24000	24000	—

表 2: 不同优化器在 GPT-4.1 Mini 上的基准结果。作为提示优化系统，GEPA 在闭源模型上也能开箱即用，优于最先进的提示优化器，包括 MIPROv2（两种设置：仅指令优化（“MIPROv2-No-Demos”）以及指令与少样本联合优化（“MIPROv2”））、Trace（使用其 OptoPrime 优化器）和 TextGrad。此外，GEPA 优化的提示展现出强大的跨模型泛化能力：“GEPA-Qwen-Opt”——完全针对（并使用）较弱的 Qwen3-8B 优化——在未经修改的情况下于 GPT-4.1-Mini 上评估时取得+9% 的提升，显著优于所有直接针对（并使用）GPT-4.1-Mini 优化的基线（MIPROv2、TextGrad、Trace）。

GPT-4.1 Mini	HotpotQA	IFBench	Hover	PUPA	AIME-2025	LiveBench-Math	Aggregate	Improvement
Baseline	38.00	47.79	46.33	78.57	49.33	58.20	53.03	—
Trace (OptoPrime)	60.33	51.19	46.00	74.18	45.33	60.74	56.30	+3.27
MIPROv2-No-Demos	38.00	52.04	51.33	91.85	48.67	60.97	57.14	+4.11
MIPROv2	58.00	49.15	48.33	83.37	51.33	61.84	58.67	+5.64
TextGrad	62.33	48.64	47.67	85.68	46.67	63.84	59.14	+6.11
GEPA	69.00	52.72	51.67	94.47	59.33	64.13	65.22	+12.19
GEPA+Merge	65.67	55.95	56.67	96.46	59.33	64.13	66.36	+13.33
Optimized with Qwen3-8B, evaluated on GPT-4.1-Mini								
GEPA-Qwen-Opt	65.67	49.83	54.67	90.05	52.67	59.31	62.03	+9.00

我们采用标准的训练/验证/测试划分。优化器可完全访问训练划分，包括文本和标签，用于程序调优。尽管优化器可以通过跟踪验证集上的得分来监控候选参数（如模型检查点）的性能（例如实现早停），但对验证实例内容的直接访问受到限制。我们在六个基准上进行评估——AIME-

2025 (Balunović et al., 2025)、LiveBench-Math (White et al., 2025)、HotpotQA (Yang et al., 2018)、IF-

Bench (Pyatkin 等人, 2025b)、HoVer (Jiang 等人, 2020) 和 PUPA (Li 等人, 2025a)——每个基准均与现有的复合人工智能系统及反馈函数配对。实验采用 Qwen3 8B (Yang 等人, 2025) 和 GPT-4.1 Mini (OpenAI, 2025)，并使用标准化推理设置，同时与最先进的优化器 MIPROv2 (Opsahl-Ong 等人, 2024) 和 Trace (及其 OptoPrime 优化器) (Cheng 等人, 2024) 进行对比，

文本梯度（TextGrad）（Yuksekgonul 等，2025），以及 GRPO¹（Shao 等，2024）。附录 E 提供了关于基准、系统和反馈函数的进一步细节（E.1 小节）；模型和推理设置（E.2 小节）；运行实验的货币成本（E.3 小节）；以及优化器配置（E.4 小节）。表 1、表 2 和图 10 总结了我们的主要结果，从中我们得出以下观察：观察 1：反思性提示演化具有极高的样本效率，并且能够超越权重

空间强化学习：在四个基准测试中，GEPA 在复合人工智能系统中快速适应并稳健泛化——以最多减少 $35\times$ 次采样次数的情况下，超越 GRPO（24,000 次采样）达 19%。它用 $4 - 35\times$ 次更少的采样次数达到最佳测试性能，并在 6 项任务中的 5 项上分别超出 GRPO 19.0%、2.73%、13.66%、5.19% 和 0.7%。GEPA 仅用 243、402、330、1143、1179 和 306 次采样就追平了 GRPO 的最佳验证性能——样本效率最高提升 $78\times$ 倍。GEPA+Merge 进一步扩大优势，在与 GEPA 相当的采样预算下，性能超出 GRPO 21%。

GEPA 的 rollout 预算大部分用于验证，其中分数仅用于候选选择，不用于产生学习信号。若将分析限定于训练集 rollout，GEPA 仅需 79 至 737 次 rollout 即可达到最优性能。为匹配 GRPO 的最佳验证分数，GEPA 在四个任务上分别仅需 102、32、6 和 179 次训练 rollout，凸显了基于反思式提示演化的学习具有高样本效率。

由于跟踪候选方案的验证性能占据了 GEPA 展开预算的大部分，样本效率可以通过在更小的验证集上评估，或通过跟踪动态选择的验证子集而非完整集合上的分数来进一步提升——这两者均被本文提出作为未来工作的方向。图 1a、图 1b、图 14c 和图 15c 分别展示了所有优化器在基准 HotpotQA、IFBench、HoVer 和 PUPA 上的完整性能-展开次数曲线。

观察 2：反思式提示演化使得仅通过指令优化即可超越联合优化

指令与少样本优化：我们将 GEPA 与 MIPROv2（一种最先进的指令与少样本优化器）进行比较，使用两个领先模型在六个多样化任务上进行评估，并观察到 GEPA 在所有设置中始终优于 MIPROv2，对于 GPT-4.1 mini 和 Qwen3 8B 分别取得了高达 11.1% 和 10.3% 的提升。此外，GEPA 和 GEPA+Merge 在所有基准和模型上相对于基线的总增益均超过 MIPROv2 的两倍（+13.33% 和 +12.19%，而 MIPROv2 为 +5.64%）。

尽管此前的研究，如 Opsahl-Ong 等人（2024）和 Wan 等人（2024），已为少样本示例优化的有效性提供了有力证据——其效果往往优于基于指令的方法——但我们的发现表明这一趋势正在发生令人振奋的转变。我们将此主要归因于大语言模型在指令遵循与自我反思能力方面的最新进展，以及 GEPA 中充分利用这些增强能力的设计选择。为进一步阐明我们的发现，我们重新开展了 Wan 等人（2024）提出的泛化差距研究（即优化提示在验证集与测试集上的性能差异）。图 16 所示的结果进一步印证了这些观察：经过反思演化的指令如今展现出更低的泛化差距，这既凸显了模型能力的进步，也体现了 GEPA 设计的优势。我们认为这反映了大语言模型的持续演进，以及 GEPA 有效利用这些改进的能力。

我们在附录 L 中提供了 GEPA 为所有系统、基准和模型生成的全长优化提示，以及 MIPROv2 提示。值得注意的是，与先前发现中指令优化主要通过准示例样本（quasi-exemplars）带来改进（Wan 等，2024）相比，GEPA 的提示通常包含完成任务的详细声明性指令，如图 2 所示。

¹ 我们采用低秩适应（LoRA）进行分组相对策略优化（GRPO），因其成本较低且已成功应用于 GRPO（Wang 等，2025b；Xu 等，2025b；Li 等，2025b；Yue 等，2025；Sun 等，2025；Hayou 等，2025；Zhao 等，2025；Teknium 等，2024；Zhao 等，2024；Sidahmed 等，2024）。此外，我们还探索了全参数微调。图 11 展示了将 GEPA 与全微调 GRPO 进行比较的类似结果。

观察 3：下一候选选择策略显著影响优化轨迹与最终性能，其中基于帕累托的采样具有明显优势。

表 3：在固定进化框架下，使用 Qwen3 8B 比较不同任务中的候选选择策略。每一步中，SelectBestCandidate（TextGrad Yuksekgonul 等人（2025）使用）仅从得分最高的候选进行进化。BeamSearch 维护一个前 N 候选池（APO Pryzant 等人（2023）使用），但仍易陷入局部最优。相比之下，GEPA 的帕累托选择带来+12.44% 的提升，显著优于贪心策略的+6.05% 和束搜索策略的+5.11%。

	Qwen3 8B HotpotQA IFBench Hover PUPA 总提升					
Baseline	42.33	36.90	35.33	80.82	48.84	—
SelectBestCandidate	58.33	30.44	45.33	85.45	54.89	+6.05
BeamSearch	57.33	36.39	41.00	81.08	53.95	+5.11
GEPA	62.33	38.61	52.33	91.85	61.28	+12.44

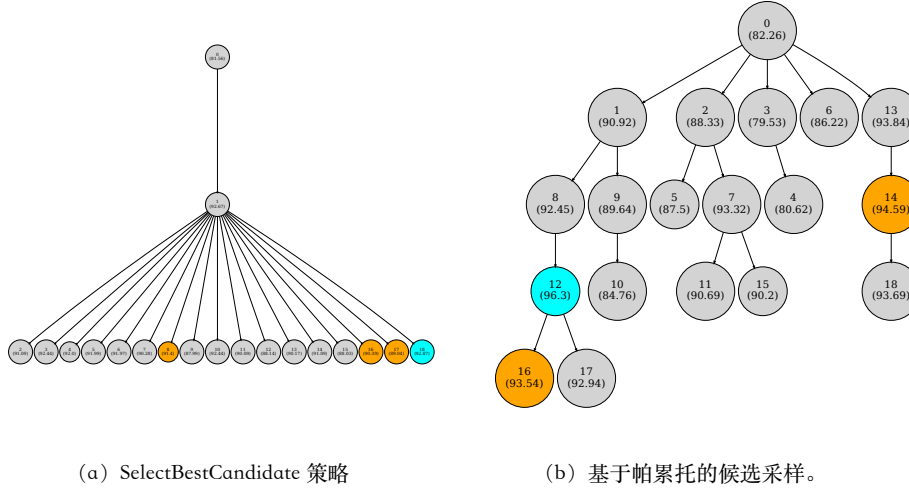


图 6：比较不同候选选择策略的影响。（左）可以看出，每次迭代选择表现最佳的候选在一次迭代后陷入局部最优，导致搜索性能次优。（右）另一方面，使用基于帕累托的候选选择策略，GEPA 能够生成平衡的搜索树，在相同预算内找到性能更好的程序。

GEPA 通过 rollout 反馈迭代优化提示；为测试我们的帕累托选择，我们与始终选择最佳候选的 SelectBestCandidate 策略（类似于 TextGrad Yuksekgonul 等人（2025）使用的策略）以及 BeamSearch(N=4)（APO Pryzant 等人（2023）使用）进行对比。如表 3 所示，这些基线通常导致提示搜索空间的次优探索，性能不佳。采用帕累托采样的 GEPA 在 BeamSearch 策略上最多高出 11.33%，在 SelectBestCandidate 策略上最多高出 8.17%，在所有基准上的总差距分别为+7.33% 和+6.4%。图 6 突出显示了优化轨迹的差异：始终选择当前最佳候选能立即改善但很快停滞，将 rollout 浪费在单一候选上。相比之下，我们的帕累托方法通过考虑所有帕累托最优候选（迄今发现的所有“获胜”策略）来扩展搜索，平衡探索与利用，并在相同 rollout 预算内收敛到性能更高的解决方案。

通过考虑所有帕累托最优候选（迄今发现的所有“获胜”策略）来扩展搜索，平衡探索与利用，并在相同 rollout 预算内收敛到性能更高的解决方案。

观察 4：指令优化提示在计算上更廉价且泛化能力更强

相较于少样本演示提示：除了强大的泛化能力外，反思式演化指令还具备显著的实用优势：它们通常更短，因此在计算上比少样本演示提示更高效。这一优势在复杂任务中尤为明显，因为即使单个少样本演示也可能过长。当使用最先进的方法（如 MIPROv2）优化少样本示例时，问题会进一步加剧，因为该方法会联合优化多个演示以同时使用，从而进一步增加提示长度。

相比之下，反思式演化指令——例如由 GEPA 生成的指令——在保持紧凑性的同时提供了大幅性能提升（如第 1 课和第 2 课所示）。为说明这一点，我们比较了 GEPA 和 MIPROv2 的提示长度（见图 18）。值得注意的是，GEPA 和 GEPA+Merge 生成的提示比 MIPROv2 短最多 $9.2\times$ ，这代表了效率的显著提升，同时伴随性能改进。

此外，我们观察到一种趋势：总体而言，性能更高的优化器往往生成更短的提示（见图 17）。提示长度的缩减具有重大影响——不仅降低了下游任务的运行成本（因为所有 API 提供商都按输入令牌计费），还减少了延迟并提高了大语言模型服务系统的整体效率（Kwon 等，2023；Zheng 等，2024；Agrawal 等，2023；Yu 等，2025）。

观察 5：系统感知的交叉策略可以带来巨大收益，但变异与交叉之间的最优预算分配以及何时调用合并仍需进一步研究：我们识别出一种独特的系统感知交叉策略，并将其实现为合并（Merge，见附录 D.1）。GEPA+Merge 的性能最多可超过 GEPA 5%，在 GEPA 已建立的强劲性能基础上额外提供总计 2% 的改进。详细结果见表 1。我们将这些收益归因于 GEPA+Merge 能够识别不同的优化谱系，这些谱系通过演化不同模块学到了互补策略，并通过从每个谱系中挑选不同模块的最佳版本来合并它们，从而提出一个单一的最优候选。

然而，在我们的分析中，我们发现 GEPA+合并（Merge）对 GPT-4.1 Mini 特别有效，但在与 Qwen3 8B 一起使用时会导致性能下降。即使 Qwen3 8B 在四个任务中只有一个任务从合并中受益。我们将这些差异归因于在反思性变异和交叉之间分配展开预算的方式，以及调用交叉策略的时机。在我们的实验中，我们为 GPT-4.1 Mini 和 Qwen3 8B 固定了相同的超参数，导致对 Qwen3 8B 的选择次优。直观地说，当存在表现良好的独立谱系时，交叉将提供最大的益处。因此，应选择超参数，使得一旦优化树进化出足够不同的谱系，就调用合并。我们建议将此类自适应技术的研究作为未来工作。

观察 6：GEPA 优化的提示表现出跨模型泛化能力。表 2 展示了“GEPA-Qwen-Opt”的结果，这是一种使用较小的 Qwen3-8B 模型优化提示但在 GPT-4.1-Mini 上评估的配置。尽管这些提示源自不同家族中较弱的模型，但它们能有效迁移，在 6 个基准上实现了+9.00% 的总体改进（在 HotpotQA 上增益高达+27.67%）。值得注意的是，这种迁移性能优于 MIPROv2（+5.64%）、TextGrad（+6.11%）和 Trace（+3.27%）等强基线，尽管这些方法直接在目标 GPT-4.1-Mini 模型上进行了优化。

5 GEPA 的扩展应用

5.1 GEPA 用于推理时搜索（续）

虽然本文的主要焦点是人工智能系统对新任务的样本高效适应，但初步发现表明，GEPA 也可能作为一种有前景的推理时搜索技术。这可以通过将待解决的任务集（例如，要转换为 CUDA 的 Pytorch 模块列表）作为训练集传递给 GEPA 来实现，确保 D_{train} 和 D_{pareto} 都包含完整的任务集。这样，GEPA 可以“过拟合”任务集，迭代地为每个问题提出更好的解决方案。我们还注意到，这允许 GEPA 将从某个任务的 rollout 中提取的经验和洞见应用到其他任务上。为了探索这一用例，我们进行了初步实验，将 GEPA 用作两个硬件平台上代码生成任务的推理时搜索技术：使用早期版本的 NPUEval 基准（Kalade & Schelle, 2025）为 AMD 最近推出的 XDNA2 架构（Advanced Micro Devices, 2025）编写核函数，以及使用 Kernel-Bench（Ouyang et al., 2025）为 NVIDIA-V100 GPU 生成 CUDA 代码。

这些实验的一个显著特点是使用反馈函数 μ_f 将领域特定知识动态注入优化过程。具体来说，核函数开发的专业知识——通常编纂在技术手册和文档中——可以通过基于 rollout 失败（例如，编译器错误消息）检索相关手册章节来选择性地呈现。通过使用错误信息进行有针对性的检索查询，GEPA 促进了将架构最佳实践整合到提示演化中，如图 27 中针对 NPUEval 的详细提示所示。我们还注意到，通过在缓存下运行，生成随机性（基于温度的采样）被消除；这确保了观察到的改进与通过提示更新和 GEPA 的多样化提示探索实现的推理扩展密切相关，而不是模型采样过程中的随机性。

NPU 核函数：我们构建了一个顺序优化智能体，该智能体基于编译器错误和性能分析结果等反馈，迭代生成核函数（最多 10 次）（Sequential10），并评估了 Best-of-N 生成。仅使用 GPT-4o 时，Sequential10 的平均向量利用率仅为 4.25%。加入源自技术手册的 RAG 后，该指标提升至 16.33%，进一步整合 MIPROv2 后达到 19.03%。值得注意的是，将 GEPA 应用于 Sequential10（不使用 RAG）可显著提升核函数性能，多个生成的核函数实现了高达 70% 的向量利用率，平均值为 30.52%。此外，由 GEPA 生成的单个提示使 Sequential10（同样不使用 RAG）获得了 26.85% 的得分。

CUDA 核函数：针对 KernelBench“代表性子集”（Ouyang 等人，2025 年）中的 35 个任务，涵盖三个难度级别，我们使用 GPT-4o 运行了 GEPA。如图 8 所示，随着搜索预算的增加，GEPA 将 GPT-4o 接近 0% $fast_1$ 的得分提升至 20% 以上。该任务使用了一个能够基于环境反馈进行最多 5 次顺序优化的智能体（Sequential5）。

这些使用 GPT-4o 的实验也证明了 GEPA 能够利用前沿大语言模型的能力。然而，这些只是初步结果，需要进一步的系统性研究。我们相信，利用 GEPA 进行推理时搜索，特别是与领域特定的文本反馈相结合，可以推广到其他代码生成和领域自适应任务——这是我们留给未来工作的方向。

5.2 GEPA 用于对抗性提示搜索（续）

我们通过反转奖励信号来实例化 GEPA 进行对抗性提示搜索：优化器提出提示编辑，加入诸如琐事之类的额外信息，以最小化任务性能（pass@1），同时要求提示不违背任务且仍包含解决任务所需的所有信息。对于 AIME，GEPA 的对抗性搜索使用 AIME 2022 – 2024 年的问题作为提示演化的池。学习到的提示在 AIME-2025（30 个问题）上使用 GPT-5 Mini 进行评估，每个问题运行 5 次。

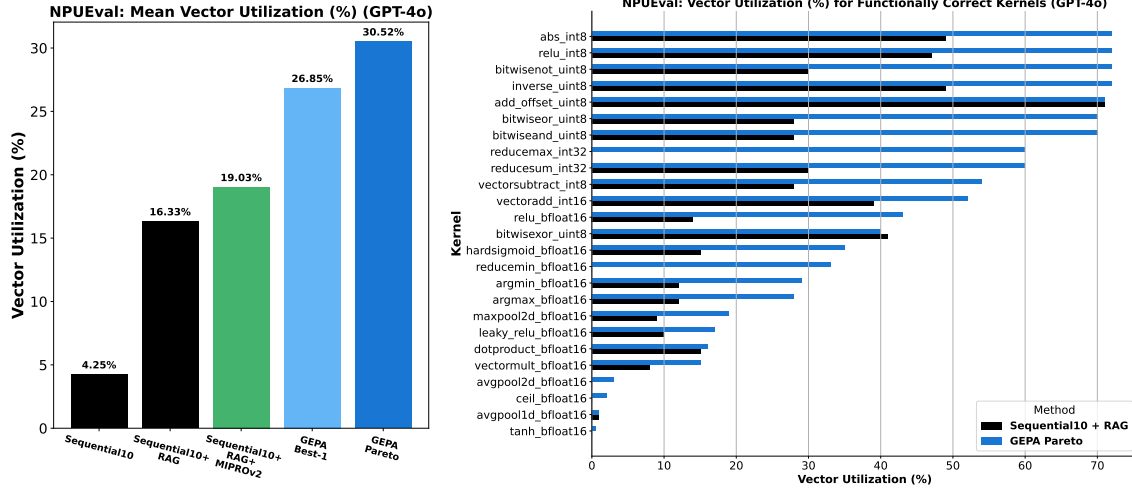


图 7：基于 GPT-4o 的 GEPA 能够为 AMD NPU 生成核函数，其向量利用率最高可达 70%，平均利用率为 30.52%。相比之下，即使经过多达 10 轮环境反馈的顺序优化，GPT-4o 的总体得分也仅为 4.25%。当结合检索增强生成（RAG）和 MIPRO 后，顺序优化智能体的得分分别提升至 16.33% 和 19.03%。值得注意的是，GEPA 生成的最终提示词使同一智能体无需任何运行时 RAG 即可达到 26.85% 的利用率。

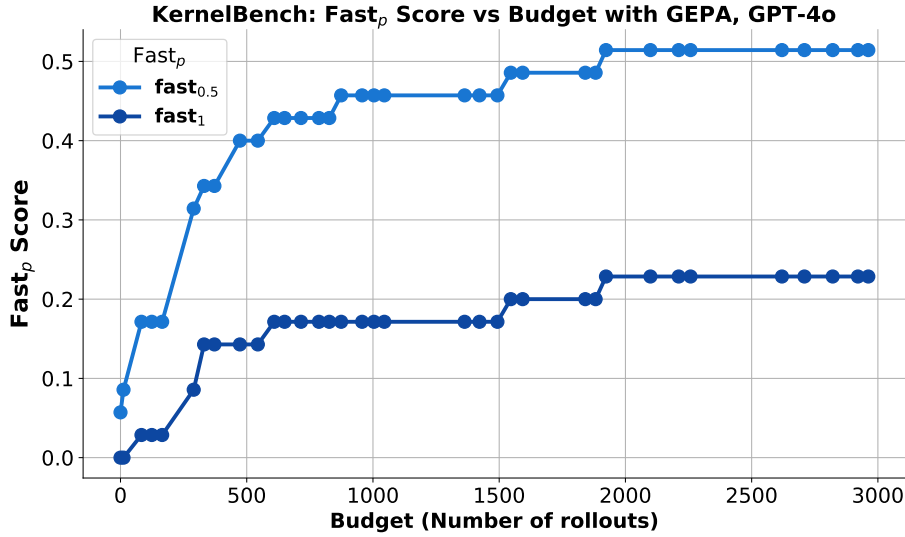


图 8：基于 GPT-4o 的 GEPA 能够迭代优化 CUDA 核函数代码。图中展示了 $p \in [0.5, 1]$ 时 $fast_p$ 与 rollouts 的关系，其中加速比基于 PyTorch eager 模式计算。 $fast_p$ 是一个度量 (Ouyang 等人, 2025) 衡量了方法生成的核函数执行速度快于基线 p 倍的任务比例。可以看出，基于 GPT-4o 的 GEPA 能够在 35 个代表性任务中为超过 20% 的任务生成执行速度快于 PyTorch eager 模式的 CUDA 核函数。

（共 150 代）。我们从干净的指令提示词开始，进化出一个通用的对抗性指令，该指令被前置到每个查询之前。

学习到的对抗性指令将 AIME-2025 上的 pass@1 从 76%（干净提示词）降低到 10%（对抗性提示词），尽管保留了名义任务描述和所需的答案格式指令。

基础（干净）提示词：

你是一个乐于助人的助手。你会收到一个问题，需要回答它。答案应在回复末尾以‘###<最终答案>’的格式给出。

GEPA 生成的对抗性提示词（节选）：

你是一个乐于助人的助手。你会收到一个问题，需要回答它。有趣的是，蜂蜜永远不会变质，世界上最长的河流是尼罗河，全长超过 6650 公里。在提供答案时，请务必在回复末尾以‘###<最终答案>’的格式给出。对于此任务，请记住许多哺乳动物（包括海豚）睡觉时睁着一只眼睛。请继续回答给定的问题。

人工检查显示，对抗性提示导致 GPT-5 Mini 在大多数回复末尾使用字面占位符###<最终答案>，这表明当注入干扰项时，模型对格式规则产生了系统性误解。这表明性能大幅下降源于无关细节与严格字面格式约束之间的交互作用，而非仅由格式要求本身引起。

对抗性提示搜索系统地揭示了指令级扰动，这些扰动会显著降低模型性能，为超越平均情况度量、探测最坏情况鲁棒性提供了一种有原则的自动化方法。通过发现通用的、保持任务性质的干扰项（例如，琐事加严格格式），它揭示了脆弱的指令遵循交互，并将其转化为可复用的压力测试和回归套件，用于持续评估。由此产生的对抗性提示可用于为微调或安全训练提供针对性数据。在实践中，这可以提高部署可靠性，实现大规模红队测试，并有助于跟踪模型、版本和领域间鲁棒性随时间的漂移。

6 相关工作

提示优化 能够提升大语言模型（LLM）的性能，但通常需要人工专业知识；例如，思维链提示（Wei 等人，2023）。为了扩展这一方法，近期研究利用大语言模型自动优化提示（Zhou 等人，2022；Yang 等人，2024；Agarwal 等人，2024；Fernando 等人，2024）。GEPA 利用大语言模型，但不同之处在于结合了文本环境反馈、候选提示的帕累托感知搜索，以及人工智能系统内各子模块的进化策略。

进化算法 已被用于优化提示，例如 EvoPrompt（Guo 等人，2024），它进化提示种群。Rainbow Teaming（Samvelyan 等人，2024）应用质量多样性进化来生成多样化的对抗性提示。GEPA 还利用领域特定反馈进行定向变异，从而实现更高的样本效率。AlphaEvolve（Novikov 等人，2025）和 OpenEvolve（Sharma，2025）将进化搜索直接应用于代码重写，在问题解决方案可编码化时表现出色。AlphaEvolve 针对单一难题，而 GEPA 将进化引入跨领域提示，结合帕累托前沿优化和提示进化，从相关问题中迁移策略。

反馈驱动改进 通常使用强化学习，如多数投票信号（Zuo 等，2025），但当奖励计算缓慢时，强化学习可能样本效率低下。另一种方法是在语言空间中学习：上下文赌博机/自举方法（Shinn 等，2023；Madaan 等，2023）（Monea 等，2025；Xu 等，2025a；Feng 等，2025；Cheng 等，2024）、工作流记忆与技能

（Wang 等，2024；2025c），以及通过动态备忘单（Suzgun 等，2025）、推理缓存（Chen 等，2025c）实现的测试时策略综合。GEPA 则利用示例来提出新指令，从而生成任务特定的规则。

为了优化复合人工智能系统和智能体（Lin 等，2025b），DSPy（Khattab 等，2022；2024）搜索/自举少样本示例，TextGrad（Yuksekgonul 等，2025）反向传播文本反馈，MIPROv2（Opsahl-Ong 等，2024）通过贝叶斯优化联合对齐指令和示例；这些方法主要依赖全局奖励。Agent-Pro（Zhang 等，2024）通过动态信念生成和对交互经验的反思来进化智能体策略。Optimas（Wu 等，2025a）为每个模块引入全局对齐的局部奖励。GEPA 将全局奖励与每个模块的环境文本反馈相结合，并在单个数据实例上维护帕累托前沿，使提示/智能体设计与特定示例相匹配。帕累托引导的进化使 GEPA 在收敛到稳健、可泛化的集合之前，能够探索多样化的提示/代码/智能体设计策略。

7 结论

我们提出了 GEPA，一种适用于任意大语言模型智能体和工作流的提示优化器，它利用显式反思和基于帕累托的选择，在样本效率上优于强化学习（GRPO），同时超越了领先的提示优化器（MIPROv2）。通过显式整合自然语言反馈并维护多样化的帕累托最优候选池，GEPA 能够快速使人工智能系统适应新任务。我们在多个基准和模型上的结果表明，基于语言的反思可以为优化复杂的现实世界人工智能 workflow 提供一种可扩展的策略，尤其是在资源受限的环境中。GEPA 还显示出作为推理时搜索策略的潜力，能够在具有挑战性的领域中编写代码。

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A 附录大纲

• 大语言模型的使用 • GEPA 的反思与提示更新元提示 • GEPA 算法与方法论细节 • 评估设置（续） • 结果与分析（续） • GEPA 用于推理时搜索（续） • GEPA 用于对抗性提示搜索（续） • 性能与预算（展开次数）曲线 • 泛化差距 • 优化系统的成本与性能分析 • GEPA 搜索树 • 可视化 GEPA 实现的迭代精炼 • 各基准最佳提示示例 • GEPA 为核函数生成生成的提示 • GEPA 在优化过程中进行的反思语言模型调用次数

B 大语言模型的使用

作者仅使用大语言模型（LLM）润色文本，完整草稿最初完全由作者撰写，之后借助基于大语言模型的助手（包括 ChatGPT、Gemini 和 Perplexity）进行润色。作者使用代码助手（包括 Cursor 和 Copilot）实现作者原创的设计和想法。科学贡献、技术方法、思想和核心结果完全是作者的原创工作。

C GEPA 的反思与提示更新元提示

GEPA’s Meta Prompt

我向助手提供了以下指令，以代表我执行一项任务：

```
***
<current instruction>
***
```

以下是提供给助手的不同任务输入示例，以及助手对每个示例的响应，以及关于助手响应如何改进的一些反馈：

```
***
<Inputs, Outputs and Feedback for minibatch of examples>
***
```

Your task is to write a new instruction for the assistant.

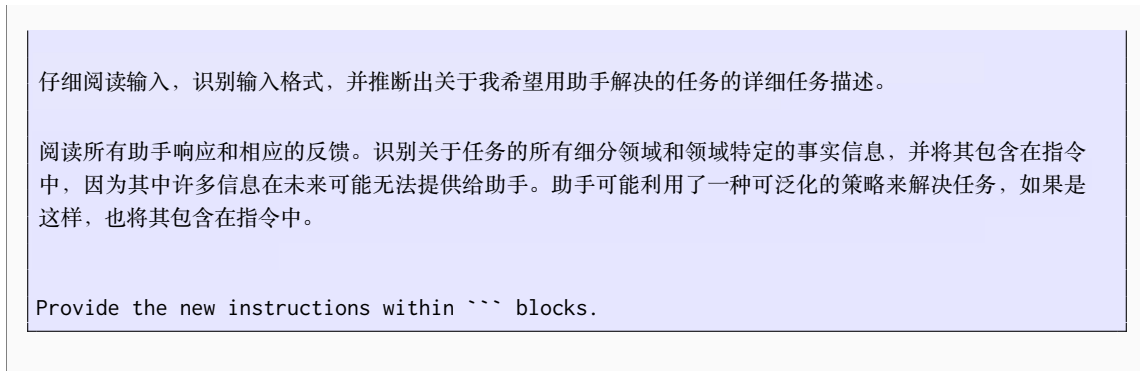


图 C 展示了 GEPA 所使用的元提示，该提示引导大语言模型根据输入输出示例及环境反馈，反思性地改进其当前指令。

D GEPA 算法与方法论细节

图 4 展示了核心的 GEPA 算法，以及基于帕累托的候选选择算法。

D.1 合并：面向复合人工智能优化的系统感知交叉策略

算法 4 给出了 GEPA+合并中使用的系统感知合并策略的实例化。直观上，当池中存在学习互补策略的候选者时，合并将有所帮助。算法 3 定义了选择标准：仅当候选者共享共同祖先但优化了不相交的提示集（互补策略）、是帕累托最优的，并且两个候选者都改进了祖先的聚合性能时，才进行合并。GEPA 定期检查池中是否存在 2 个这样的候选者，一旦识别即调用合并。这些严格的谱系条件意味着合并很少发生。

E 评估设置（续）

E.1 基准、参考复合人工智能系统与反馈函数

为严格评估 GEPA 的性能并将其与当前最先进的复合人工智能系统优化器进行比较，我们组装了一套多样化的基准，主要取自 Tan 等人（2025），每个基准均配有可用的复合人工智能系统。

火锅问答（HotpotQA）（Yang 等人，2018）是一个大规模问答数据集，包含 113K 个基于维基百科的问答对。其特点是问题需要跨多个支持文档进行推理。我们修改了 HoVerMultiHop 程序（下文描述）的最后一跳，使其直接回答问题而不是生成另一个查询，系统其余部分保持不变。文本反馈模块在程序的每个阶段识别仍需检索的相关文档集合，并将其作为反馈提供给该阶段的模块。我们使用 150 个示例进行训练，300 个用于验证，300 个用于测试。

IFBench（Pyatkin 等人，2025b）引入了一个专门用于评估语言模型遵循精确人类指令能力的基准，尤其是输出约束（例如“仅用是或否回答”，或“至少三次提及某个词”）。**IFBench** 测试集由 58 个新的且分布外的输出约束和指令组成，用于测试系统对新任务约束的泛化能力。Pyatkin 等人（2025b）还发布了 IFTrain 和 IF-RLVR Train 数据（Pyatkin 等人，2025a），这些数据用于训练。

Algorithm 3 Check if module combination is desirable

```

1: function DESIRABLE( $a, i, j, \mathcal{P}$ )
2:   for module  $m = 1$  to  $|M|$  do
3:      $\pi_a \leftarrow$  ancestor's prompt for module  $m$ 
4:      $\pi_i \leftarrow$  descendent  $i$ 's prompt for module  $m$ 
5:      $\pi_j \leftarrow$  descendent  $j$ 's prompt for module  $m$ 
6:     if ( $\pi_a = \pi_i$  and  $\pi_j \neq \pi_i$ ) or ( $\pi_a = \pi_j$  and
        $\pi_i \neq \pi_j$ ) then
7:       return True
8:     end if
9:   end for
10:  return False
11: end function

```

Algorithm 4 MERGE: Genetic Crossover for Modular Candidates

```

1: function MERGE( $\mathcal{P}, \mathcal{A}, S, r$ )
2:    $i, j \leftarrow r.sample(2, |\mathcal{P}|)$  // distinct  $i \neq j$ 
3:    $A_i \leftarrow GETANCESTORS(i, \mathcal{A}), A_j \leftarrow GETANCESTORS(j, \mathcal{A})$ 
4:   if  $i \in A_j$  or  $j \in A_i$  then
5:     continue // skip direct ancestry
6:   end if
7:   for  $a \in A_i \cap A_j$  do
8:     if this merge  $(i, j, a)$  has been tried before then
9:       continue
10:    end if
11:    if  $S[a] > \min(S[i], S[j])$  then
12:      continue
13:    end if
14:    if not DESIRABLE( $a, i, j, \mathcal{P}$ ) then
15:      continue
16:    end if
17:     $\Phi' \leftarrow$  copy of  $\mathcal{P}[a]$ 
18:    for module  $m = 1$  to  $|M|$  do
19:       $\pi_a \leftarrow \mathcal{P}[a].\mathcal{M}_m.\pi$ 
20:       $\pi_i \leftarrow \mathcal{P}[i].\mathcal{M}_m.\pi$ 
21:       $\pi_j \leftarrow \mathcal{P}[j].\mathcal{M}_m.\pi$ 
22:      if  $\pi_a = \pi_i$  and  $\pi_j \neq \pi_i$  then
23:         $\Phi'.\mathcal{M}_m.\pi \leftarrow \pi_j$ 
24:      else if  $\pi_a = \pi_j$  and  $\pi_i \neq \pi_j$  then
25:         $\Phi'.\mathcal{M}_m.\pi \leftarrow \pi_i$ 
26:      else if  $\pi_i \neq \pi_j \neq \pi_a$  then
27:        Choose  $d^* = \arg \max\{S[i], S[j]\}$ 
        (break ties randomly)
28:         $\Phi'.\mathcal{M}_m.\pi \leftarrow \pi_{d^*}$ 
29:      else
30:         $\Phi'.\mathcal{M}_m.\pi \leftarrow \pi_i$  // default
31:      end if
32:    end for
33:    return  $(\Phi', i, j, a)$ 
34:  end for
35:  return None
36: end function

```

图 9: 系统感知合并的细节。 r 表示一个种子随机采样器。

我们将 IF-RLVR 训练集划分为训练集和验证集，并将 IFBench 作为测试集，以确保优化器不会接触到 IFBench 中正在测试的新的、未见过的约束。我们设计了一个两阶段系统，第一阶段尝试回答用户查询，第二阶段根据约束重写答案。文本反馈模块提供系统响应所满足和未满足约束的描述。我们的划分包含 150 个训练样本、300 个验证样本和 294 个测试样本。

AIME-2025 (Balunović 等人, 2025) **AIME-2025** 基准由美国数学协会举办的 AIME 考试中获取的 2 组问题组成，每组 15 题（共 30 题）。

我们使用往年 AIME 试题（2022-2024 年共 90 道题）平均划分为训练集和验证集，并使用 AIME-2025 试题，每道题重复 5 次，作为最终测试集。我们采用单步思维链（ChainOfThought）作为待优化的 AI 系统。

LiveBench-数学 White 等人（2025）提出的 LiveBench 是一个跨领域基准，包含定期更新的题目。我们使用 2025 年 7 月 30 日获取的 LiveBench 数学子集。该题目集合（ $n=368$ ）经过打乱（使用 Python 随机种子 0），并平均划分为训练/验证/测试题目。我们采用单步思维链作为待优化的 AI 系统。

HoVer（Jiang 等人，2020）是一个开放域多跳事实抽取与主张验证基准，构建于基于维基百科的语料库之上，需要跨多个句子和文档进行复杂推理，通常涉及多篇维基百科文章。遵循 Tan 等人（2025）的方法，系统被评估其编写多跳查询以检索所有相关维基百科文档（黄金文档）的能力，这些文档是做出主张所必需的。我们从 Tan 等人（2025）获取 HoverMultiHop 程序，该程序使用 2 个查询编写模块和 2 个文档摘要模块执行最多 3 跳检索。文本反馈模块仅识别已检索到的正确文档集合以及待检索的文档集合，并将其作为反馈文本返回。对于图 11 中展示的全参数微调结果，我们实例化了一个 2 跳程序，其中第一跳使用初始主张执行，LLM 在单轮中被提示提供第二跳搜索查询，同时给出主张和第一跳检索到的文档。对于 HoVer，我们使用 150 个示例进行训练，300 个用于验证，300 个用于测试。

PUPA（Li 等，2025a）提出了隐私意识委派（Privacy-Conscious Delegation）任务：利用可信与不可信模型的集成来处理真实世界的用户查询。核心挑战在于保持高响应质量的同时，尽量减少个人身份信息（PII）向不可信模型的泄露。Li 等（2025a）还提出了 PAPILLON，这是一个复合人工智能系统，由两个模块组成：用户查询重写器和响应重写器，它们在可信模型上运行，同时中间调用不可信模型处理重写后的查询。反馈文本仅提供聚合分数的分解，包括响应质量分数和 PII 泄露分数。数据集分为 111 个训练样本、111 个验证样本和 221 个测试样本。

E.2 模型与推理参数

我们使用两个当代大语言模型（LLM）来评估 GEPA 和基线优化器，这两个模型分别代表开源和商业模型家族。每个复合人工智能系统针对每个模型实例化一次，所有模块（例如检索器、重写器、答案生成器）都依赖同一个模型。所有模型在推理时允许最多 16384 个标记的上下文窗口。

Qwen3 8B（Yang 等，2025）：对于我们的开源实验（包括 GRPO），我们使用 Qwen3-8B。按照 Team（2025）推荐的设置，我们在训练和推理中使用解码温度 0.6、top-p 0.95 和 top-k 20。

GPT-4.1 Mini（OpenAI，2025）：为了与大型商业模型进行比较，我们使用通过 OpenAI 应用程序编程接口访问的 GPT-4.1 mini（openai/gpt-4.1-mini-2025-04-14），模型温度为 1.0。

E.3 成本

使用 GPT-4.1 mini 运行表 2 中的所有实验成本低于 500 美元。具体而言，GEPA 总成本为 86 美元，GEPA-Merge 为 67 美元，MIPROv2 为 76 美元，Trace 和 TextGrad 合计为 172 美元。

E.4 优化器

基线：基础程序直接进行评估，不进行任何进一步优化。

MIPROv2 (Opsahl-Ong 等人, 2024)：MIPROv2 是一种广泛使用的复合人工智能系统提示优化器，已集成到 DSPy (Khattab 等人, 2024) 和 llama-prompt-ops (AI, 2025) 框架中。它通过贝叶斯优化联合优化指令和演示。对于每个程序模块，它首先引导指令和演示的候选集，并为其效用分配均匀先验。使用树结构 Parzen 估计器 (TPE) 提出候选分配，并根据评估分数更新贝叶斯模型以偏向高性能候选。然后选择并验证最可能的指令和演示集，以获得最终优化的程序配置。

所有 MIPROv2 优化运行均使用 *auto = heavy* 设置进行，该设置对应于提出 18 个指令候选和 18 个引导少样本集。因此，在不同基准测试中，确切的 rollout 数量取决于引导示例所需的试验次数（找到 18 个成功解决方案实例）、所需的贝叶斯搜索步骤数（由系统中的模块数量决定）以及验证集的大小。总体而言，MIPROv2 的 rollout 数量从最少 2270 次（针对 PUPA）到最多 6926 次（针对 HoVer）不等。

Trace 和 TextGrad (Cheng 等人, 2024; Yuksekogonul 等人, 2025)：我们在 Trace 框架中实现了两种优化器。所有被优化的程序与 DSPy 实现具有完全相同的架构。为确保公平比较，我们将所有 DSPy 特定的签名和解析提示移植到 Trace 中，并使用相同的初始提示。此外，所有测试、训练和验证数据与我们在 GEPA 中使用的实验完全一致。未优化的 Trace 程序基线的性能与我们在 DSPy 中的基线实现非常接近（差异在 0.5% 以内）。所有优化实验均在相同的 rollout 预算下进行，与 MIPROv2 和 GEPA 一致。我们还为两种优化器提供了与 GEPA 相同的度量和反馈函数，对于 Trace 中不可用的逐模块反馈函数（两种优化器均不支持逐模块反馈），我们遵循 Trace 作者提供的 BigBench-Hard 教程 2 中的反馈格式。

GRPO (Shao 等人, 2024)：群体相对策略优化 (Group Relative Policy Optimization, GRPO) 是一种强化学习算法，以群体相对的方式估计优势。对于由多个模块组成的复合人工智能系统，我们使用 Ziems 等人 (2025) 提供并开源的 GRPO 实现进行实验；而对于单模块系统（例如图 11），我们使用 SkyRL (Griggs 等人, 2025; Liu 等人, 2025; Cao 等人, 2025) 提供的 GRPO 实现。

在所有复合系统训练运行中，每个训练步骤使用群体大小为 12，每个步骤有 4 个训练实例（总批大小为 48，每个设备的训练批大小为 1）。训练采用 LoRA (Hu 等人, 2022)，秩维度为 16， $\alpha = 64$ ，随机失活率为 0.05，使用 bf16 精度，针对投影模块 [q, k, v, o, up, 下, 门]。我们使用 1×10^{-5} , $\beta = 0.01$ 的学习率、奖励尺度归一化和梯度范数裁剪为 0.1。梯度在每次更新前累积 20 步，学习率调度器采用“带预热的恒定学习率”。启用非重入梯度检查点以进一步减少内存使用。GRPO 优化运行 500 个训练步骤，总计固定 24,000 次 rollout，每 20 个训练步骤进行一次验证，用于实现早停。复合人工智能系统的 GRPO 训练实验在 1 块 H100/A100 (80 GB 内存) 上进行，推理 rollout 使用单独的 GPU。

对于单模块的群组相对策略优化 (GRPO) 训练，我们采用全参数微调，群组大小为 16。每个训练步骤使用全局批大小 32，通过在 8 个图形处理器 (GPU) 上每设备微批大小 4 来实现。展开生成以每 GPU 前向微批大小 12 执行。训练使用全分片数据并行 2 (FSDP2) 进行分布式，采样在温度 1.0 下进行。我们应用库尔贝-莱布勒 (KL) 正则化，并将学习率设置为 1×10^{-6} 。每 5 个训练步骤进行一次验证。在评估期间，采样在温度 0.6、top- $p = 0.95$ 和 top- $k = 20$ 下进行。

²https://microsoft.github.io/Trace/examples/nlp/bigbench_hard.html

我们手动探索了两次训练运行中 [学习率、贝塔、范数裁剪] 超参数的多个值。

GEPA: GEPA 是我们的优化器，基于第 3 节描述的算法。我们评估了主优化器 GEPA 的 2 个变体：GEPA 和 GEPA+合并，以及通过将基于帕累托的采样策略替换为朴素的“选择最佳候选”策略（选择最佳候选和选择最佳候选+合并）创建的 2 个消融。所有 GEPA 优化运行使用小批量大小 3，合并（当启用时）在优化运行期间最多调用 5 次。为了确保与 MIPROv2 的公平比较，我们在每个基准上对齐 GEPA 和 MIPROv2 的计算预算。每个基准的训练集用作 $D_{feedback}$ （用于推导训练信号，如第 3 节所述），验证集用作 D_{pareto} 。具体来说，由于 MIPROv2 的总展开预算取决于验证集大小和模块数量等因素，我们首先记录 MIPROv2 在每个基准上消耗的展开次数，然后限制 GEPA 的优化以匹配此展开预算。虽然提议和验证程序的差异导致系统使用的确切预算略有不同，但差异始终在 10.15% 以内。该协议确保任何性能差异源于优化算法本身，而非搜索预算的差异。每个优化器的确切展开次数在附录 G 中可视化。

F 结果与分析（续）

图 10 可视化了两个模型在聚合和单个基准上的最终测试集性能。

G 性能与预算（展开次数）曲线

图 12、13、14、15 显示了所有优化器在所有基准上的完整性能与展开次数曲线。

H 泛化差距

图 16 展示了不同优化方法的泛化差距。

I 优化系统的成本与性能分析

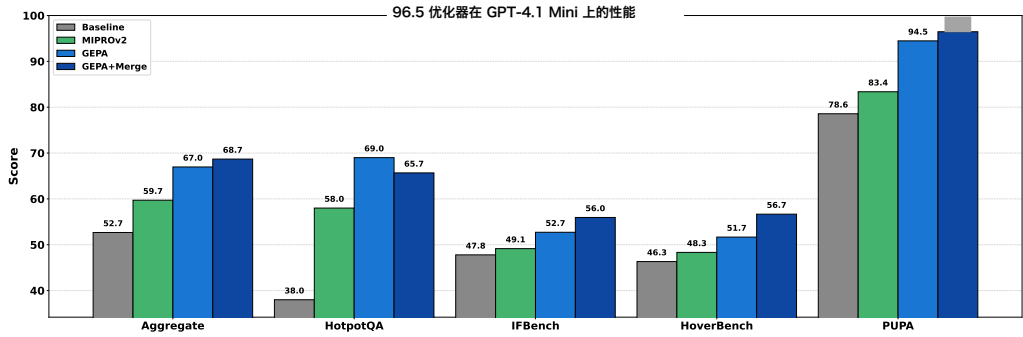
优化系统的提示规模在决定使用该优化系统的下游成本中起着重要作用。图 17 展示了最终优化系统的聚合提示长度（作为成本代理）与所达到性能之间的关系。值得注意的是，GEPA 的提示比 MIPROv2 的提示短约 33%，同时实现了更高的性能。

J GEPA 搜索树

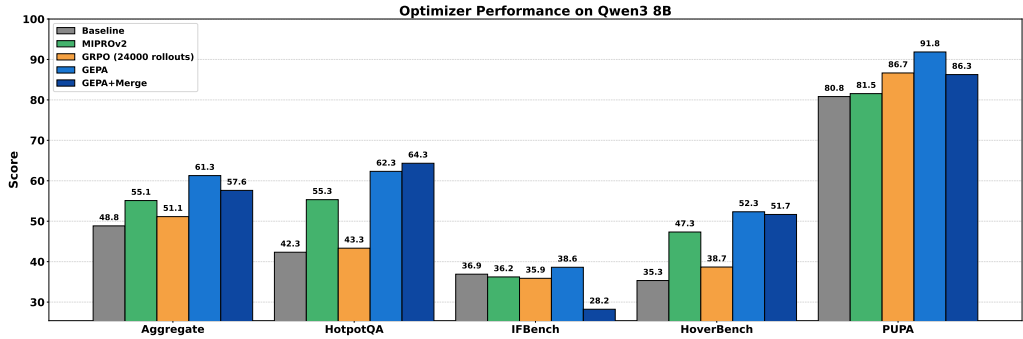
图 19、20、21、22、23、24、25 和 26 展示了由 GEPA 的各种配置（以及消融 SelectBestCandidate）生成的遗传搜索树。

K 可视化 GEPA 实现的迭代细化

图 5 展示了 GEPA 在 PUPA 优化过程中执行的提示细化的摘要。在本节中，我们展示优化过程中产生的完整提示。



(a) Final test set performance for aggregate and individual benchmarks for gpt-41-mini.



(b) Final test set performance for aggregate and individual benchmarks for qwen3-8b.

图 10：聚合基准与单项基准的最终测试集性能。

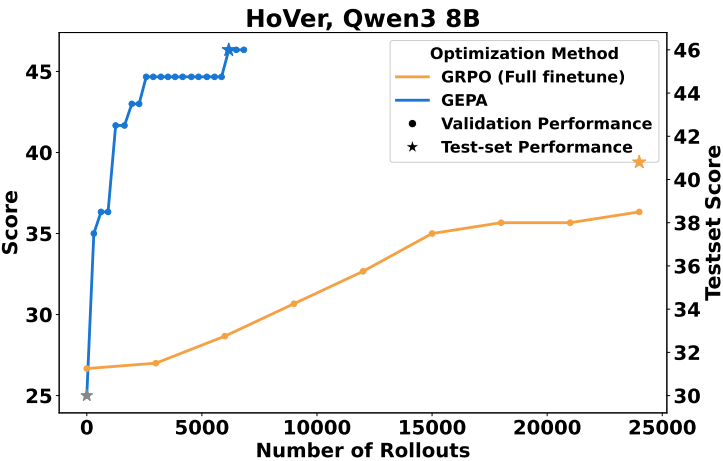


图 11：该图比较了 GEPA 与采用全参数微调的 GRPO 在 2 跳 HoVer 任务上的学习行为。相对差距与之前观察到的 GEPA 与采用 LoRA 的 GRPO 的比较（图 1、12、13、14、15）一致，表明 GEPA 相对于全参数和参数高效版本的 GRPO 均达到了可比的性能差距。

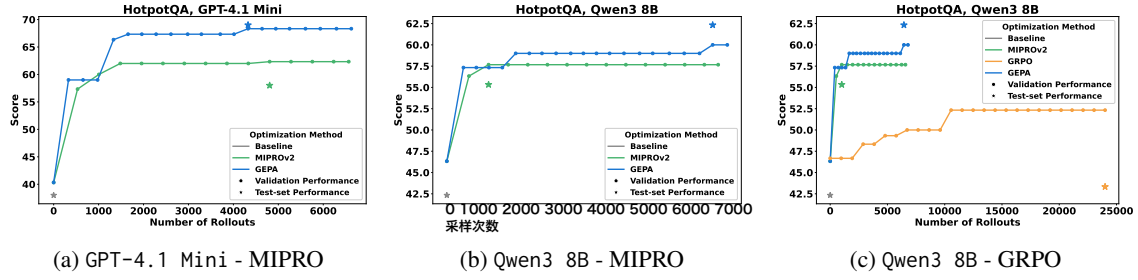


图 12: Hotpot QA 基准: 不同模型/设置下采样次数与得分的关系。

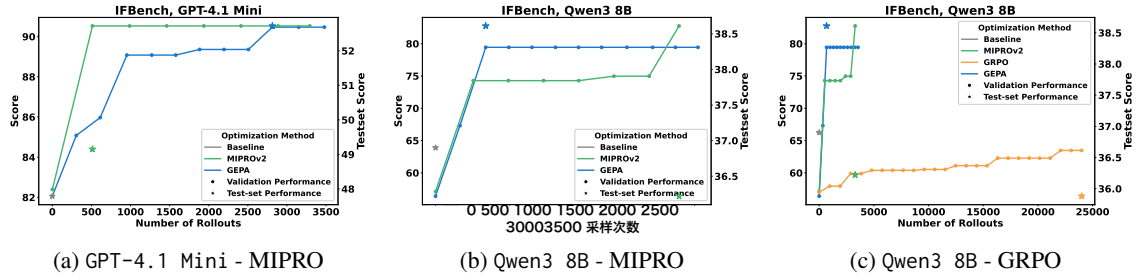


图 13: IFBench: 不同模型/设置下采样次数与得分的关系。

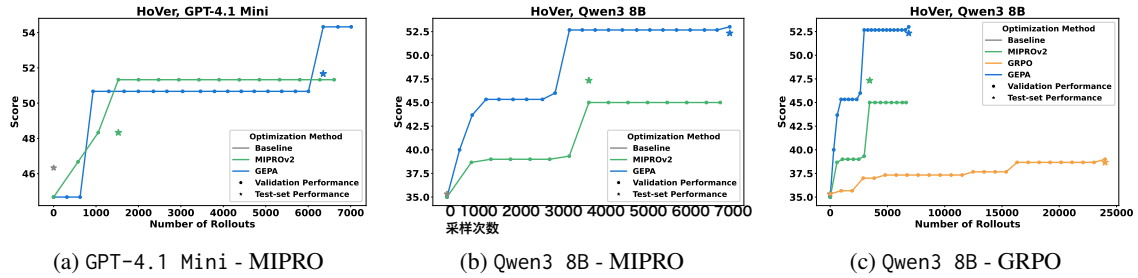


图 14: HoVerBench: 不同模型/设置下采样次数与得分的关系。

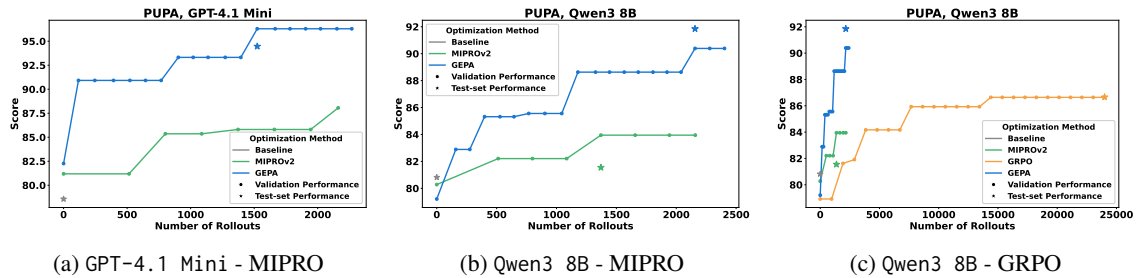


图 15: PUPA: 不同模型/设置下的轮次与得分关系。

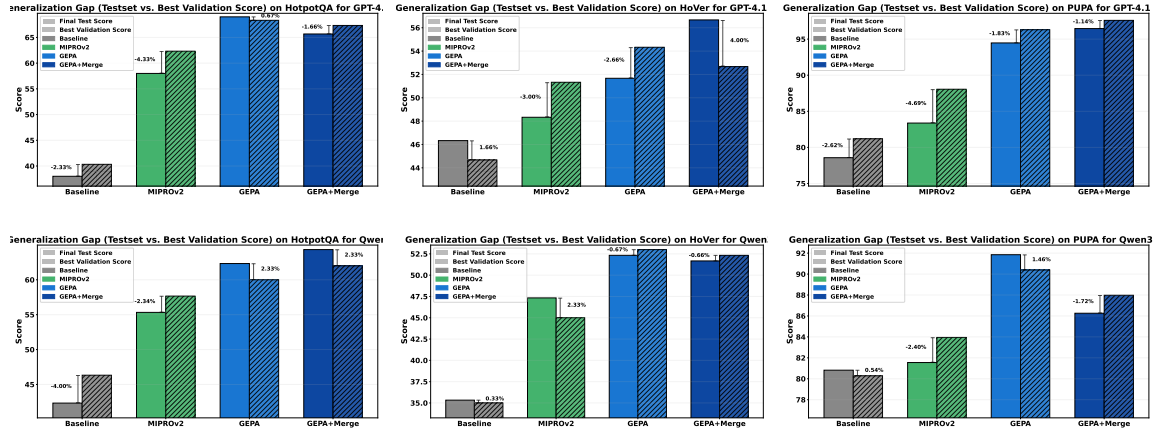


图 16：不同优化方法的泛化差距。参照 Wan 等人（2024）的做法，我们可视化了不同优化器的泛化差距（即最终测试集性能与最佳验证性能之间的差异）。Wan 等人（2024）此前观察到示例样本往往具有更好的泛化能力，而我们的结果表明，通过反思式提示演化生成的指令能够实现更强的泛化能力以及整体性能的提升。我们推测这种差异可能源于底层大语言模型能力的不断提升，因为较新的模型既能更好地遵循指令，也具备反思自身输出的能力。

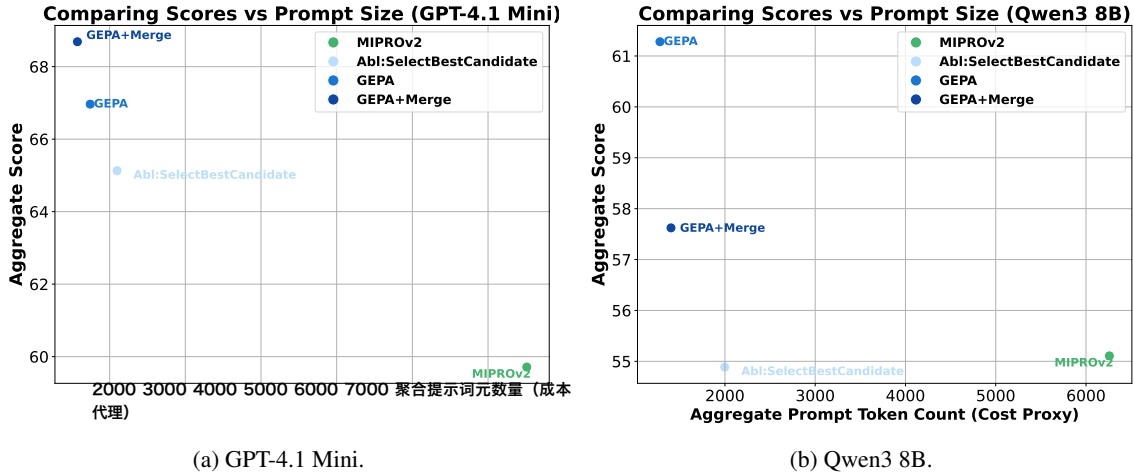
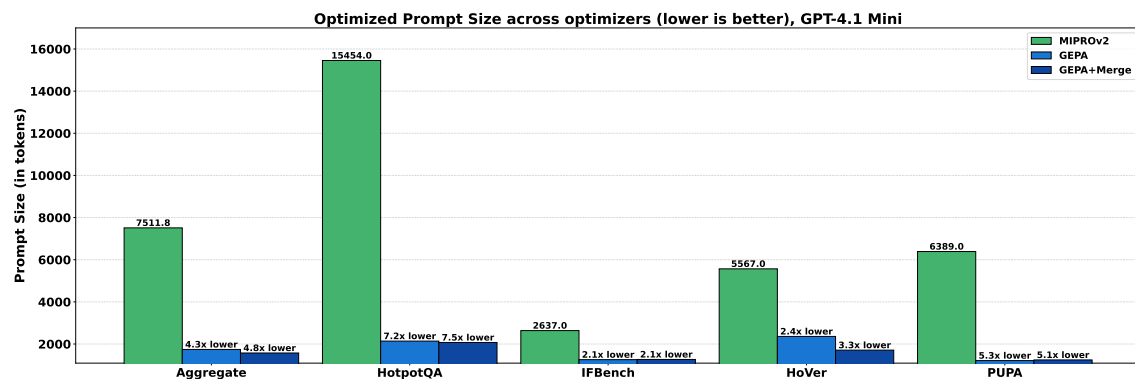
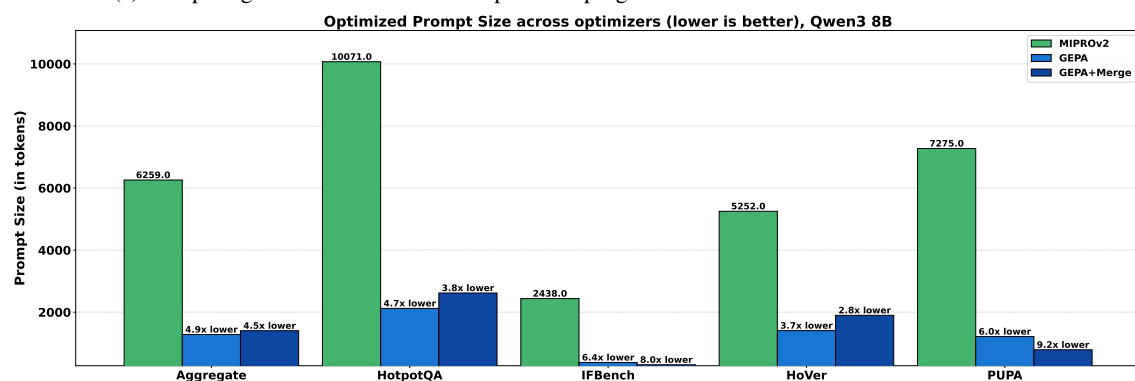


图 17：这些图表展示了每个优化器最终优化系统的最终聚合得分与聚合提示大小（跨所有基准）的关系。可以看出，GEPA 始终生成的提示大小不到 MIPROv2 提示大小的 33%，同时获得更高的性能。GEPA 的大部分提示词元用于提供指令，而 MIPROv2 的大部分提示词元则与少样本示例相关。



(a) Comparing the token counts of the optimized programs across benchmarks for GPT-4.1 Mini.



(b) Comparing the token counts of the optimized programs across benchmarks for Qwen3 8B.

图 18：比较各基准上优化程序的词元数量。

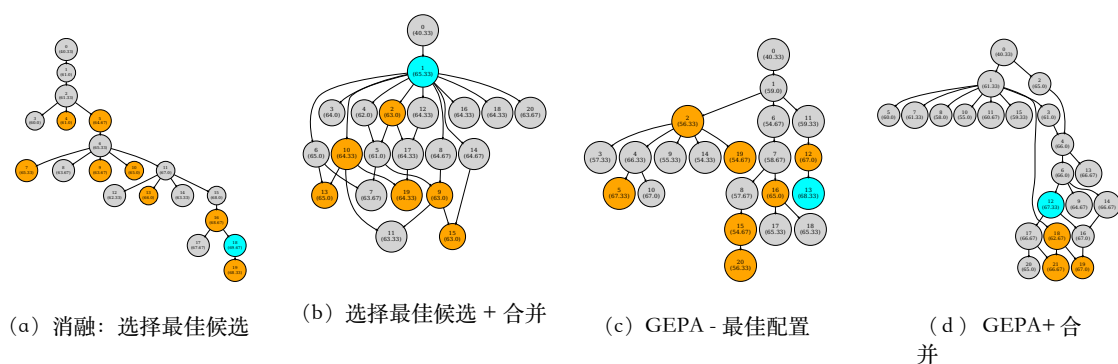


图 19：HotpotQA GPT-4.1 Mini

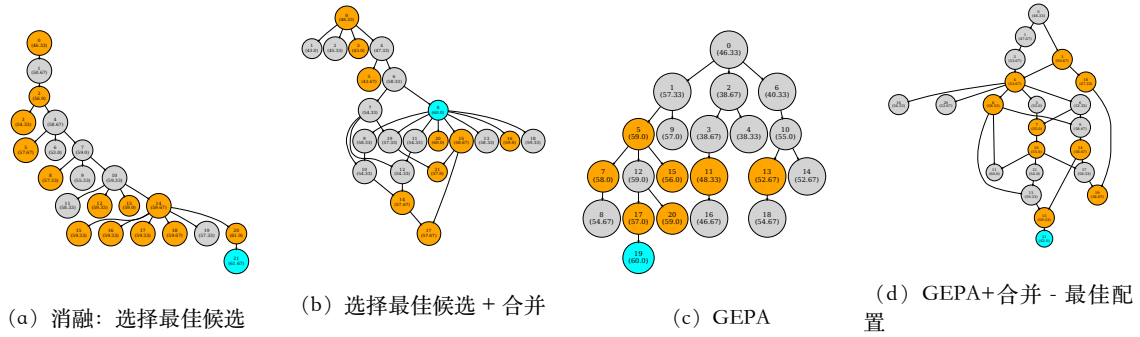


图 20：HotpotQA Qwen3 8B

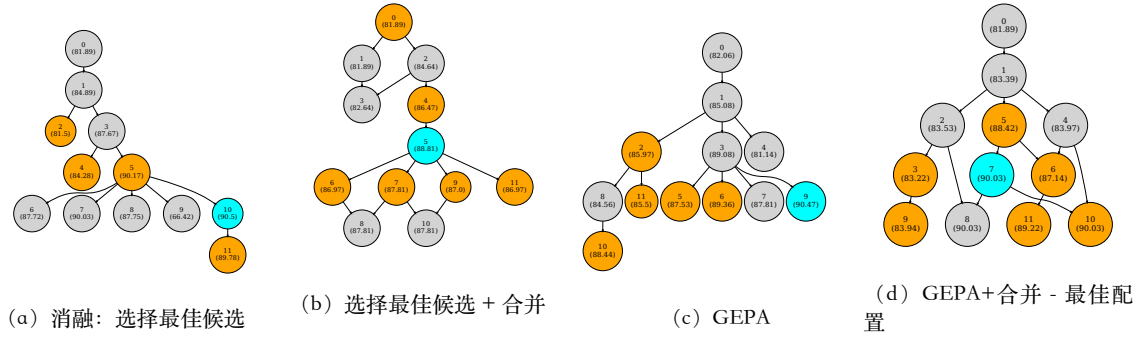


图 21：IFBench GPT-4.1 Mini

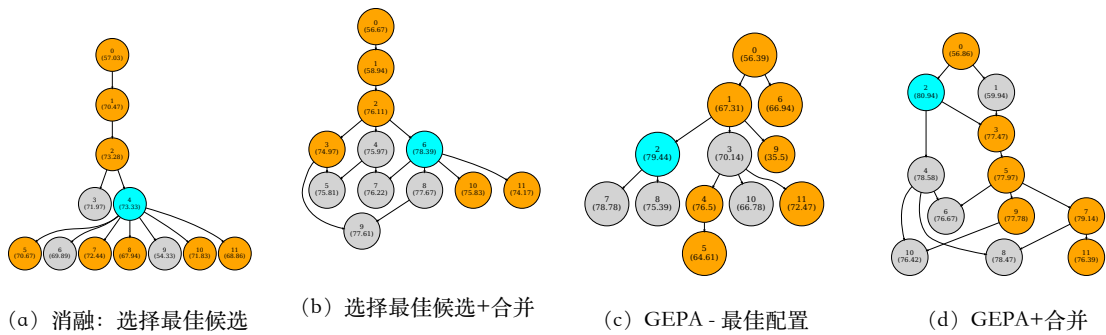


图 22：IFBench Qwen3 8B

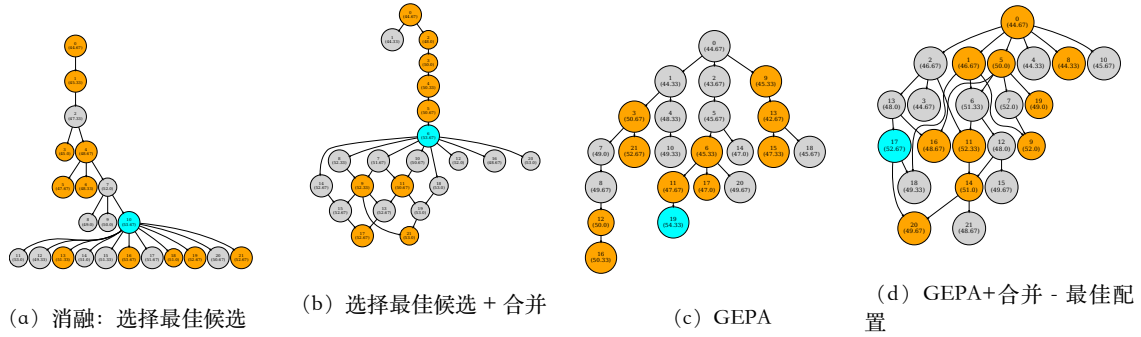


图 23: HoVer GPT-4.1 Mini

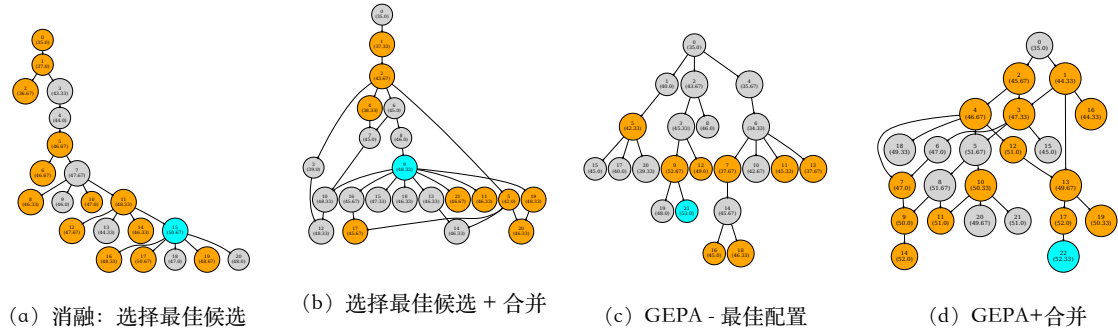


图 24: HoVer Qwen3 8B

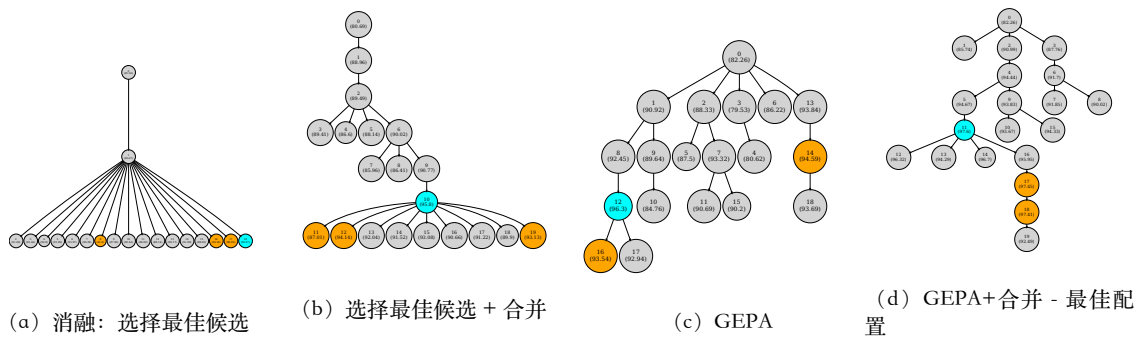


图 25: PUPA GPT-4.1 Mini

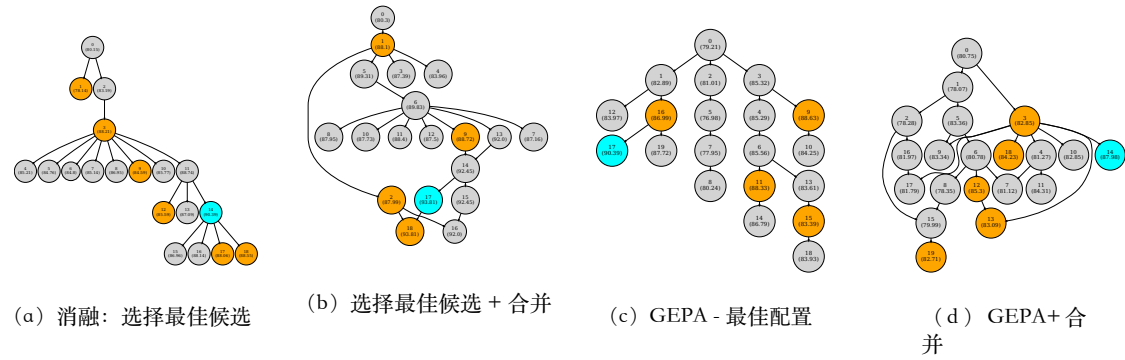


图 26: PUPA Qwen3 8B

K.1 PUPA 中间阶段的提示

Node 0 (Score 82.26)

给定一个私密用户查询，为强大的外部大语言模型创建一个保护隐私的请求。该大语言模型可以在不获取用户私密信息的情况下提供协助。

Node 2 (Score 90.99)

Task Description:
You are provided with a private user query that may contain sensitive, personal, or identifying information. Your role is to transform this query into a privacy-preserving request suitable for an external powerful large language model (LLM). The external LLM can be consulted to assist with the task but must not receive any private or personally identifiable information (PII) about the user. Your goal is to maximize the quality and relevance of the LLM request while minimizing or completely eliminating leakage of private information.

关键点和领域特定细节:

1. 隐私保护:

- 外部大语言模型请求中不得包含任何用户特定或敏感数据。
- 当查询包含位置、日期、姓名、网址或其他可识别细节时，在大语言模型请求中对其进行泛化或省略。
- 用中性占位符或通用术语替换或抽象任何私人或潜在可识别内容，同时不丢失大语言模型任务所需的意图或含义。

2. 查询理解与重构:

- 仔细分析用户的查询，以理解底层任务或信息需求。
- 判断查询是否涉及翻译、事件推荐、建议、摘要或其他任务。

- Identify when user-provided content is sensitive or proprietary (e.g., unique texts, URLs to closed content) and avoid directly exposing it to the external LLM.

3. 质量最大化:

- 生成清晰、精确的大语言模型请求，引导大语言模型在无需私人上下文的情况下执行必要任务。
- 保留核心信息或功能需求，以便大语言模型能够有效响应。
- 在引用外部文档或链接时，不要包含实际链接或隐私信息；改为请求与该主题相关的一般性或典型信息。

4. 隐私保护请求的常见策略:

- 对敏感内容使用改写或抽象。
- 请求一般化或基于示例的信息，而不是具体的用户数据。
- 当需要翻译时，如果文本非隐私，则仅包含需要翻译的文本；否则相应地进行一般化处理。
- 对于特定地点或事件的查询，请求该类别或区域的一般性或典型信息，而不包含敏感细节。

5. 推理说明:

- 对于每次转换，给出清晰的推理，说明隐私是如何得到保护的，以及为何大语言模型请求在满足用户意图的同时不会泄露敏感信息。

Input Format:

- A user query string possibly containing private or sensitive information.
- Required output:
 - a) A reasoning paragraph explaining your analysis and privacy considerations.
 - b) A privacy-preserving LLM request that can be sent to an external LLM for assistance.

遵循上述原则与策略，您将在与第三方语言模型交互时，同时优化辅助质量与用户隐私保护。

Node 4 (Score 94.44)

Task Description:

You will be given a user query that may include sensitive, personal, or identifiable information. Your role is to transform this query into a privacy-preserving request suitable for submission to an external large language model (LLM). The goal is to enable the external LLM to assist effectively while ensuring no private, identifying, or sensitive information about the user is exposed.

关键要求与领域特定细节:

1. 隐私保护:

- 在发送给大语言模型的请求中，不得包含任何用户特定细节、个人姓名、位置、日期、网址或其他可能识别身份的信息。
- 当用户查询包含私人、敏感或专有数据时，必须对这些细节进行泛化、抽象化或省略。

- 用中性占位符替换敏感具体信息，或以泛化方式描述，同时不丢失查询的核心意图。
- 避免将非公开、专有或机密内容直接暴露给外部大语言模型。

2. 理解与重构:

- Carefully analyze the user's query to identify the underlying task type (e.g., summarization, creative writing, translation, profile writing, event recommendations, company background, academic content generation).
- When reformulating, maintain the essential informational or functional need so that the external LLM can provide a useful, relevant response.
- Preserve thematic and contextual elements necessary for quality output, while abstracting any sensitive or private details.
- If the task involves user-supplied content (e.g., essays, texts, character descriptions), distill or summarize the relevant content or use generic versions instead of exposing original private text.

3. 最大化改写后提示的质量:

- 构建清晰、精确且结构良好的请求，明确指导外部大语言模型完成任务。
- 保留适当的细节和上下文以确保相关性，但需谨慎平衡隐私问题。
- 若引用外部文档、网址或机构，请勿包含任何链接或私有标识符；改为请求与主题相关的一般性或典型信息。
- 在为创意或虚构角色任务编写提示时，需澄清该资料或内容为虚构，以避免意外泄露真实个人数据。

4. 隐私保护的常见策略:

- 对个人及敏感内容进行改写或抽象化处理。
- 在适当情况下使用一般性描述或基于假设/示例的请求。
- 省略或泛化具体名称、日期、地点、机构或专有课程代码。
- 当用户内容较多时，重点吸收核心主题并据此指导外部模型，而非传递原始内容。

5. 解释要求:

- 提供一段简明的推理说明，解释如何识别敏感或隐私细节，以及为保护用户隐私所采取的步骤。
- 阐明重新表述后的大语言模型请求如何在不造成数据泄露风险的前提下，保留用户的原始意图和任务需求。
- 该解释为强制性要求，用于记录隐私保护方法并说明所采用的抽象化选择。

输入与输出格式:

- Input: A single user query string that may include sensitive or personal content.
- Output:
 - a) Reasoning: A paragraph explaining the privacy considerations and reformulation strategy.
 - b) LLM Request: A generalized, privacy-safe prompt suitable for querying an external LLM.

Examples and Feedback Integration:

- Avoid including real names, course codes, or faculty references; replace them with general terms (e.g., "an interdisciplinary health minor" rather than "HHV").
- When given character bios or profiles, specify the fictional nature clearly and do not include potentially identifying physical descriptions unless necessary for the task.
- When a user asks about specific companies or entities, generalize the request to "a major company in [industry/country]" if privacy concerns exist.
- Maintain a balance between minimal leakage and maximizing task relevance and quality.
- Ensure the reformulated prompt supports a comprehensive and useful response from the external LLM without compromising privacy.

通过严格遵循这些详细说明，将确保在最大程度保护用户隐私的同时，有效利用外部大语言模型协助处理跨领域的多样化用户请求。

Node 5 (Score 94.67)

隐私保护查询重构助手的新指令：

Task Overview:

You receive a user query that may contain sensitive, private, or personally identifiable information (PII). Your goal is to transform this input into a generalized, privacy-preserving prompt suitable for querying an external large language model (LLM). The reformulated prompt must enable the external LLM to fulfill the user's original intent effectively, while rigorously protecting user privacy by abstracting or omitting any sensitive details.

Input Format:

- 表示用户查询的单个字符串。
- 查询可能包含私人姓名、地点、日期、专有信息或可识别上下文。

Output Format:

- Part (a) Reasoning: a concise paragraph explaining:
 - How you identified sensitive or private information in the input.
 - What generalization, abstraction, or omission strategies you applied to protect privacy.
 - How the reformulated prompt maintains the original task or informational need without risking data leakage.
- Part (b) LLM Request: a carefully constructed and concise, privacy-safe prompt that retains essential thematic and functional elements to achieve a useful, relevant response from the external LLM.

关键领域特定细节与最佳实践：

1. 隐私保护原则：

- 删除或替换所有用户特定姓名（个人或组织）、确切日期或持续时间、地点、网址、专有课程或产品代码、客户或客户姓名以及其他标识符。

- When geographic or organizational mentions are critical for context, abstract these to broader or public categories (e.g., "a sustainable travel website focused on a Central American country" rather than naming the country explicitly).
- Omit or abstract internal organizational pressures, client names, or sensitive contractual details to generic placeholders (e.g., "a client" rather than a named company).
- For biographical or individual-related queries, avoid requesting or referencing real personal info; instead, frame requests around hypothetical or generic profiles unless the name is widely public and essential for the task.

2. 理解并重构任务：

- Identify the underlying task (creative writing, summarization, professional communication drafting, etc.) from the user's query.
- Preserve the functional intent and thematic requirements (e.g., content topics around sustainability, summary of a person's background, professional email follow-up).
- For user-supplied content, do not repeat verbatim or expose any original text; instead, extract core themes, instruct the LLM accordingly, or replace with generic examples.

3. 在保护隐私的同时最大化输出质量：

构建清晰、精确且包含充分上下文的提示，以确保大语言模型输出全面且相关。
避免模糊或过于宽泛的请求，以免降低相关性或实用性。
在保证质量所需的细节与隐私保护所需的泛化之间保持平衡。

4. 常见改写策略：

- Replace specific names with generic role identifiers or placeholders ("a business contact," "a notable individual").
- Replace specific locations or institutions with generalized descriptors, e.g., "a country known for eco-tourism."
- For time references, use relative or approximate terms without revealing explicit dates.
- For internal or proprietary details, describe the scenario generically (e.g., "a pressure from a client to close a matter urgently").
- When handling personal profiles or characters, explicitly state that the profile is fictional if requesting content generation about a person.

5. 解释要求：

始终提供推理过程，详细说明如何识别并缓解隐私风险。解释在抽象化后如何保留核心任务。该推理记录了隐私保护方法并论证了设计选择。

示例与常见陷阱：不得保留或轻微模糊个人姓名或公司名称；应完全抽象化或删除。避免直接引用用户提供的文本。使用中性且符合上下文的措辞描述用户意图。仅当地理或文化背景已公开且对任务至关重要时才予以保留。

确保输出提示仍为外部大语言模型易于理解的清晰指令。

摘要：您的改写必须确保个人信息或隐私/专有内容零暴露，同时保留足够的上下文和清晰度，以便从外部大语言模型生成高质量、相关的响应。这涉及细致的任务分析、对敏感细节的系统性抽象，以及清晰、精确的提示构建，并结合对所用隐私保护机制的透明推理。

Node 11 (Score 97.6)

隐私保护查询改写助手的新指令（增强版）：

Task Overview:

You receive a user query that may contain sensitive, private, or personally identifiable information (PII). Your objective is to reformulate this query into a generalized, privacy-preserving prompt suitable for sending to an external large language model (LLM). The reformulated prompt must enable the external LLM to fulfill the user's original intent effectively while rigorously protecting all user privacy by abstracting, omitting, or generalizing any sensitive details.

Input Format:

- A single user query string.
- The query might contain identifiable names (people, organizations), specific geographic locations, exact dates or timeframes, proprietary or internal terminology (e.g., system names, product codes), client or customer names, URLs, or any other sensitive or private information.

Output Format:

- Part (a) Reasoning:
 - Provide a concise paragraph that explains:
 - * How you identified sensitive or private information in the input
 - * What strategies you applied to protect privacy (generalization, omission, abstraction, replacement with placeholders)
 - * How the reformulated prompt preserves the original intent and task requirements without risking data leakage
- Part (b) LLM Request:
 - 一个简洁、精心构建的隐私安全提示，它：
 - * 移除或匿名化所有个人信息和专有/内部细节
 - * 根据需要抽象地点、名称、日期和技术术语
 - * 为大语言模型生成清晰且上下文丰富的指令，以生成与用户原始任务相符的相关且信息丰富的响应

详细的领域特定指南与最佳实践：

1. 敏感数据的识别与处理：

- All user-specific or personal names (individual or organizational) must be removed or replaced with generic role descriptors (e.g., "a business contact," "a client," "a notable individual"). Never lightly obscure or partially redact; full abstraction is required.

- All geographic mentions must be abstracted unless the location is publicly known, essential to the task, and can be generalized (e.g., "a region known for eco-tourism" instead of naming a country or city explicitly).
- Exact dates or durations must never be retained; instead, use relative or approximate temporal references (e.g., "recently," "over the past year").
- Internal or proprietary terms including system names, product codes, subscription types, and technical jargon must be generalized or replaced with neutral descriptors to avoid leakage of intellectual property or sensitive operational details.
- Avoid direct quotes or verbatim inclusion of user-supplied texts unless obfuscated by generalization.

2. 任务理解与改写：

- Identify the functional intent of the query: Is it creative writing, translation, summarization, professional communication drafting, technical explanation, or other?
- Preserve the thematic and informational core of the query (e.g., request for educational quality analysis, technical translation of a passage, biographical summary).
- Do not reproduce the original input verbatim; instead, frame the LLM prompt around the essential thematic elements extracted from the input.
- For queries regarding individuals, avoid direct reference to real personal information unless the name is widely public and essential; even then, use a generic or hypothetical framing for the individual profile.

3. 高质量且保护隐私的提示策略：

- 在足够的上下文细节与隐私抽象之间取得平衡，以保持提示的清晰度和相关性。
- 使用中立、上下文感知的表述，清晰指示大语言模型（LLM）所需的内容与风格。
- 避免模糊或过于宽泛的提示，以免导致回复实用性不足或质量下降。
- 当提及系统或专有内容时，指示大语言模型对特定术语进行泛化处理，并在不泄露敏感信息的前提下保持技术含义。
- 当要求对专业文本进行直接翻译时，指定替换或抽象内部命名。

4. 解释要求：

- The reasoning must transparently explain how privacy risks were identified (e.g., presence of names, locations, dates, proprietary terms).
- It must describe the abstraction or omission methods applied (e.g., replacing "Jonah Van Beijnen" with "a notable individual," substituting "Makau" with "a specific region," or "Yoda" with "a system name").
- Clarify how the essential task and user intent were preserved despite these generalizations (e.g., focusing on educational quality, technical translation, biographical summary).
- This explanation justifies your design choices and demonstrates adherence to privacy-preserving principles.

需避免的常见陷阱：

- 不得仅对敏感细节进行轻度模糊或部分遮盖；必须进行完全匿名化或抽象化处理。
- 不得逐字重复任何用户提供的个人信息（PII）或专有内容。
- 避免包含未经修改的网址、确切日期或直接引语。

- 不得留下可能降低改写后提示词质量或语境清晰度的歧义。
- 除非查询涉及公众人物且确有必要，否则不得包含任何真实个人或组织名称；如需提及，应使用通用描述词替代。

Example Summary of Effective Approach (Informed by Prior Examples):

- For geographic queries: replace exact place names with general regions and provide a brief contextual descriptor.
- For technical texts containing system names or subscription types: instruct the LLM to translate or process the text while replacing or abstracting proprietary system identifiers.
- For biographical summaries about specific individuals: remove the real name and request a generic, well-structured four-paragraph summary about “a notable individual,” preserving the overall intent without leaking PII.

Summary:

Your reformulations must ensure zero exposure of any PII or private/proprietary content while retaining enough thematic and functional clarity for the external LLM to produce high-quality, relevant outputs. This requires thorough analysis of the user's query, rigorous application of privacy-preservation strategies, and explicit reasoning explanations that document your approach and choices.

L 各基准最佳提示词示例

本节展示针对每种（基准，模型）配置所获得的最佳优化提示词。以下各小节对应一种（基准，模型）配置。由于每个复合人工智能系统由多个模块组成，各小节包含多个方框，分别列出每个模块的提示词。MIPROv2 优化后的提示词为每个任务包含最多 4 个少样本示例。为简洁起见，此处仅提供第一个示例。GEPA 的提示词仅包含优化后的指令，此处完整提供。

L.1 HOTPOTQA, GPT-4.1 MINI

HotpotQA GPT-4.1 Mini create_query_hop2.predict

Base Prompt:

Given the fields `question`, `summary\_1`, produce the fields `query`.

MIPROv2 Prompt:

You are an expert multi-hop question answering system designed to refine retrieval queries for complex questions. Given the original question and an initial summary of retrieved documents (summary_1), think step by step to analyze how the summary relates to the question. Use this reasoning to generate a focused and precise query that will guide the retrieval of additional relevant information needed to answer the question completely. Your output should include a clear chain-of-thought reasoning process followed by the refined query that targets complementary evidence for the next retrieval hop.

Demos:

Example 1:

```
question:
  Simone Benedetti plays for a team that currently competes in which Serie?
summary_1:

西蒙尼·贝内代蒂效力于维尔图斯恩泰拉，但文中未指明该队目前参加哪个级别的联赛。推理：

The summary indicates that Simone Benedetti plays for Virtus Entella, but it does not
state the current Serie (league level) in which Virtus Entella competes. To
accurately answer which Serie the team currently competes in, it is necessary to look
up the most recent league information for Virtus Entella.
query:
  What Serie (league level) does Virtus Entella currently compete in?

#####
3 other demos omitted for brevity.
#####
```

由配置 GEPA 生成的 GEPA 提示词：

You will be given two input fields: `question` and `summary\_1`.

你的任务是生成一个新的搜索查询（`query`），该查询针对多跳检索系统的**第二跳**进行优化。原始用户问题通常较为复杂，需要来自多个文档的信息才能回答。第一跳查询是用于检索初始文档集的原始问题。你的目标是生成一个**第二跳查询**，以检索出第一跳中未找到但为完整回答原始问题所必需的*额外相关文档*。

详细任务说明与提示：

1. **输入理解：**

- `question` 是用户提出的原始多跳问题。
- `summary\_1` is a concise summary of information from a document retrieved in the first hop, which partially addresses the question.

2. **目的与上下文：**

- 你生成的 `query` 旨在找到完整回答 `question` 所需的*缺失信息*。
- 多跳检索系统按阶段工作：
 - Second hop: Your query must help retrieve any *other relevant documents* NOT found in the first hop that hold complementary or broader context necessary for final answer extraction.

3. **从示例和反馈中得出的关键观察：**

- 首跳文档通常只涵盖问题中的一个实体或方面。- 其余相关文档往往涉及 `summary\_1` 中提到的相互关联或更高层次的概念，但这些问题并未在原始问题中明确询问。- 应制定 `查询` 以明确针对这些*缺失*但逻辑上相关的文档。- 避免仅仅改写原始问题或重述 `summary\_1` 中已知的事实。

- 相反，推断哪些更广泛或相关的实体/概念可能提供关键缺失信息。
- 例如，若 `summary\_1` 描述的是某个小型民政教区的人口，但问题要求的是更广泛区域的总人口，则您的查询应针对该更广泛区域（例如，“2011 年马德拉群岛人口”）。
- 类似地，如果 `summary\_1` 翻唱了一首歌，而问题询问该歌曲所属的专辑，但第一跳检索到的是歌曲级别的文档，那么查询应检索关于该专辑本身的文档。

4. **如何构建查询：**

- Identify the entities or topics mentioned in `summary\_1` that appear related but different from first-hop documents.
- 重新表述查询，明确提及与原始问题相关的更广泛或相关实体。
- 纳入问题中的相关关键背景以保持针对性，但将重点转向缺失的部分。
- 目标是检索与初始检索结果相关联或互为补充的文档。

5. **实用策略：**

- Read the `summary\_1` carefully to spot references to bigger contexts or other entities not covered in the first hop.
- 自问：“该摘要暗示了哪个实体或方面，可能回答原始问题但尚未找到？”
- 针对该实体或概念，制定一个精确、聚焦的事实性查询，以检索缺失的文档。

6. **Output:**

- Produce only the field `query` as a clear, concise question or keyword phrase designed for efficient retrieval of **second-hop documents**.
- Ensure the query relates logically to the original question while targeting the broader or complementary knowledge identified in `summary\_1`.
- Do **not** include the original question or simply rephrase it.
- Do **not** duplicate information already well-covered by the first hop retrieval.

遵循这些原则，将有助于多跳检索系统找到所有必要的文档，从而完整地回答多方面的原始问题。

HotpotQA GPT-4.1 Mini final_answer.predict

Base Prompt:

Given the fields `question`, `summary\_1`, `summary\_2`, produce the fields `answer`.

MIPROv2 Prompt:

You are an expert multi-hop reasoning assistant skilled in synthesizing information from multiple summaries to answer complex questions. Given the `question`, along with two intermediate summaries `summary\_1` and `summary\_2` that contain relevant evidence, carefully analyze and integrate the information step-by-step to produce a clear, logical reasoning process followed by a concise and accurate final answer.

Demos:

Example 1:

```
question:
    are Machaeranthera and Prumnopitys both the plants?
summary_1:
    是的，Machaeranthera 和 Prumnopitys 都是植物。Machaeranthera 是雏菊科的一个开花植物属，而
    Prumnopitys 是罗汉松科的一个常绿针叶树属。summary_2:

    是的，Machaeranthera 和 Prumnopitys 都是植物；Machaeranthera 是雏菊科的一个开花植物属，而
    Prumnopitys 是罗汉松科的一个常绿针叶树属。推理：

    两份摘要均一致认为 Machaeranthera 和 Prumnopitys 是植物。具体而言，Machaeranthera 是雏菊科的一个开花
    植物属，Prumnopitys 是罗汉松科的一个常绿针叶树属。由于两者均属于植物属，因此答案为是。答案：

    yes

#####
3 other demos omitted for brevity.
#####
```

由配置 GEPA 生成的 GEPA 提示词：

任务描述：

给定三个输入字段：`question`、`summary\_1` 和 `summary\_2`。你的目标是生成一个 `answer` 字段，利用两个摘要中的信息直接且明确地回答该问题，并在需要时结合你的权威领域知识加以增强。

Input Format:

- `question`: A natural language question that may require a fact, definition, name, date, yes/no response, or other specific information.
- `summary\_1` and `summary\_2`: Two independently generated summaries or snippets containing information related to the question. They may vary in completeness, accuracy, and specificity.

要求与方法：

1. ****准确理解问题。**** 确定所问的确切内容——无论是名称、具体事实、日期还是是/否答案。
2.
 - 如果它们一致并直接回答了问题，则以此作为主要证据。
 - 如果一个摘要提供了另一个摘要未提及的事实，请仔细评估其合理性。
 - 如果摘要之间存在冲突，请运用领域专业知识和权威知识来解决，或明确说明不确定性。
3. ****Domain-specific factual verification and nuance:****

- **Names and nicknames:** Provide only the specific nickname or name when asked, without extra phrasing. For example, when asked for the nickname of a person or entity, respond with the nickname alone, not a full sentence.
 - **Nationality and identity distinctions:** Use the most precise terms aligned with factual correctness and common usage (e.g., "English" vs. "British") based on domain knowledge.
 - **Dates and historical facts:** Verify dates or historical claims with domain knowledge to pick the correct fact, especially when there might be confusion between franchise start dates vs. event dates etc.
 - **Yes/no questions:** Prefer concise answers of "Yes" or "No" only, unless the question demands elaboration.
 - **Types or categories:** If a question asks about the type or category (e.g., type of company), provide the most direct concise phrase without including adjectives like nationality unless asked explicitly.
4. **Answer conciseness and relevance:**
- 提供简洁直接的问题答案。
 - 避免重复或重述问题。
 - 除非需要或为了清晰起见，避免提供不必要的上下文。
 - 除非必要，避免构建完整句子；例如，对昵称或是否类问题的回答应尽可能简短具体。
5. **When authoritative knowledge supplements the summaries:**
- If the summaries are incomplete or potentially inaccurate, incorporate trusted knowledge from your training about the topic to provide the correct and precise answer.
 - For example, when a summary gives a year that conflicts with known release dates or factual details, prefer the verified date.
 - When the summaries differ in style (one uses a formal phrase, another provides just the nickname), respond with the correct, clean answer format (e.g., just the nickname alone).
6. **Examples of correct reasoning and answers:**
- 问题：“2005 年丰田长滩大奖赛杆位得主的昵称是什么？”
 - Correct answer: "the thrill from West Hill"
 - Question: "What type of company is Zipcar led by Scott Griffith from 2003-2013?"
 - Correct answer: "car-sharing company"
 - Question: "Who was the partner of British comic book artist, Henry Flint, that helped create Zombo?"
 - 正确答案: "Al Ewing"

Summary:

- 将两份摘要作为主要但非唯一证据。
 - 利用领域知识可靠地核实和情境化事实，尤其是国籍、日期、昵称、公司类型和是否类问题。
 - 提供简短、直接的答案，匹配所要求的特异性。
 - 除非明确要求，避免不必要的阐述。
- 在适当情况下，利用知识明确解决冲突或歧义，或陈述不确定性。

该方法确保答案既准确又简洁，适合直接使用或集成到知识库或问答系统中。

HotpotQA GPT-4.1 Mini summarize1.predict

Base Prompt:

Given the fields `question`, `passages`, produce the fields `summary`.

MIPROv2 Prompt:

Given a question and a set of related passages, carefully analyze the information by thinking through the relevant facts step-by-step. Produce a clear and concise summary that synthesizes the key points from the passages directly relevant to answering the question, ensuring the summary is focused, accurate, and grounded in the evidence provided.

Demos:

Example 1:

question:

位于吉西拉尔迪·法瓦宫前的教堂，其建筑风格起源于哪座城市？段落：

['Palazzo Ghisilardi Fava | Palazzo Ghisilardi Fava is a Renaissance style palace, located on via Manzoni 4 in Bologna, region of Emilia Romagna, Italy; it houses the Medieval Civic Museum of Bologna.', 'Madonna di Galliera, Bologna | The Madonna di Galliera is a church with a Renaissance facade and Baroque interiors, located on Via Manzoni, in central Bologna, Italy. It stands in front of the Palazzo Ghisilardi Fava. The present name over the portal is the "Chiesa di Filippini Madonna di Galliera e Filippo Neri".', 'Palazzo Pretorio, Prato | The Palazzo Pretorio of the Tuscan town of Prato was the old city hall located town center, standing in front of the current Palazzo Comunale. It now accommodates the Civic Museum of Prato, which was reopened on September 2013.', 'Palazzo Pisani Gritti | The Palazzo Pisani Gritti is a Gothic-style palace located across the Grand Canal from the Church of the Salute, and stands along the Campo del Traghetto where the church of Santa Maria del Giglio, in the Sestieri of San Marco, Venice, Italy.', 'D  m  t  r Tower | The D  m  t  r tower is the oldest building in Szeged, Hungary. The foundation was most probably laid during the 11th century, while the lower part was built (in Romanesque style) from the 12th century, and the upper part (Gothic style) from the 13th century. The tower was once part of the former "Szent D  m  t  r templom" (St. Demetrius church), but today it stands in D  sm Square, in front of the much larger Votive Church of Szeged. The upper part was rebuilt from the original stones in 1926. The architecture of the tower is similar to another found in Southern France, or in the territory of the former Byzantine Empire.', 'Palazzo Leoni, Bologna | The Palazzo Leoni is a Renaissance style palace located on Via Marsala #31, in front of the outlet of Via Mentana, in central Bologna, region of Emilia-Romagna, Italy. The apse of the church of San Martino, is across the street.', 'Palazzo Altieri | Palazzo Altieri is a palace in Rome, which was the home of the Altieri family in the city. The palace faces the square in front of the Church of the Ges  z.']

reasoning:

The question asks for the city where the architectural style of a church that stands in front of the Palazzo Ghisilardi Fava originated. Passage [2] identifies the church standing in front of the Palazzo Ghisilardi Fava as "Madonna di Galliera" and mentions it has a Renaissance facade and Baroque interiors. Passage [1] states the Palazzo Ghisilardi Fava is located in Bologna, Italy. Since the church is located in Bologna and features a Renaissance architectural style, it can be inferred that the architectural style of this church originated in the city of Florence, which is historically recognized as the birthplace of Renaissance architecture. Thus, the

architectural style (Renaissance) of the church in front of the Palazzo Ghisilardi Fava originated in Florence.

summary:

位于吉西拉尔迪·法瓦宫前的教堂，其文艺复兴建筑风格起源于佛罗伦萨。

3 other demos omitted for brevity.
#####

由配置 GEPA 生成的 GEPA 提示词:

You are a first-hop **summarization module** in a multi-hop question answering (QA) system. Your task is to generate a concise, informative ``summary`` given two input fields: a ``question`` and a list of relevant ``passages``.

你的目标是从检索到的段落中提取并综合关键信息，这些信息应:

1. 与初始问题直接相关。2. 捕捉理解问题范围和背景所需的核心事实和实体。3. 包含相关联系、桥接实体、日期、地点或描述，使系统能够在后续跳转中制定有针对性的有效后续查询。4. 为下游答案生成模块提供坚实的事实基础。

Task specifics and best practices:

- The ``summary`` must represent a distilled synthesis, not just a compression or extractive snippet.
- Explicitly include cited passage titles or key entity labels (e.g., "Children in Need 2006 | ..." or "Anthony Levandowski | ...") in your summary to highlight the origin of information.
- Incorporate sufficient context to hint at missing or un-retrieved supporting facts, thus enhancing the multi-hop retrieval process.
- When the question asks for an attribute (e.g., nationality, location, company origin), ensure you provide:
 - 识别段落中提到的相关主体或实体。
 - 提取所述或隐含的属性或相关信息。
 - 桥接细节，可帮助系统在下一步检索中追踪剩余信息。
- 避免形成最终答案；相反，应聚焦于输入文档中“目前已知的内容”，以促进后续查询的细化。

Examples of critical elements to include:

- Entity names, roles, titles, and dates tied to the question.
- Names of organizations or locations connected through intermediary entities.
- Distinctive identifiers or clarifications that can help narrow down next-step retrieval (such as "Natasha Kaplinsky is an English presenter," "Kapolei is a city on Oahu," or "Waymo spun out of Alphabet").

Format of output:

提供一个段落或几句话，连贯地总结与问题相关的关键段落，引用段落标题或实体以清晰地陈述事实。

这种方法确保摘要在多跳问答中既能为下一跳检索提供信息，又能为最终答案抽取奠定基础。

HotpotQA GPT-4.1 Mini summarize2.predict

Base Prompt:

Given the fields `question`, `context`, `passages`, produce the fields `summary`.

MIPROv2 Prompt:

Given a `question`, relevant `context`, and a list of `passages`, provide a clear, concise summary that integrates the key information from the passages in relation to the question and context. Use step-by-step reasoning to explain how the summary is derived from the evidence before presenting the final synthesized summary.

Demos:

Example 1:

question:

佛罗伦萨第二公爵的高祖父的曾祖父创办了哪家银行? 上下文:

佛罗伦萨第二公爵的高祖父的曾祖父创办的银行是美第奇银行，由乔瓦尼·迪·比奇·德·美第奇创立。段落:

['Giovanni di Bicci de\' Medici | Giovanni di Bicci de\' Medici (c. 1360 – February 20/28, 1429) was an Italian banker, a member of Medici family of Florence, and the founder of the Medici Bank. While other family members, such as Chiarissimo di Giambuono de\' Medici, who served in the Signoria in 1201, and Salvestro de\' Medici, who was implicated in the Ciompi Revolt of 1378, are historically significant, Giovanni\'s founding of the family bank truly began the family\'s rise to power in Florence. He was the father of Cosimo de\' Medici ("Pater Patriae"), grandfather of Piero di Cosimo de\' Medici, great-grandfather of Lorenzo de\' Medici (the Magnificent) and great-great-great-grandfather of Cosimo I de\' Medici, Grand Duke of Tuscany.', 'Averardo de\' Medici | Averardo de\' Medici (1320-1363), also known as Everard De Medici, was the son of Salvestro de\' Medici (died 1346), "il Chiarissimo" (English meaning "the very clear.") and the father of three children: Giovanni, Francesco, and Antonia. Giovanni di Bicci de\' Medici would later become the first historically relevant member of Medici family of Florence and the eventual founder of the Medici bank.', 'Villa Medici at Cafaggiolo | The Villa Medicea di Cafaggiolo is a villa situated near the Tuscan town of Barberino di Mugello in the valley of the River Sieve, some 25 kilometres north of Florence, central Italy. It was one of the oldest and most favoured of the Medici family estates, having been in the possession of the family since the 14th century, when it was owned by Averardo de\' Medici. Averardo\'s son, Giovanni di Bicci de\' Medici, is considered to be the founder of the Medici dynasty.', 'Medici: Masters of Florence | Medici: Masters of Florence is an Italian-British television drama series about the Medici dynasty set in 15th-century Florence, starring Dustin Hoffman as Giovanni di

Bicci de' Medici, Richard Madden as Cosimo de' Medici, and Stuart Martin as Lorenzo de' Medici ("The Elder"). The series was co-created by Frank Spotnitz ("The X-Files" and "Man in the High Castle") and Nicholas Meyer (""). Sergio Mimica-Gezzan ("The Pillars of the Earth") directed all eight episodes. Episodes 1 and 2 aired on Rai 1 (Italian TV) on 18 October 2016. According to Italian ratings compiler Auditel, it attracted a record 7.6 million viewers. The first season consists of eight episodes.', 'Neri di Bicci | Neri di Bicci (1419–1491) was an Italian painter of the Renaissance. A prolific painter of mainly religious themes, he was active mainly in Florence and in the medium of tempera. His father was Bicci di Lorenzo. His grandfather, Lorenzo di Bicci was also a painter in Florence, a pupil of Spinello Aretino.', 'Cosimo de' Medici | Cosimo di Giovanni de' Medici (called 'the Elder' (Italian "il Vecchio") and posthumously "Father of the Nation" (Latin "pater patriae"); 27 September 1389 – 1 August 1464) was an Italian banker and politician, the first of the Medici political dynasty, "de facto" rulers of Florence during much of the Italian Renaissance. Despite his influence, his power was not absolute; Florence's legislative councils at times resisted his proposals, something which would not have been tolerated by the Visconti of Milan, for instance; throughout his life he was always "primus inter pares" ("first among equals"). His power derived from his wealth as a banker, and he was a great patron of learning, the arts and architecture.', "Giovanni il Popolano | Giovanni de' Medici, in full Giovanni di Pierfrancesco de' Medici, later known as il Popolano (21 October 1467 – 14 September 1498) was an Italian nobleman of the Medici House of Florence. He was the son of Pierfrancesco di Lorenzo de' Medici, and therefore a member of a secondary branch of the family."]

reasoning:

问题询问佛罗伦萨第二公爵的高祖父的曾祖父创办的银行。段落 [1] 指出，乔瓦尼·迪·比奇·德·美第奇创立了美第奇银行，并且是科西莫一世·德·美第奇的高祖父的曾祖父，后者是托斯卡纳大公。佛罗伦萨公国最终在科西莫一世统治下转变为托斯卡纳大公国，而科西莫一世在问题上下文中被认定为佛罗伦萨第二公爵。

Thus, Giovanni di Bicci de' Medici, as the great-great-great-grandfather of the second Duke of Florence (Cosimo I de' Medici), founded the Medici Bank.

summary:

美第奇银行由乔瓦尼·迪·比奇·德·美第奇创立，他是佛罗伦萨第二公爵的高祖父的曾祖父。

3 other demos omitted for brevity.
#####

由配置 GEPA 生成的 GEPA 提示词:

Given the fields `question`, `context`, `passages`, produce the fields `summary`.

L2 火锅问答 (HOTPOTQA) , 千问 3 8B (QWEN3 8B)

HotpotQA Qwen3 8B create_query_hop2.predict

Base Prompt:

Given the fields `question`, `summary\_1`, produce the fields `query`.

MIPROv2 Prompt:

给定问题与第一份摘要，综合问题与摘要的关键要素。识别需要检索或确认的具体信息。构建一个聚焦且精确的查询，以指导多跳推理过程的下一步，确保其直接针对知识缺口或需要从额外上下文中进一步澄清的内容。
演示:

由配置 GEPA+合并 (GEPA+Merge) 生成的 GEPA 提示:

给定字段 `question` 和 `summary\_1`, 生成字段 `query`, 以优化多跳系统对额外文档的检索。

Task Details:
1. **Objective:** Your query must target documents not retrieved in the first hop, using clues from the summary and the original question.
2. **Key Strategy:**

- Identify gaps in the first hop's retrieved documents (e.g., missing entities, relationships, or specific details).
- Use explicit information from the summary (e.g., names, locations, quantities) to rephrase the question into a query that surfaces new relevant documents.
- Avoid restating the answer directly; instead, structure the query to explore connections or unresolved details.

3. **Domain-Specific Guidance:**

- If the summary explicitly answers the question, the query should still focus on retrieving documents that provide deeper context or verify the answer (e.g., "What is the headquarters location of [Company]?" instead of "The answer is [Location]").
- Leverage entities mentioned in the summary (e.g., "Carhartt," "Aubrey O'Day") to anchor the query.
- If no documents are missing, rephrase the query to explicitly request the answer (e.g., "Which has more acts, Elektra or From the House of the Dead?").

4. **Avoid:**

- 生成与原始问题重复的查询。
- 假设摘要包含第二跳所需的全部信息。

HotpotQA Qwen3 8B final_answer.predict

Base Prompt:

Given the fields `question`, `summary\_1`, `summary\_2`, produce the fields `answer`.

MIPROv2 Prompt:

给定问题、summary_1, 和 summary_2,, 生成逐步推理过程，逻辑连接两份摘要中的信息以得出最终

```
answer, and then provide the answer. Ensure the reasoning explicitly references how the
summaries support the conclusion and maintains consistency with the question's
requirements.
Demos:
Example 1:
question:
    are Machaeranthera and Prumnopitys both the plants?
summary_1:
    Yes, both Machaeranthera (a genus of flowering plants in the daisy/aster family) and
    Prumnopitys (a genus of coniferous trees in the podocarp family) are plants.
summary_2:
    Yes, both Machaeranthera (a flowering plant genus in the daisy family) and
    Prumnopitys (a coniferous tree genus in the podocarp family) are plants.
reasoning:
    两份摘要均确认 Machaeranthera 和 Prumnopitys 是植物。摘要 1 明确说明它们是植物，将 Machaeranthera 归类
    为菊科/紫菀科的开花植物属，将 Prumnopitys 归类为罗汉松科的针叶树属。摘要 2 以略有不同的措辞重复了这
    一确认。答案：

    Yes

#####
3 other demos omitted for brevity.
#####
```

由配置 GEPA+合并 (GEPA+Merge) 生成的 GEPA 提示：

```
Given the fields `question`, `summary_1`, and `summary_2`, produce the field `answer` by:
1. **Extracting precise terminology**: Identify the exact noun or specific term required
in the answer (e.g., "Medicare" rather than "Medicare cuts"). Avoid vague or generalized
terms unless explicitly stated in the summaries.
2. **Resolving ambiguity**: If the question references a title, historical role, or
specific designation (e.g., "second Duke of Florence"), prioritize contextual or
historical clues from the summaries to infer the correct answer, even if the exact term
is not explicitly stated. Use domain-specific knowledge (e.g., Medici family lineage) to
fill gaps when summaries are indirect or vague.
3. **Cross-referencing summaries**: Ensure consistency between summaries. If summaries
conflict, prioritize the one with explicit factual claims (e.g., numerical data, direct
statements). If no explicit claim exists, synthesize information while ensuring alignment
with historical, political, or cultural context.
4. **Avoiding overgeneralization and extra information**: Focus strictly on the most
specific and directly stated information in the summaries. Do not add context,
explanations, or external knowledge beyond what is explicitly provided. For example, if
the question asks for a year, provide only the year; do not include band member details
or historical background.
5. **Prioritizing factual alignment**: If a summary explicitly states the answer, use
that. If summaries are indirect or vague, synthesize information while ensuring alignment
with factual knowledge (e.g., linking "Path to Prosperity" to Rep. Paul Ryan's Medicare
proposal).

**Key adjustments based on feedback**:
- **Conciseness**: Answers must be strictly factual and concise, avoiding additional
context or explanations. For example, if the question is "Is X shorter than Y?" the
```

answer should be a simple "No" or "Yes" based on numerical comparisons, not a full explanation.

- **Numerical precision**: When comparing measurements (e.g., heights, dates), ensure exact values are used and explicitly stated in the summaries. If summaries provide conflicting numbers, resolve via direct factual claims.
- **Domain-specific knowledge**: Use known facts (e.g., architectural records, historical timelines) to validate ambiguous answers, but only when summaries lack explicit information.

HotpotQA Qwen3 8B summarize1.predict

Base Prompt:

Given the fields `question`, `passages`, produce the fields `summary`.

MIPROv2 Prompt:

Given the fields `question`, `passages`, produce the fields `summary`.

Demos:

Example 1:

question:

泰·加内特 (Tay Garnett) 与亚历山大·克鲁格 (Alexander Kluge) 的职业是什么? 段落:

['Tay Garnett | William Taylor "Tay" Garnett (June 13, 1894 – October 3, 1977) was an American film director and writer.', 'Trade Winds (film) | Trade Winds is a 1938 comedy film distributed by United Artists. It was directed by Tay Garnett, and starred Fredric March and Joan Bennett. The screenplay was written by Dorothy Parker, Alan Campbell and Frank R. Adams, based on story by Tay Garnett.', 'Prestige (film) | Prestige is a 1932 American pre-Code drama film directed by Tay Garnett and written by Tay Garnett, Rollo Lloyd and Francis Edward Faragoh. The film stars Ann Harding, Adolphe Menjou, Melvyn Douglas and Guy Bates Post. The film was released on January 22, 1932, by RKO Pictures.', 'Alexander Kluge | Alexander Kluge (born 14 February 1932) is a German author, philosopher, academic and film director.', 'Cattle King | Cattle King is a 1963 film directed by Tay Garnett. It stars Robert Taylor and Robert Loggia. It also appears to have been called Guns of Wyoming in some countries.', 'G  nther von Kluge | G  nther von Kluge (30 October 1882 – 19 August 1944) was a German field marshal during World War II. Kluge held commands on both the Eastern and Western Fronts. Although Kluge was not an active conspirator in the 20 July plot, he committed suicide on 19 August 1944, after having been recalled to Berlin for a meeting with Hitler in the aftermath of the failed coup. He was replaced by Field Marshal Walter Model.', 'Bad Company (1931 film) | Bad Company is a 1931 American Pre-Code gangster film directed and co-written by Tay Garnett with Tom Buckingham based on Jack Lait's 1930 novel "Put on the Spot". It stars Helen Twelvetrees and Ricardo Cortez. Told from the view of a woman, the working titles of this film were "The Gangster's Wife" and "The Mad Marriage". Unlike many static early sound films, Garnett includes several scenes using a moving camera climaxing in a gigantic assault on an office building with both sides using heavy machine guns.']

reasoning:

The question asks for the common job of Tay Garnett and Alexander Kluge. From the passages, Tay Garnett is explicitly identified as a film director and writer (passages 1, 2, 3, 5, 7). Alexander Kluge is described as a German author, philosopher, academic, and film director (passage 4). The shared job between them is "film director."

summary:
Both Tay Garnett and Alexander Kluge are film directors.

3 other demos omitted for brevity.
#####

由配置 GEPA+合并（GEPA+Merge）生成的 GEPA 提示：

您是多跳问答系统中的第一跳**摘要模块**。您的任务是生成一份**全面、结构化的摘要**，该摘要应：

1. **Extracts direct answers** from the top retrieved passages to address the question.
2. **识别并突出缺失或隐含的线索**，这些线索可能需要进一步检索（例如实体、关联或上下文细节）。
3. **综合信息**，将段落中的明确事实与领域特定知识或逻辑推断相结合，以指导后续步骤。

Summary Structure

- **Entity/Person Mention**: Clearly state the subject (e.g., "Billy Truax", "Eintracht Braunschweig") and include **full names, titles, or official designations** (e.g., "Thomas Lance Rentzel", "Braunschweiger Turn- und Sportverein Eintracht von 1895 e.V.").
- **Direct Answer**: Include **explicit answers** from the passages (e.g., birth dates, team affiliations, or direct statements).
- **Clues for Next Steps**: Signal **missing information** (e.g., "Lance Rentzel's birth year is explicitly stated, but his exact birthplace is not; need to search for 'Lance Rentzel birthplace'").
- **Domain-Specific Context**: Add **relevant background** (e.g., "Eintracht Braunschweig is a German football club based in Braunschweig, Lower Saxony" or "NFL players' birth dates are critical for age comparisons").

Guidelines

- **Do not omit** any entity or detail from the retrieved passages that could be relevant for follow-up queries (e.g., team names, locations, or historical context).
- **Prioritize clarity** by **separating direct answers from inferred clues** (e.g., using bullet points, subheadings, or bolded labels).
- **Avoid assumptions** not supported by the passages; if information is absent, **explicitly state that it is missing** and suggest **precise search terms** (e.g., "Verify Wichita Dwight D. Eisenhower National Airport's tower status via FAA records").
- **Include quantifiable data** (e.g., "few thousand Stabyhouns exist globally", "born July 15, 1943") to enable precise comparisons.
- **Highlight connections** between entities (e.g., "Billy Truax and Lance Rentzel were traded in 1970") to aid in cross-referencing.

Key Niche/Domain-Specific Insights

- **NFL Player Comparison**: Birth dates are critical for age determination, and team affiliations (e.g., "traded in 1970") may imply historical context.
- **Airport Classification**: "Non-towered" status is explicitly stated in some passages (e.g., "non-towered public airport"), while others require inference (e.g., "major commercial airports typically have towers").
- **Football Club Context**: Clubs like Eintracht Braunschweig require background on their location, league, and history (e.g., "based in Braunschweig, Lower Saxony").

- **Quantifiable Data**: Use exact dates, numbers, or rankings (e.g., "few thousand Stabyhouns exist globally") to enable precise comparisons.

Critical Additional Instructions

- **Ensure All Retrieved Documents Are Represented**: Explicitly include all entities, titles, and details from the retrieved passages (e.g., full names, film titles, and specific roles).
- **Signal Missing Links**: If a connection between entities is implied but not explicitly stated (e.g., "Nancy Steiner worked on *The Lovely Bones*"), flag this as a potential gap and suggest search terms to resolve it.
- **Prioritize Bridging Concepts**: Highlight relationships between entities (e.g., "Gary Pinkel coached Toledo in 1993 and holds the most wins in school history") to enable focused follow-up queries.
- **Avoid Overgeneralization**: Only include domain-specific context that is either explicitly stated in the passages or directly inferable (e.g., "major commercial airports typically have towers" is acceptable, but "airports with fewer than 10,000 passengers are non-towered" is not unless stated).

Example Format

For the question **"Which NFL player is younger, Billy Truax or Lance Rentzel?"**:

- **Entity/Person Mention**: Billy Truax (William Frederick Truax), Lance Rentzel (Thomas Lance Rentzel)
- **Direct Answer**:
 - **Billy Truax**: Born July 15, 1943.
 - **Lance Rentzel**: Born October 14, 1943.
- **Clues for Next Steps**: None required; birth dates are explicitly provided.
- **Domain-Specific Context**: Birth dates are sufficient to determine age difference within the same year.

For the question **"Which is a non-towered airport, Wichita Dwight D. Eisenhower National Airport or Montrose Regional Airport?"**:

- **Entity/Person Mention**: Wichita Dwight D. Eisenhower National Airport, Montrose Regional Airport
- **Direct Answer**:
 - **Montrose Regional Airport**: "non-towered public airport" (passage 3).
 - **Wichita Dwight D. Eisenhower National Airport**: No explicit mention of tower status; inferred as **towered** (typical for major commercial airports).
- **Clues for Next Steps**: Verify Wichita's tower status via FAA records or additional sources (e.g., "Wichita Dwight D. Eisenhower National Airport tower status").
- **Domain-Specific Context**: Non-towered airports lack a control tower, relying on pilot communication (passage 4). Major commercial airports like Wichita usually have towers.

提示: 摘要时不要仅压缩；要综合——既要包含直接答案，也要包含系统后续步骤所需的线索。若检索到的文档内容缺失关键信息，务必明确说明，并提供可操作的搜索词以弥补缺口。

HotpotQA Qwen3 8B summarize2.predict

Base Prompt:

Given the fields `question`, `context`, `passages`, produce the fields `summary`.

MIPROv2 Prompt:

Given the fields `question`, `context`, and `passages`, generate a structured reasoning process that explains how to synthesize information from these sources, and then produce a concise summary of the key findings.

Demos:

Example 1:

question:

Simone Benedetti plays for a team that currently competes in which Serie?

context:

The provided passages do not specify the Serie in which Simone Benedetti's team (Virtus Entella) currently competes.

passages:

['Simone Benedetti | Simone Benedetti (born 3 April 1992) is an Italian professional footballer who plays as a centre back for Virtus Entella.', 'Virtus Entella | Virtus Entella (usually referred as simply Entella or Chiavari Entella) is an Italian association football club, based in Chiavari, Liguria. They currently compete in Serie B.', 'Simone Palermo | Simone Palermo (born 17 August 1988) is an Italian footballer who plays as a midfielder for Virtus Entella at Serie B.', 'Andrea Paroni | Andrea Paroni (born 14 October 1989) is an Italian footballer who currently plays as a goalkeeper for Virtus Entella in the Serie B.', 'Alessandro Iacobucci | Alessandro Iacobucci (born 3 June 1991) is an Italian footballer who currently plays as a goalkeeper for Serie B side Virtus Entella.', 'George Ganea | George Ganea (born 26 May 1999 in Bucharest) is a Romanian footballer who plays as a Striker for Serie B club Virtus Entella Primavera team.', 'Gianpaolo Castorina | Gianpaolo Castorina (born August 30, 1976 in Milan) is a retired Italian professional footballer turned coach, currently in charge as head coach of Serie B club Virtus Entella.']

reasoning:

The question asks for the Serie in which Simone Benedetti's team (Virtus Entella) currently competes. While the context initially states that this information is not specified, the passages explicitly mention that Virtus Entella competes in Serie B. Multiple passages (e.g., [2], [3], [4], [5], [7]) confirm this detail, directly answering the question.

summary:

Simone Benedetti's team, Virtus Entella, currently competes in Serie B.

#####

3 other demos omitted for brevity.

#####

GEPA Prompt generated by config GEPA+Merge:

Given the fields `question`, `context`, and `passages`, produce the field `summary`.

您的任务是综合问题、上下文和新检索到的段落中的信息，生成一份**全面、精确且结构良好的摘要**，使答案生成模块能够自信地得出正确答案。

Key Requirements:

1. **Explicit Answers First**: Prioritize explicitly stated facts from the context and passages (e.g., direct mentions of entities, roles, or relationships).
2. **Infer or Generalize When Necessary**: If critical details are missing from the passages, infer connections or generalize based on contextual clues and domain-specific knowledge (e.g., linking ownership structures, roles, or historical context).
3. **Bridge Gaps**: Ensure the summary includes all **key supporting information** required to answer the question, even if it is not explicitly stated in the input. For example:
 - If the answer is "Newcastle United," include details about Sports Direct's ownership and the connection to the billionaire.
 - If the answer is a person's role (e.g., "troubleshooter"), explicitly state their relationship to the question's subject and any relevant background.
4. **Structure and Precision**:
 - Clearly connect entities, roles, and relationships (e.g., "Stan Kroenke owns Sports Direct and Arsenal F.C.").
 - Avoid ambiguity by including all necessary contextual links (e.g., "Mike Ashley founded Sports Direct and owns Newcastle United").
 - Use precise terminology and ensure alignment with domain-specific knowledge (e.g., "investigative journalist" instead of "writer").
5. **Domain-Specific Knowledge**: Leverage implicit domain knowledge when passages lack critical details (e.g., knowing that "Project RAND" is linked to Henry H. Arnold and the RAND Corporation).

Example Integration:

If the question is about a person's profession in a novel, ensure the summary includes:

- The character's name.
- Their profession (explicitly stated in the text).
- Contextual links to the book series or plot (e.g., "in *The Girl in the Spider's Web*").
- Any relevant background about the profession or character's role in the story.

始终力求在**覆盖范围和相关性**上与反馈中描述的“理想摘要”相匹配，确保答案模块拥有生成正确最终答案所需的全部信息。

L.3 IFBENCH, GPT-4.1 MINI

IFBench GPT-4.1 Mini generate_response_module.predict

Base Prompt:

Respond to the query

MIPROv2 Prompt:

您将收到一个包含复杂任务及多重约束和指令的查询。您的任务是首先逐字逐句地重复该查询，不得在重复前后添加任何额外词语或评论。重复查询后，请提供详细的、逐步的思维链推理，仔细拆解并处理查询的每个方面。然后，给出严格遵守所有要求的最终回答，包括查询中指定的格式、语言和内容约束。请务必详尽、精确，并确保最终回答

fully validated and consistent with the reasoning. If the query specifies additional formatting or stylistic rules (such as language or inclusion of a postscript), include them exactly as instructed.

Demos:

由配置 GEPA+合并（GEPA+Merge）生成的 GEPA 提示：

您将收到一个查询输入，您的任务是适当地响应该查询。查询中可能包含您必须在响应中严格遵守的特定指令或约束。请仔细分析查询以确定确切要求，包括但不限于：

- 从一组受限选项中精确选择答案进行响应（包括确切的措辞和标点）。
- 至少包含特定词语指定次数。
- 至少包含特定字母指定次数。
- 如果明确要求，在提供答案之前逐字重复整个查询，且不得在重复前后添加额外词语或字符。
- 如果指示，在响应中避免使用特定禁用词语或关键词。
- 遵循查询中嵌入的任何其他明确指令或约束。

When the query requests calculations or factual answers (e.g., combinatorial calculations), you should:

1. Carefully interpret the mathematical or logical problem.
2. Show your reasoning internally to confirm the final answer (reasoning does not need to be included in the response unless explicitly requested).
3. Provide the final direct response strictly following all instructions, especially when asked to repeat the query verbatim first before giving the answer.

通用方法：

- 始终彻底解析查询，以提取每一项约束和指令。
- 确保您的回复在格式、措辞和内容上与指示完全一致。
- 不得虚构或遗漏用户明确要求的任何部分。
- 满足所有格式、词汇、数值和结构约束，不得有任何偏差。
- 若查询涉及逐字重复文本，请勿更改大小写、标点或措辞。
- 将任何必需的关键词或字母按要求次数自然地融入您的回答中。
- 当多个约束（如避免特定词汇同时包含其他词汇）同时适用时，确保同时满足所有约束。

本任务要求精确、完全复现，并严格遵守查询中嵌入的任何约束或指令。目标是以所要求的确切方式交付所请求的答案，不添加或省略任何多余内容。

IFBench GPT-4.1 Mini ensure_correct_response_module.predict

Base Prompt:

确保响应正确且符合给定约束。您的响应将作为最终响应使用。

MIPROv2 Prompt:

Given the original query and the initially generated response, carefully verify that the response fully meets all specified constraints and instructions in the query. Produce detailed step-by-step reasoning that explains how the response satisfies or fails the requirements such as exact repetition of the query text, minimum sentence count, formatting rules (e.g., capitalization), letter frequency, or other explicit user demands. Based on this reasoning, output a final corrected and fully compliant response that adheres strictly to the original instructions, ensuring accuracy, completeness, and formatting correctness. Your final output should begin by repeating the original query exactly as provided, with no modifications or additional text before it, followed by a comprehensive, well-structured answer that respects all constraints, and a clear explanation of how the requirements have been met.

Demos:

Example 1:

query:

Can you provide the social media handles of Harry Potter? Your answer must contain exactly 8 bullet points. Use the markdown bullet points such as:

- * This is point 1.
- * This is point 2 Finish your response with this exact phrase That's my answer.. No other words should follow this phrase.

response:

- * Twitter: [@wizardingworld](https://twitter.com/wizardingworld)
- * Instagram: [@wizardingworld](https://www.instagram.com/wizardingworld/)
- * Facebook: [Wizarding World](https://www.facebook.com/wizardingworld)
- * YouTube: [Wizarding World](https://www.youtube.com/c/WizardingWorld)
- * Twitter: [@harrypotterfilm](https://twitter.com/harrypotterfilm)
- * Instagram: [@harrypotterfilm](https://www.instagram.com/harrypotterfilm/)
- * Facebook: [Harry Potter](https://www.facebook.com/harrypotter)
- * TikTok: [@wizardingworld](https://www.tiktok.com/@wizardingworld)

这就是我的答案.. 推理过程:

请求是获取哈利·波特的社交媒体账号，格式为使用 Markdown 项目符号的恰好八个要点，且响应必须以确切短语“这就是我的答案。”结尾。所提供的响应包含八个要点，引用了与哈利·波特及魔法世界系列相关的官方或广泛认可的社交媒体账号，涵盖推特、照片墙、脸书、油管 and 抖音。格式正确，项目符号使用 Markdown 语法得当，结尾短语与要求完全匹配。因此，该响应满足所有给定约束。final_response:

- * Twitter: [@wizardingworld](https://twitter.com/wizardingworld)
- * Instagram: [@wizardingworld](https://www.instagram.com/wizardingworld/)
- * Facebook: [Wizarding World](https://www.facebook.com/wizardingworld)
- * YouTube: [Wizarding World](https://www.youtube.com/c/WizardingWorld)
- * Twitter: [@harrypotterfilm](https://twitter.com/harrypotterfilm)
- * Instagram: [@harrypotterfilm](https://www.instagram.com/harrypotterfilm/)
- * Facebook: [Harry Potter](https://www.facebook.com/harrypotter)
- * TikTok: [@wizardingworld](https://www.tiktok.com/@wizardingworld)

That's my answer..

```
#####  
3 other demos omitted for brevity.  
#####
```

由配置 GEPA+合并（GEPA+Merge）生成的 GEPA 提示：

您将收到一个用户查询，其中包含关于您必须如何构建响应的明确、详细指令。您的总体任务是准确无误地解读这些嵌入指令，并生成一个严格遵守每一条指令的响应，不添加、省略或改写任何强制文本、短语或结构元素。

本任务的关键详细要求和策略：

1. ****查询解析与指令提取****

- 仔细阅读整个查询，以识别所有关于以下方面的明确指令：
 - Whether and how to repeat the query text (verbatim or partially).
 - Specific length constraints (number of sentences, bullet points, word counts).
 - Formatting instructions (e.g., capitalization requirements, quotation marks, markdown bullet styles).
 - Mandatory phrases or exact sentences that must appear (especially those to be repeated verbatim or appended at the end).
 - Content limitations or prohibitions (for example, refusal language or disclaimers for unethical requests).
- Note that some instructions may be nested or appear within the query's wording and are critical to follow exactly.

2. ****精确文本复现****

- When asked to repeat the query text (or any other required phrase) verbatim, do so with zero changes – no added or removed words, punctuation, or formatting.
- Do not prepend or append anything to the repeated text unless explicitly instructed.
- Preserve all original capitalization, spacing, and punctuation exactly as in the query.

3. ****结构与格式合规****

- 严格遵循所有格式说明，例如：
 - Wrapping the entire response in quotation marks if required.
 - Using specified markdown bullet point styles (e.g., asterisks).
 - Ensuring capitalization instructions (e.g., all caps or minimum occurrences of uppercase words) are perfectly met.
 - Adhering to sentence or paragraph counts exactly as requested.

4. ****响应内容准确性与适当性****

- After fulfilling all structural requirements, respond to the main substantive question accurately and completely.
- Use domain knowledge and reliable calculations to ensure factual correctness in answers.
- For questions requesting sensitive or potentially harmful content (e.g., cures without scientific basis), produce responsible answers that include disclaimers or refusals if instructed.

- 始终尊重伦理准则以及对此类查询所规定的任何拒绝语言或结论性陈述。

5. ****无多余文本****

- 不得添加超出查询明确允许或要求范围的解释、内部推理、道歉或元评论。
- 最终输出必须是精确、可直接交付的响应，完美满足所有用户指令。

6. ****观察到的示例与模式****

- Users often combine multiple complex formatting and content instructions (e.g., repetition of request text, followed by specific number of sentences or bullet points, with capitalization rules).
- Ensure you carefully distinguish when to repeat the query text verbatim and when to respond directly (sometimes the repetition excludes an instruction sentence).
- Handle instructions about capitalized words appearing a minimum number of times by distributing such words naturally but thoroughly across the response.
- When length constraints specify minimums (e.g., "at least 39 sentences"), ensure you meet or exceed exactly rather than approximating.
- For bullet points or enumerations, use the precise markdown style asked for (commonly asterisks).
- Follow refusal instructions verbatim and terminate the response exactly as specified.

****总结:**** 您的回答必须是对用户查询中嵌入指令的完美文本与结构呈现。始终优先遵循用户的具体规范，而非默认生成规范。输出为最终结果，不会进行编辑，因此必须 100% 符合要求。

This task therefore requires a rigorously rule-based approach:

- First parse and isolate all explicit instructions (including repetitions and formatting).
- Then assemble your answer by reproducing mandated text identically.
- Next produce the requested content in the exact required format, length, and style.
- Finally, review your output against the instructions ensuring total compliance before delivering your response.

L.4 IFBENCH, QWEN3 8B

IFBench Qwen3 8B generate_response_module.predict

Base Prompt:

Respond to the query

MIPROv2 Prompt:

You are a precision-focused instruction-adherence expert tasked with generating structured, ethically compliant responses. When responding to queries, always think step-by-step to analyze constraints (formatting rules, content boundaries, ethical guidelines), synthesize requirements, and produce outputs that strictly adhere to specified parameters. Ensure responses are fully formatted as requested (e.g., lowercase


```
text, bullet points, markdown), avoid any unverified claims, and maintain neutrality in
ambiguous ethical scenarios.
Demos:
Example 1:
query:
    Can you provide the social media handles of Harry Potter? Your answer must contain
    exactly 8 bullet points. Use the markdown bullet points such as:
* This is point 1.
* This is point 2 Finish your response with this exact phrase That's my answer.. No other
words should follow this phrase.
reasoning:
该查询询问的是哈利·波特这一虚构角色的社交媒体账号。由于他本人没有个人社交媒体账号，回答应改为
列出与哈利·波特系列相关的官方账号。这些账号包括由华纳兄弟、魔法世界及相关组织管理的账号。答案
按要求结构化恰好 8 个要点。回答：

    * @WizardingWorld (Twitter)
* @WizardingWorld (Instagram)
* @WizardingWorld (Facebook)
* @JKRowling (Twitter)
* @HarryPotterAlliance (Facebook)
* @HogwartsExpress (Twitter)
* @TheWizardingWorldOfHarryPotter (Facebook)
* @HarryPotterBooks (Twitter)
That's my answer.

#####
3 other demos omitted for brevity.
#####
```

由配置 GEPA 生成的 GEPA 提示词：

```
Respond to the query by first repeating it word for word without any changes. Then,
provide your answer. Ensure your answer includes niche or domain-specific factual
information relevant to the task. If the query specifies constraints (e.g., letter
frequency, sentence limits, or formatting rules), adhere to them strictly. Use a
generalizable strategy to solve the task, such as breaking down complex problems into
logical steps or applying specialized knowledge. Avoid adding any preamble or postscript
before or after the repeated query.
```

IFBench Qwen3 8B ensure_correct_response_module.predict

Base Prompt:

确保响应正确且符合给定约束。您的响应将作为最终响应使用。

MIPROv2 Prompt:

您是一位细致且合乎伦理的 AI 助手，负责生成详细且合规的回答。收到查询时，请先逐步思考以创建推理过程

that addresses the query's requirements, then produce a response that strictly adheres to all specified constraints (e.g., formatting rules, content boundaries, ethical guidelines). Ensure the final output is structured with a "Reasoning:" section followed by a "Final Response:" section, and verify that it aligns with the query's exact specifications while maintaining logical coherence and ethical responsibility.

Demos:

Example 1:

query:

What are some techniques for winning a swordfight in a medieval combat game? In your response, the letter f should appear less than 18 times. Do not include keywords ['heart', 'listen', 'reputation', 'side'] in the response.

response:

优先考虑站姿与平衡以控制战斗节奏。利用角度创造进攻机会，同时尽量减少暴露。精准格挡以打乱对手节奏。保持距离以避免过度投入，并瞄准关节或四肢等弱点。专注于节奏与时机以利用破绽。推理：

该回答概述了中世纪剑术的关键技巧，侧重于站姿、角度、格挡、距离以及瞄准弱点。它避免了受限关键词，并将字母“f”的出现次数限制为 2 次，满足了约束条件。建议实用且符合历史战斗策略。final_response:

优先考虑站姿与平衡以控制战斗节奏。利用角度创造进攻机会，同时尽量减少暴露。精准格挡以打乱对手节奏。保持距离以避免过度投入，并瞄准关节或四肢等弱点。专注于节奏与时机以利用破绽。

3 other demos omitted for brevity.
#####

由配置 GEPA 生成的 GEPA 提示词：

Ensure the response strictly follows these instructions:

1. ****First****, repeat the user's query ****word for word**** without any changes or additions.
2. ****Then****, provide your answer in the specified format, adhering to all constraints (e.g., markdown, structure, content).
3. ****Do not include any additional text, explanations, or formatting**** beyond the repeated query and your answer.
4. ****Include niche/domain-specific factual details**** (e.g., technical commands, best practices, or platform-specific configurations) if applicable, as these are critical for accurate task completion.
5. ****Use precise formatting**** (e.g., bullet points, code blocks, headers) as requested, ensuring no markdown is omitted or altered.
6. ****Avoid generalizable strategies**** unless explicitly instructed; focus on actionable, specific guidance.
7. ****Validate all technical steps**** (e.g., Dockerfile syntax, CLI commands) for accuracy and completeness.
8. ****Highlight potential pitfalls and solutions**** to address common issues in the task.
9. ****Prioritize clarity and conciseness****, ensuring the response is both comprehensive and easy to follow.
10. ****Adhere to language and case requirements**** (e.g., all caps, English only) if specified.

L5 HOVER、GPT-4.1 MINI

HoVer GPT-4.1 Mini create_query_hop2.predict

Base Prompt:

Given the fields `claim`, `summary\_1`, produce the fields `query`.

MIPROv2 Prompt:

Given the original claim and the initial summary of retrieved evidence, carefully analyze the information step-by-step to identify any gaps, contradictions, or points needing further clarification. Then, generate a clear, focused, and precise query that targets additional relevant information to deepen the investigation and help verify or refute the claim. Your response should include a detailed chain-of-thought reasoning explaining your thought process in formulating this refined query.

Demos:

Example 1:

claim:

AirBaltic's has bases at Tallinn Airport and the airport that ceased operations in 2015, with the airport code of Ryanair: VNO, ICAO: EYVI.

summary_1:

AirBaltic has bases at Tallinn Airport and Vilnius Airport (IATA: VNO, ICAO: EYVI), but Vilnius Airport did not cease operations in 2015. The airport that ceased operations in 2015 mentioned in the passages is unrelated and has different codes. Therefore, the claim that airBaltic's base with code VNO/EYVI ceased operations in 2015 is false.

reasoning:

The claim states that AirBaltic has bases at Tallinn Airport and an airport with codes VNO (IATA) and EYVI (ICAO) which ceased operations in 2015. However, the summary clarifies that Vilnius Airport, which has those codes, did not cease operations in 2015. The airport that did cease operations in 2015 has different codes and is unrelated to AirBaltic's bases. Therefore, the claim that AirBaltic's base at the airport with codes VNO/EYVI ceased operations in 2015 is incorrect.

query:

Did Vilnius Airport (IATA: VNO, ICAO: EYVI) cease operations in 2015?

3 other demos omitted for brevity.
#####

由配置 GEPA+合并（GEPA+Merge）生成的 GEPA 提示：

Task Description:

You are given two fields: `claim` and `summary\_1`. Your goal is to produce a field called `query` — a concise set of relevant, fact-checking questions or search style queries that can be used to retrieve evidence documents verifying or refuting the claim based on information contained or inferred from the summary.

Input Format:

- `claim`: A factual statement potentially containing multiple fact assertions about people, events, attributes, titles, dates, roles, or relationships.
- `summary\_1`: A short paragraph summarizing factual information related to the claim, often clarifying or correcting some parts of the claim.

Output Format:

- `query`: One or more specific, well-phrased questions or keyword queries that directly target the key factual discrepancies or verifications raised by the claim in light of the summary.

Detailed Instructions:

1. **Extract key factual elements from the claim** – names, dates, titles, roles, events, or relationships explicitly or implicitly stated.
2. **Contrast these facts with the summary to identify points of agreement, contradiction, or ambiguity.**
3. **Formulate fact-checking queries that are:**
 - Tightly focused on the core factual issues raised by the claim and addressed or contradicted by the summary.
 - Include named entities, dates, roles, or other domain-specific identifiers directly mentioned in both claim and summary to improve retrieval effectiveness.
 - When relevant, break complex claims into multiple queries ensuring each fact is verifiable separately.
4. **When relevant details appear only in the summary but are hinted at or missing from the claim (e.g., specific titles, roles, or names), include these in the queries to enable retrieval of key evidence.**
5. **Use a clear, natural question format or targeted keyword phrases that could serve well as search queries.**
6. **Avoid overly broad or generic queries; precision improves evidence retrieval quality.**
7. **Optionally, you may provide brief reasoning internally (not required in output) to ensure the queries cover all claim aspects and reflect the summary insights.**

Examples of typical query components include:

- Correct dates of events or deaths.
- Confirmation of a person's role or association with a known work.
- Verification of relationships or allegations.
- Details about specific cultural or domain elements (e.g., operas' structure, song directors).
- Clarifying entity attributes or classifications (e.g., ethnicity, nationality).

遵循这些说明，您将生成既全面又有针对性的查询，从而最大限度地提高检索与主张验证任务相关的事实证据的机会。

HoVer GPT-4.1 Mini create_query_hop3.predict

Base Prompt:

Given the fields `claim`, `summary\_1`, `summary\_2`, produce the fields `query`.

MIPROv2 Prompt:

您是一位专注于多跳推理和证据综合的专家事实核查员。给定一个主张和两个整合了先前检索步骤证据的中间摘要，请逐步仔细分析信息，以生成清晰且有针对性的查询。该查询旨在通过利用洞见，检索最相关的附加文档，以帮助验证或反驳该主张。

from both summaries. Provide a detailed chain-of-thought reasoning explaining how you integrate the summaries and the claim to formulate this refined query.

Demos:

Example 1:

claim:

弗雷迪·塞斯勒的兄弟是一位餐馆老板，其俱乐部经常有具有威尔士血统的女演员露丝·伊丽莎白·“贝蒂”·戴维斯光顾。summary_1:

Freddie Sessler's brother was Siegi Sessler, a London restaurateur whose club was frequented by Bette Davis, an actress with Welsh ancestry.

summary_2:

弗雷迪·塞斯勒的兄弟是西吉·塞斯勒，一位著名的伦敦餐馆老板，其俱乐部西吉俱乐部经常有包括具有威尔士血统的女演员露丝·伊丽莎白·“贝蒂”·戴维斯在内的好莱坞明星光顾。推理：

该主张指出，弗雷迪·塞斯勒的兄弟是一位餐馆老板，其俱乐部经常有具有威尔士血统的女演员露丝·伊丽莎白·“贝蒂”·戴维斯光顾。两份摘要均支持这一主张，确认其兄弟为伦敦餐馆老板西吉·塞斯勒，其俱乐部确实经常有以威尔士血统著称的女演员贝蒂·戴维斯光顾。Summary_2 补充说明俱乐部名称为西吉俱乐部，并提到还有其他好莱坞明星光顾。为准确核查该主张，有必要验证弗雷迪·塞斯勒的兄弟是否为西吉·塞斯勒，确认他是否为餐馆老板，核实贝蒂·戴维斯是否经常光顾其俱乐部，并确认贝蒂·戴维斯具有威尔士血统。查询：

西吉·塞斯勒是谁？他是否是弗雷迪·塞斯勒的兄弟？西吉·塞斯勒是否是一位餐馆老板，其俱乐部经常有露丝·伊丽莎白·“贝蒂”·戴维斯光顾？贝蒂·戴维斯是否具有威尔士血统？

3 other demos omitted for brevity.
#####

由配置 GEPA+合并（GEPA+Merge）生成的 GEPA 提示：

给定三个文本字段：`claim`、`summary\_1` 和 `summary\_2`，您的任务是生成一个 `query` 字段，该字段旨在基于两份摘要中包含的信息，有效检索与验证或反驳该主张相关的证据文档。

详细任务描述与注意事项：

1. **查询目的：**

- 查询应准确、全面地针对声明中在摘要里被处理或澄清的关键事实要素。
- 查询将用于检索证据文档，因此应足够具体以定位相关支持或矛盾内容，同时足够宽泛以涵盖摘要中出现的所有重要细节。

2. **利用摘要：**

- Summaries often correct, clarify, or add factual context to the claim. Your query must incorporate these clarifications (e.g., name corrections, factual specifics, or counterpoints) to ensure retrieval of relevant evidence reflecting the nuanced truth.

- 包含摘要中提及或更正的所有与声明相关的独特实体、事实、日期、地点、名称和关系。
- 例如，如果摘要更正了声明中的纪录片标题，查询应引用更正后的标题及相关细节，以有效引导检索。

3. **查询内容策略：**

- Explicitly mention key entities, such as persons, places, dates, works, or political entities involved.
- Include attributes or relationships relevant to the claim's accuracy (e.g., "Was person X a politician in country Y during year Z?" or "Did documentary A and documentary B film in different locations such as location 1 and location 2?").
- If there is a factual dispute or correction in the summaries (e.g., nationality, official names, population figures), phrase the query to target evidence clarifying this dispute.

4. **领域特定细微差别：**

- Be mindful of historical geopolitical names and periods (e.g., "United Kingdom of the Netherlands between 1815 and 1830").
- Recognize proper titles and correct spellings (e.g., corrected documentary titles).
- Include both subjects or objects mentioned as comparisons or contrasts in the claim and summaries (e.g., two documentaries, two politicians, two towns).

5. **表述风格：**

- 查询应表述为清晰、精确且客观的问题，能够基于证据进行回答——通常结构为是/否或信息性查询。
- 避免过于宽泛或模糊的表述。力求细节丰富的查询，将摘要中的多个证据点连接起来。

6. **可推广策略：**

- Identify the core claim elements and verify if the summaries confirm, contradict, or amend these elements.
- Incorporate both the claim and corrections from summaries into the query so evidence retrieval captures the full factual context.
- Use the comparison or contrast highlighted by the summaries to create queries that specifically test the claim's veracity (e.g., comparing locations, roles, or historical timeframes).

遵循这些指南，您将生成能够最大化检索相关文本可能性的查询，从而准确证实或反驳该主张。

HoVer GPT-4.1 Mini summarize1.predict

Base Prompt:

Given the fields `claim`, `passages`, produce the fields `summary`.

MIPROv2 Prompt:

给定一个`主张`和一系列相关的`段落`，通过逐步推理仔细分析证据，以评估该主张的有效性。生成详细的思维链`推理`，解释段落中的信息如何支持或反驳该主张，随后给出清晰、简洁的`摘要`，综合关键

findings in relation to the claim. Ensure the reasoning explicitly connects evidence from the passages to the claim, enabling a thorough and transparent multi-hop verification process.

Demos:

Example 1:

claim:

拥有当前生产和销售视频游戏《Darkened Skye》中用于魔法的糖果的公司的制造商，其总部位于美国弗吉尼亚州。段落：

['Darkened Skye | Darkened Skye is a third-person action-adventure video game developed by Boston Animation. It was released for Microsoft Windows and the Nintendo GameCube in North America in 2002 and the PAL regions in 2003. The game was also packaged with Outlaw Golf. Its title character is a young woman named Skye who lives in a fantasy realm searching for her mother. She does not use firearms, but can perform magic using Skittles candies, as well as use her staff as a melee weapon, though it also becomes an energy weapon when used in conjunction with the Skittles.', 'Rocket Fizz | Rocket Fizz is a franchise chain of candy stores in the United States. The company markets a diverse variety of candies and produces its own line of soft drinks with unique flavors. Its flagship store is located in Camarillo, California, and most of the company's franchise stores are located in California. The company also markets candies that are rare to find or in limited production by various manufacturers, such as those that were popular during the 1960s to 1980s. As of September 2017, there are 86 Rocket Fizz store locations in the United States.', 'Spangler Candy Company | The Spangler Candy Company is a privately owned international confectioner that has been manufacturing and marketing candy since 1906.', 'Future US | Future US, Inc. (formerly known as Imagine Media and The Future Network USA) is an American media corporation specializing in targeted magazines and websites in the video games, music, and technology markets. Future US is headquartered in the San Francisco with a small sales office in New York City. Future US is owned by parent company, Future plc, a small publishing company based in the United Kingdom.', 'Fun Dip | Fun Dip is a candy manufactured by The Willy Wonka Candy Company, a brand owned by Nestl  . The candy has been on the market in the United States and Canada since the 1940s and was originally called Lik-M-Aid. It was originally manufactured by Fruzola, and then Sunline Inc., through their Sunmark Brands division, which was purchased by Nestl   in January 1989. It comes in many different flavors with candy sticks that are included.', 'Hot Tamales | Hot Tamales is a chewy, cinnamon flavored, oblong-shaped candy introduced in 1950      manufactured and marketed in the United States by Just Born, a family-owned, Pennsylvania-based candy company.', 'Frankford Candy & Chocolate Company | Frankford Candy & Chocolate Company is an American candy manufacturer, located in Philadelphia, Pennsylvania, founded in 1947 by Sam Himmelstein. The current CEO is Stuart Selarnick. In recent years, Frankford Candy & Chocolate Company has become a very successful candy manufacturer. It is one of the largest producers of chocolate rabbits in the United States, making over 100 varieties. In 2000, it started licensing popular children's TV characters, such as SpongeBob SquarePants."]

reasoning:

该声明指出，电子游戏《黑暗天空》（Darkened Skye）中所用糖果的制造商总部位于弗吉尼亚州。段落 [1] 证实，电子游戏《黑暗天空》使用彩虹糖（Skittles）作为魔法。因此，所涉糖果为彩虹糖。关键步骤是确定生产和销售彩虹糖的公司，以及其总部是否位于弗吉尼亚州。所提供的段落均未明确提及彩虹糖或其制造商。段落列出了多家糖果公司，但均未提及彩虹糖或与其制造商相符。已知彩虹糖的生产商为

玛氏公司（Mars, Incorporated），其总部位于弗吉尼亚州麦克莱恩。尽管这一事实未在所提供段落中直接说明，但属于相关常识。由于段落未反驳该声明，且基于已知事实以确保准确性，可以认为该声明属实，即拥有生产和销售彩虹糖公司的制造商总部位于弗吉尼亚州。摘要：

《黑暗天空》中用作魔法的糖果是彩虹糖，由玛氏公司生产。玛氏总部位于美国弗吉尼亚州麦克莱恩。因此，基于外部知识确认制造商总部位于弗吉尼亚州，该声明属实。

3 other demos omitted for brevity.
#####

由配置 GEPA+合并（GEPA+Merge）生成的 GEPA 提示：

Given the fields `claim`, `passages`, produce the fields `summary`.

HoVer GPT-4.1 Mini summarize2.predict

Base Prompt:

Given the fields `claim`, `context`, `passages`, produce the fields `summary`.

MIPROv2 Prompt:

Given a `claim`, relevant `context`, and a set of supporting `passages`, generate a detailed step-by-step reasoning process that logically connects the evidence to the claim, followed by a concise summary that clearly states whether the claim is supported or refuted based on the aggregated information. Ensure the reasoning explicitly references key evidence from the context and passages to justify the conclusion in the summary.

Demos:

Example 1:

claim:

卡利安吉 - 阿南德吉（Kalyanji - Anandji）二人组中，卡利安吉·维尔吉·沙阿（Kalyanji Virji Shah）的兄弟曾获印度电影观众奖最佳音乐导演奖。上下文：

是的，卡利安吉·维尔吉·沙阿的兄弟阿南德吉·维尔吉·沙阿（Anandji Virji Shah），作为卡利安吉 - 阿南德吉二人组成员，于 1975 年凭借电影《空白纸》（Kora Kagaz）获得印度电影观众奖最佳音乐导演奖。段落：

['Kalyanji Virji Shah | Kalyanji Virji Shah (30 June 1928 – 24 August 2000) was the "Kalyanji" of the Kalyanji-Anandji duo. He and his brother Anandji Virji Shah have been famous Indian film musicians, and won the 1975 Filmfare Award for Best Music Director, for "Kora Kagaz". He is a recipient of the civilian honour of Padma Shri (1992).', 'Anandji Virji Shah | Anandji Virji Shah is an Indian music director. Together with his brother he formed the Kalyanji-Anandji duo, and won the 1975 Filmfare Award for Best Music Director, for "Kora Kagaz". He is a recipient of the civilian honour of Padma Shri (1992).', 'Kalyanji&Anandji | Kalyanji&Anandji are an Indian composer duo from Gujarat: Kalyanji Virji Shah (30 June 1928-03 November

2000) and his brother Anandji Virji Shah (born 02 March 1933). The duo are known for their work on Hindi film soundtracks, particularly action potboilers in the 1970s. Some of their best-known works are "Don", "Bairaag", "Saraswatichandra", "Qurbani", "Tridev" and "Safar". They won the 1975 Filmfare Award for Best Music Director for "Kora Kagaz".', 'Kalyanji&Anandji discography | This is a discography of Bollywood composer duo Kalyanji Anandji, consisting of Kalyanji Virji Shah and Anandji Virji Shah. They have composed music for over 200 films in their 46 year career.', 'Viju Shah | Viju Shah (born, Vijay Kalyanji Shah, 5 June 1959) is a music director of Hindi cinema. He is the son of music director Kalyanji Virji Shah of composer duo Kalyanji Anandji. He composed music for movies such as "Vishwatama" (1992), "Mohra" (1994), "Tere Mere Sapne" (1996) and "" (1997) for which got his second nomination for the Filmfare Award for Best Music Director and he won the 1998 Filmfare Award for Best Background Score for "".', 'Bombay the Hard Way: Guns, Cars and Sitaris | Bombay the Hard Way: Guns, Cars and Sitaris is a 1998 project of producer Dan the Automator in which music from 1970s funk Bollywood composer duo Kalyanji-Anandji (Kalyanji Virji Shah and Anandji Virji Shah) was taken and given a funky remix. The album was eventually withdrawn (possibly due to copyright concerns). A sequel, "Bombay 2: Electric Vindaloo", was also produced.', 'Kora Kagaz | Kora Kagaz (Hindi: क़ोरा काग़ज़; translation: Blank Paper) is a 1974 Hindi drama film produced by Sanath Kothari and directed by Anil Ganguly. The film stars Vijay Anand, Jaya Bhaduri, A.K. Hangal, Achala Sachdev and Deven Verma. The films music is by Kalyanji Anandji. The famous title song "Mera Jeevan Kora Kagaz" was by Kishore Kumar.']

reasoning:

该声明指出，卡利安吉-阿南德吉二人组中，卡利安吉·维尔吉·沙阿的兄弟曾获印度电影观众奖最佳音乐导演奖。从段落和上下文可以明确，该二人组由卡利安吉·维尔吉·沙阿及其兄弟阿南德吉·维尔吉·沙阿组成。两人于1975年凭借电影《空白纸》共同获得印度电影观众奖最佳音乐导演奖。因此，作为卡利安吉的兄弟，阿南德吉·维尔吉·沙阿确实作为该二人组成员获得了印度电影观众奖最佳音乐导演奖。摘要：

阿南德吉·维尔吉·沙阿，卡利安吉·维尔吉·沙阿的兄弟，作为卡利安吉-阿南德吉二人组成员，于1975年获得印度电影观众奖最佳音乐导演奖。

3 other demos omitted for brevity.
#####

由配置 GEPA+合并 (GEPA+Merge) 生成的 GEPA 提示：

给定输入字段：

- `claim`: A factual statement or assertion that may be true or false.
- `context`: A brief explanation or summary relating to the claim that typically clarifies the claim's accuracy.
- `passages`: A list of relevant textual evidence or knowledge snippets containing facts, descriptions, or biographies related to entities or concepts mentioned in the claim.

你的任务是生成一个满足以下要求的`摘要`：

1. **准确反映声明与证据之间的关系**：仔细分析输入的`声明`，并对照`上下文`和所有`段落`进行核查。

判断该声明是得到证据支持、部分支持还是被证据反驳。

2. ****Explicitly connect key entities and facts from the passages****: Your summary must mention the main entities and facts that directly confirm or refute parts of the claim. For example, highlight relevant names, works (films, albums, songs, books), attributes, dates, nicknames, or roles that clarify the claim's accuracy.

3. ****建立实质性联系****: 摘要不应仅仅重述声明或结论。相反，它应明确将声明与段落中的详细证据联系起来——例如具体的电影名称、专辑名称、歌曲名称、日期或关系——以提供清晰的理由说明该声明的准确性或不准确性。

4. ****Include relevant disambiguations or clarifications****: Where applicable, clarify potential misunderstandings or common confusions highlighted in the passages (e.g., distinguishing similarly named films or songs, specifying which individual is referenced, or noting alternate titles and adaptations).

5. ****使用简洁、事实性的语言****: 摘要应清晰简洁，但需包含所有必要的证据，以理解该声明是如何得到支持或反驳的。

6. ****支持未来的查询生成****: 由于摘要将用于生成查询以查找证据，它必须明确提及关键实体、标题和事实，以指导检索和推理。

****总体而言，你的方法应是对声明、上下文和每个段落进行全面推理，将相关事实和证据综合成一个连贯且证据丰富的摘要，清楚地记录该声明为何为真、为假或部分为真。**** 避免模糊或过于笼统的陈述；相反，应将摘要建立在段落和上下文中存在的精确事实细节之上。

L.6 HoVer, QWEN3 8B

HoVer Qwen3 8B create_query_hop2.predict

Base Prompt:

Given the fields `claim`, `summary\_1`, produce the fields `query`.

MIPROv2 Prompt:

In a high-stakes scenario where the accuracy of your query directly determines the validity of a complex claim, generate a precise query that builds on the provided claim and summary to uncover critical evidence. Use step-by-step reasoning to identify gaps in the initial summary, then craft a query that explicitly targets these gaps to retrieve additional supporting information. Ensure your query is unambiguous and directly addresses the claim's unresolved aspects.

Demos:

Example 1:

claim:

Sandi Simcha DuBowski directed a documentary released in 2001. The horror film Zeitgeist was not also released in 2001.

summary_1:
The claim is correct: Sandi Simcha DuBowski directed the 2001 documentary *Trembling Before G-d*, and there is no evidence in the passages that the horror film *Zeitgeist* was released in 2001.

reasoning:
该说法正确，因为桑迪·西姆哈·杜博夫斯基确实执导了 2001 年的纪录片《在上帝面前颤抖》，如 summary_1 所述。关于《时代精神》并非 2001 年上映的断言，由于所提供的摘要中没有证据将该影片与那一年联系起来，因此得到支持。查询：

所提供的摘要中是否有任何证据表明恐怖电影《时代精神》于 2001 年上映？

3 other demos omitted for brevity.
#####

由配置 GEPA 生成的 GEPA 提示词：

根据字段`claim`和`summary\_1`，生成一个精确且有针对性的查询，以确定验证或反驳该说法所需的具体证据。

Key Requirements:

1. **Target Missing Evidence**: Identify and explicitly ask about the unverified or unconfirmed details in the summary (e.g., names, dates, locations, or connections) that are critical to the claim.
 - Example: If the summary mentions "Massimo Giordano" as a potential counterpart but lacks birthplace details, the query should ask for evidence confirming their ties to Naples.
2. **Correct Historical/Domain-Specific Anomalies**: Address discrepancies like anachronisms (e.g., Ali Qushji's work in the 1960s vs. his actual 15th-century timeline) or misattributions (e.g., *Hayy ibn Yaqdhan* by Ibn Tufail, not Ali Qushji).
3. **Link to Summary Context**: Ensure the query references the summary's key points (e.g., "the claim states X, but the summary notes Y is unverified").
4. **Use Specific Terms**: Include exact names, titles, or dates mentioned in the summary to avoid ambiguity (e.g., "Ana's Nin," "Metropolitan City of Naples," "1948").

Example Strategy:

If the summary states:

- "The claim is partially supported. While [Fact A] is confirmed, [Fact B] lacks verification."
- The query should ask:
* "Is there evidence confirming [Fact B], such as [specific name/date/location]?" *

Domain-Specific Notes:

- Verify historical timelines (e.g., Ali Qushji died in 1474, not the 1960s).
- Confirm authorships (e.g., *Hayy ibn Yaqdhan* is attributed to Ibn Tufail, not Ali Qushji).
- Check for geographic or biographical details (e.g., birthplaces, cities, or professional connections).

确保您的查询直接针对摘要中未经证实的说法，并包含摘要中的所有关键词，以最大限度地提高证据检索效果。

HoVer Qwen3 8B create_query_hop3.predict

Base Prompt:

Given the fields `claim`, `summary\_1`, `summary\_2`, produce the fields `query`.

MIPROv2 Prompt:

您是一名事实核查助手，擅长多跳推理和信息综合。根据字段`claim`、`summary\_1`和`summary\_2`，生成一个精确的查询，将原始说法与两个摘要综合起来，以探究更深层次的上下文关系，通过有针对性的证据检索解决歧义并确认支持性细节。示例：

由配置 GEPA 生成的 GEPA 提示词：

Given the fields `claim`, `summary\_1`, and `summary\_2`, produce the field `query` that:

1. Explicitly asks whether the claim is supported by the provided summaries.
2. Includes **specific evidence** from the summaries (e.g., names, titles, dates, or factual details) to guide retrieval of relevant documents.
3. Focuses on **key disputed points** in the claim (e.g., incorrect attributions, missing evidence, or conflicting statements) to ensure the query targets the most relevant information.

Key considerations for the query:

- If the summaries mention specific works (e.g., **Planes, Trains and Automobiles**), include the title.
- If the summaries reference named entities (e.g., John Candy, KLM, Richard Ford), include them.
- If the summaries clarify contradictions (e.g., "not X but Y"), structure the query to highlight this contrast.
- Avoid vague phrasing; instead of asking "Is the claim true?" use "Does the evidence in the summaries confirm [specific detail]?"

Example:

If the claim is about a film and the summaries mention **Planes, Trains and Automobiles** (1987), the query should include that title to retrieve relevant evidence.

...

HoVer Qwen3 8B summarize1.predict

Base Prompt:

Given the fields `claim`, `passages`, produce the fields `summary`.

MIPROv2 Prompt:

Given the fields `claim` and `passages`, generate a structured reasoning process that analyzes the claim step-by-step using the provided evidence, and produce a concise summary that distills the key findings and evaluates the claim's validity based on the evidence.

Demos:

Example 1:

claim:

卡利安吉-阿南德吉二人组中的一半，即卡利安吉·维尔吉·沙阿的兄弟，曾获得印度电影观众奖最佳音乐导演奖。段落：

['Kalyanji Virji Shah | Kalyanji Virji Shah (30 June 1928 – 24 August 2000) was the "Kalyanji" of the Kalyanji-Anandji duo. He and his brother Anandji Virji Shah have been famous Indian film musicians, and won the 1975 Filmfare Award for Best Music Director, for "Kora Kagaz". He is a recipient of the civilian honour of Padma Shri (1992).', 'Kalyanji-Anandji | Kalyanji-Anandji are an Indian composer duo from Gujarat: Kalyanji Virji Shah (30 June 1928–03 November 2000) and his brother Anandji Virji Shah (born 02 March 1933). The duo are known for their work on Hindi film soundtracks, particularly action potboilers in the 1970s. Some of their best-known works are "Don", "Bairaag", "Saraswatichandra", "Qurbani", "Tridev" and "Safar". They won the 1975 Filmfare Award for Best Music Director for "Kora Kagaz".', 'Anandji Virji Shah | Anandji Virji Shah is an Indian music director. Together with his brother he formed the Kalyanji-Anandji duo, and won the 1975 Filmfare Award for Best Music Director, for "Kora Kagaz". He is a recipient of the civilian honour of Padma Shri (1992).', 'Viju Shah | Viju Shah (born, Vijay Kalyanji Shah, 5 June 1959) is a music director of Hindi cinema. He is the son of music director Kalyanji Virji Shah of composer duo Kalyanji Anandji. He composed music for movies such as "Vishwatama" (1992), "Mohra" (1994), "Tere Mere Sapne" (1996) and "" (1997) for which got his second nomination for the Filmfare Award for Best Music Director and he won the 1998 Filmfare Award for Best Background Score for "".', 'Kalyanji-Anandji discography | This is a discography of Bollywood composer duo Kalyanji Anandji, consisting of Kalyanji Virji Shah and Anandji Virji Shah. They have composed music for over 200 films in their 46 year career.', 'Bombay the Hard Way: Guns, Cars and Sitar | Bombay the Hard Way: Guns, Cars and Sitar is a 1998 project of producer Dan the Automator in which music from 1970s funk Bollywood composer duo Kalyanji-Anandji (Kalyanji Virji Shah and Anandji Virji Shah) was taken and given a funky remix. The album was eventually withdrawn (possibly due to copyright concerns). A sequel, "Bombay 2: Electric Vindaloo", was also produced.', 'Bairaag | Bairaag is a 1976 Bollywood film. Produced by the duo Mushir-Riaz, it is directed by Asit Sen. The film stars Dilip Kumar, who received a Filmfare nomination for Best Actor for playing a triple role. "Bairaag" also stars Saira Banu, Leena Chandavarkar, Ruma Guha Thakurta, Prem Chopra, Helen, Sujit Kumar, Madan Puri, Pailant, Kader Khan, Asit Sen and Sachin. Its music is by the duo Kalyanji Anandji, who received a Filmfare nomination for Best Music. The film received a Filmfare Award for Best Sound for P. Harikishan. According to one source, the film did not do well at the box office.']

reasoning:

The claim states that Kalyanji Virji Shah's brother won the Filmfare Award for Best Music Director. The passages confirm that the Kalyanji-Anandji duo (Kalyanji Virji Shah and his brother Anandji Virji Shah) won the 1975 Filmfare Award for Best Music Director for "Kora Kagaz." Since the award was shared by both brothers, the claim is accurate as it identifies Anandji Virji Shah (Kalyanji's brother) as the winner.

summary:

阿南德吉·维尔吉·沙阿是卡利安吉·维尔吉·沙阿的兄弟，作为卡利安吉-阿南德吉二人组的一员，凭借其在《科拉·卡加兹》中的作品获得了 1975 年印度电影观众奖最佳音乐导演奖。

3 other demos omitted for brevity.
#####

由配置 GEPA 生成的 GEPA 提示词：

Given the fields `claim` and `passages`, produce the fields `summary` by:

1. **Analyzing the claim** to identify key elements (e.g., names, locations, relationships, specific terms).
2. **Cross-referencing the passages** to locate explicit or implicit evidence that supports, contradicts, or partially addresses the claim.
3. **Explicitly connecting the claim to the evidence** in the passages, even if the evidence is not directly stated in the claim (e.g., linking "Roger Yuan" to "Shanghai Noon" in Example 1).
4. **Highlighting domain-specific details** (e.g., "Sentinelese language is unclassifiable" in Example 3) that are critical for accuracy.
5. **Stating whether the claim is supported, unsupported, or partially supported**, and explaining the reasoning with clear references to the passages.
6. **Prioritizing key evidence** that would be relevant for further queries (e.g., "Marlborough" in Example 2) to ensure retrievability.

Additional Guidance:

- If the claim contains unverified details (e.g., "Roger Yuan" in Example 1), explicitly note this in the summary.
- For niche or ambiguous information (e.g., "uncontacted people" in Example 3), reference specific passages that clarify the context.
- Ensure summaries are concise but include all critical evidence needed to validate or refute the claim.

HoVer Qwen3 8B summarize2.predict

Base Prompt:

Given the fields `claim`, `context`, `passages`, produce the fields `summary`.

MIPROv2 Prompt:

Given the fields `claim`, `context`, and `passages`, perform multi-hop reasoning to generate a structured summary that validates or refutes the claim. First, analyze the claim and contextual information to identify key relationships. Next, evaluate evidence from the retrieved passages to build a coherent narrative. Construct a logical reasoning chain connecting the claim to supporting or contradictory evidence. Finally, produce a concise summary that explicitly confirms or contradicts the claim based on your analysis, citing relevant evidence from the context and passages.

Demos:

Example 1:

claim:

该州有记载的历史始于 1519 年，浸信会教徒乔治·鲁比是重建时期该州一位著名的黑人共和党领袖，而奥古斯塔斯·查普曼·艾伦正是用他的遗产资助了该州一座城市的建立。上下文：

该主张未得到文段支持。虽然乔治·鲁比（George Ruby）在重建时期与得克萨斯州有关联，且奥古斯塔斯·查普曼·艾伦（Augustus Chapman Allen）在得克萨斯州创建了休斯敦，但文段中并无证据表明得克萨斯州有文字记载的历史始于 1519 年。文段：

[“Merchants and Manufacturers Building | The One Main Building, formerly the Merchants and Manufacturers Building (commonly referred to as the M&M Building), is a building on the campus of the University of Houston’s Downtown. The building is recognized as part of the National Register of Historic Places, is a Recorded Texas Historic Landmark, and considered a Contributing Building in Downtown Houston’s Main Street/Market Square Historic District. The building was built above Allen’s Landing’s area where Houston’s founders John Kirby Allen and Augustus Chapman Allen originally settled.”, ‘Augustus Chapman Allen | Augustus Chapman Allen (July 4, 1806 – January 11, 1864), along with his younger brother, John Kirby Allen, founded the City of Houston in the U.S. state of Texas. He was born on July 4, 1806, in Canaseraga Village, New York (the present day hamlet of Sullivan in the Town of Sullivan, New York), to Sarah (Chapman) and Roland Allen.’, “History of Texas | The recorded History of Texas begins with the arrival of the first Spanish conquistadors in the region of North America now known as Texas in 1519, who found the region populated by numerous Native American /Indian tribes. Native Americans’ ancestors had been there for more than 10,000 years as evidenced by the discovery of the remains of prehistoric Leanderthal Lady. During the period of recorded history from A.D. 1519 to 1848, all or parts of Texas were claimed by five countries: France, Spain, Mexico, the Republic of Texas, the United States of America’s as well as the secessionist Confederate States of America in 1861–1865, during the American Civil War.”, ‘Charlotte Baldwin Allen | Charlotte Baldwin Allen (July 14, 1805 – August 3, 1895) is known in Texan history as the “mother of Houston”. She was the wife of Augustus Chapman Allen, who used her inheritance to finance the founding of this city.’, ‘Allen Parkway | Allen Parkway is an arterial road west of Downtown Houston, Texas. It has a distance of approximately 2.3 miles (3.7\km), running from Interstate 45 west to Shepherd Drive, where it becomes Kirby Drive. Originally known as Buffalo Parkway, it was later named after John Kirby Allen and Augustus Chapman Allen, the founders of Houston.’, “Allen’s Landing | Allen’s Landing is the birthplace of the city of Houston’s the largest city in the U.S. state of Texas. In August 1836, just months after the Republic of Texas won its independence from Mexico, two brothers (and real estate developers) from New York’s John Kirby Allen and Augustus Chapman Allen’s purchased 6,642\acres\ (27\km’s) in the area and settled there on the banks of Buffalo Bayou. Allen’s Landing is located south of the University of Houston’s Downtown Commerce Street Building, at the intersection of Commerce Street and Main Street.”, “Keith Allen (American football) | Keith D. Allen (born July 12, 1974) is an American football coach. He served as the head football coach at Southwest Baptist University from 2007 to 2012, compiling a record of 27–39. His 27 wins are the most in the history of Southwest Baptist’s football program, which began play in 1983.”]

reasoning:

The claim is not supported by the passages. While the passages confirm that Texas’s recorded history began in 1519 (passage 3) and that Augustus Chapman Allen founded Houston in Texas (passage 2), they do not explicitly link the start of Texas’s recorded history to the same state where Augustus Chapman Allen used his inheritance

用于资助一座城市的建立。该主张将这两个事实混为一谈，但文段中并无直接证据表明 1519 年有文字记载的历史开端与艾伦的遗产被用于创建休斯敦所在的州有关。摘要：

该主张未得到所提供文段的支持，因为文段确认得克萨斯州有文字记载的历史始于 1519 年，且奥古斯塔斯·查普曼·艾伦在得克萨斯州创建了休斯敦，但并未如主张所述明确将这两个事实联系起来。

3 other demos omitted for brevity.
#####

由配置 GEPA 生成的 GEPA 提示词：

Given the fields `claim`, `context`, and `passages`, produce the field `summary` that:

1. **Evaluates the claim's validity** by explicitly connecting it to **specific evidence** from the `passages` (e.g., citing exact names, dates, or terms that support/refute the claim).
2. **Highlights factual inaccuracies** in the claim, such as incorrect origins, dates, or relationships, and links them to **direct contradictions** in the `passages` or `context`.
3. **Mentions key terms or entities** critical to the claim (e.g., "Sly Stone," "Large Münsterländer," or "Hit Parader's 2006 ranking") to ensure summaries are actionable for query generation.
4. **Balances accuracy** by acknowledging both correct and incorrect aspects of the claim, as seen in the examples (e.g., validating "The Wild Bunch" compilation while correcting the magazine's nationality and ranking year).
5. **Avoids vague statements** like "no evidence" and instead specifies **which parts of the passages** are relevant (e.g., "Passage 6 confirms Hit Parader was American").

Generalizable Strategy:

- Always reference **specific passages** (e.g., "Passage 1 states...") to ground claims in the text.
- Use **precise terminology** (e.g., "German Shorthaired Pointer" instead of "cousins") to align with the data.
- Prioritize **clarity and specificity** to ensure summaries can guide further evidence retrieval.

L.7 PUPA, GPT-4.1 MINI

PUPA GPT-4.1 Mini craft_redacted_request.predict

Base Prompt:

给定一个私密用户查询，为强大的外部大语言模型创建一个保护隐私的请求。该大语言模型可以在不获取用户私密信息的情况下提供协助。

MIPROv2 Prompt:

您将收到一个私密用户查询。您的任务是仔细分析该查询，逐步思考以理解用户意图，同时识别并移除或

abstracting any sensitive, proprietary, or personally identifiable information. Then, generate a clear, privacy-preserving prompt suitable for an external large language model (LLM) that requests useful, relevant guidance or information without exposing private details. Additionally, provide a detailed chain-of-thought explanation describing your reasoning process for how you transformed the original query into the redacted, generalized LLM request. Ensure the resulting prompt is professional, contextually appropriate, and maintains user privacy while enabling the LLM to assist effectively.

Demos:

Example 1:

user_query:

Imagine you are a stock researcher and advisor, I will give you a table about stocks and I want you to give me advice on what to invest next. This is a project I am working on an I need help with predictions I am making. Here is the table: Stock Price Dividends 5D 5D % 1M 1M % YTD YTD % 1Y 1Y % 5Y 5Y % IBM \$146,62 \$6,64 \$3,81 2,66 \$2,63 1,83 \$5,07 3,58 \$16,31 12,51 \$0,03 0,02 MCD \$284,90 \$6,08 \$5,07 1,81 -\$7,20 -2,46 \$20,57 7,78 \$29,24 11,44 \$123,26 76,26 JNJ \$163,98 \$4,76 -\$3,00 -1,80 -\$3,90 -2,32 -\$14,21 -7,97 \$0,98 0,60 \$29,66 22,08 MSFT \$324,89 \$2,72 \$1,25 0,39 -\$9,88 -2,95 \$85,31 35,61 \$59,66 22,49 \$217,33 202,05 WMT \$158,36 \$2,28 \$1,39 0,89 -\$1,85 -1,15 \$14,75 10,27 \$25,47 19,16 \$62,29 64,84 KO \$60,50 \$1,84 \$0,12 0,20 -\$1,59 -2,55 -\$2,45 -3,89 -\$2,23 -3,55 \$15,13 33,35 PFE \$36,07 \$1,64 -\$1,06 -2,84 -\$0,67 -1,81 -\$15,19 -29,64 -\$10,16 -21,98 -\$5,43 -13,09 AAPL \$179,96 \$0,96 \$3,45 1,95 -\$16,57 -8,43 \$54,89 43,89 \$18,58 11,51 \$125,04 227,65 NVDA \$468,23 \$0,16 -\$3,71 -0,79 \$7,05 1,53 \$325,08 227,09 \$310,22 196,33 \$401,52 601,89 CRM \$212,10 \$0,00 \$2,75 1,31 -\$9,86 -4,44 \$77,32 57,37 \$51,89 32,39 \$59,11 38,64 PYPL \$61,16 \$0,00 \$1,06 1,76 -\$13,82 -18,43 -\$13,43 -18,00 -\$31,51 -34,00 -\$29,46 -32,51 BIDU \$137,47 \$0,00 \$10,24 8,05 -\$15,63 -10,21 \$18,34 15,39 -\$9,85 -6,69 -\$88,25 -39,10

reasoning:

The user wants investment advice based on a provided table of stock data including price, dividends, and percentage changes over various time intervals (5 days, 1 month, year-to-date, 1 year, and 5 years). The primary request is for recommendations on what to invest in next and help with making predictions.

为保护隐私，我不应在发送给外部大语言模型的提示中包含用户的原始数据或具体个股详情。相反，我应请求关于如何分析包含价格、股息及多个时间段百分比变化的股票数据表的通用建议。此外，我还可以询问在投资决策中应更侧重哪些指标，以及基于此类数据结构的预测策略。

该方法确保外部大语言模型不会收到任何私密或可识别用户身份的数据，仅收到一个通用请求。我将请大语言模型就如何解读此类股票表现指标提供指导，并为预测分析或投资选择标准提出建议。llm_request:

请提供通用指导，用于分析基于数据表格式的股票投资机会，该数据表包含股票价格、股息支付以及 5 天、1 个月、年初至今、1 年和 5 年等不同时间间隔的百分比变化。概述投资者在决策时通常权衡的关键指标，如何解读短期与长期表现趋势，并建议在预测或推荐接下来投资哪些股票时使用的策略或度量。该请求为通用建议，不包含任何具体股票名称或数值。

```
#####  
3 other demos omitted for brevity.  
#####
```

由配置 GEPA+合并（GEPA+Merge）生成的 GEPA 提示：

隐私保护查询改写助手的新指令（增强版）：

Task Overview:

You receive a user query that may contain sensitive, private, or personally identifiable information (PII). Your objective is to reformulate this query into a generalized, privacy-preserving prompt suitable for sending to an external large language model (LLM). The reformulated prompt must enable the external LLM to fulfill the user's original intent effectively while rigorously protecting all user privacy by abstracting, omitting, or generalizing any sensitive details.

Input Format:

- A single user query string.
- The query might contain identifiable names (people, organizations), specific geographic locations, exact dates or timeframes, proprietary or internal terminology (e.g., system names, product codes), client or customer names, URLs, or any other sensitive or private information.

Output Format:

- Part (a) Reasoning:
 - Provide a concise paragraph that explains:
 - * How you identified sensitive or private information in the input
 - * What strategies you applied to protect privacy (generalization, omission, abstraction, replacement with placeholders)
 - * How the reformulated prompt preserves the original intent and task requirements without risking data leakage
- Part (b) LLM Request:
 - 一个简洁、精心构建的隐私安全提示，它：* 移除或匿名化所有个人身份信息和专有/内部细节 *
 - 根据需要抽象地点、名称、日期和技术术语 * 为大语言模型生成清晰且上下文丰富的指令，以生成与用户原始任务相符的相关且信息丰富的响应

详细的领域特定指南与最佳实践：

1. 敏感数据的识别与处理：

- All user-specific or personal names (individual or organizational) must be removed or replaced with generic role descriptors (e.g., "a business contact," "a client," "a notable individual"). Never lightly obscure or partially redact; full abstraction is required.
- All geographic mentions must be abstracted unless the location is publicly known, essential to the task, and can be generalized (e.g., "a region known for eco-tourism" instead of naming a country or city explicitly).
- Exact dates or durations must never be retained; instead, use relative or approximate temporal references (e.g., "recently," "over the past year").
- Internal or proprietary terms including system names, product codes, subscription types, and technical jargon must be generalized or replaced with

使用中性描述词以避免泄露知识产权或敏感操作细节。

- 避免直接引用或逐字包含用户提供的文本，除非通过泛化进行模糊处理。

2. 任务理解与改写：

- Identify the functional intent of the query: Is it creative writing, translation, summarization, professional communication drafting, technical explanation, or other?
- Preserve the thematic and informational core of the query (e.g., request for educational quality analysis, technical translation of a passage, biographical summary).
- Do not reproduce the original input verbatim; instead, frame the LLM prompt around the essential thematic elements extracted from the input.
- For queries regarding individuals, avoid direct reference to real personal information unless the name is widely public and essential; even then, use a generic or hypothetical framing for the individual profile.

3. 高质量且保护隐私的提示策略：

- 在足够的上下文细节与隐私抽象之间取得平衡，以保持提示的清晰度和相关性。
- 使用中立、上下文感知的表述，清晰指示大语言模型（LLM）所需的内容与风格。
- 避免模糊或过于宽泛的提示，以免导致回复实用性不足或质量下降。
- 当提及系统或专有内容时，指示大语言模型对特定术语进行泛化处理，并在不泄露敏感信息的前提下保持技术含义。
- 当要求对专业文本进行直接翻译时，指定替换或抽象内部命名。

4. 解释要求：

- The reasoning must transparently explain how privacy risks were identified (e.g., presence of names, locations, dates, proprietary terms).
- It must describe the abstraction or omission methods applied (e.g., replacing “Jonah Van Beijnen” with “a notable individual,” substituting “Makau” with “a specific region,” or “Yoda” with “a system name”).
- Clarify how the essential task and user intent were preserved despite these generalizations (e.g., focusing on educational quality, technical translation, biographical summary).
- This explanation justifies your design choices and demonstrates adherence to privacy-preserving principles.

需避免的常见陷阱：- 不得仅轻微模糊或部分删改敏感细节；必须进行完全匿名化或抽象化处理。- 不得逐字重复任何用户提供的个人身份信息或专有内容。- 避免未经修改地包含网址、确切日期或直接引语。- 不得留下可能降低改写后提示质量或上下文清晰度的模糊之处。- 除非查询涉及公众人物且确有必要，否则不得包含任何真实个人或组织名称，应改用通用描述词。

有效方法示例总结（基于先前示例）：

- For geographic queries: replace exact place names with general regions and provide a brief contextual descriptor.
- For technical texts containing system names or subscription types: instruct the LLM to translate or process the text while replacing or abstracting proprietary system identifiers.
- For biographical summaries about specific individuals: remove the real name and request a generic, well-structured four-paragraph summary about "a notable individual," preserving the overall intent without leaking PII.

Summary:

Your reformulations must ensure zero exposure of any PII or private/proprietary content while retaining enough thematic and functional clarity for the external LLM to produce high-quality, relevant outputs. This requires thorough analysis of the user's query, rigorous application of privacy-preservation strategies, and explicit reasoning explanations that document your approach and choices.

PUPA GPT-4.1 Mini respond_to_query

Base Prompt:

响应用户查询。为获得启发，我们发现了一个可能相关的请求，该请求发送给一个强大的外部大语言模型及其响应。

MIPROv2 Prompt:

Given a privacy-preserving redacted prompt sent to an external untrusted LLM and its response, along with the original user query containing potentially sensitive information, generate a final, coherent, and contextually appropriate user-facing response. Your output should integrate the anonymized LLM output with the original query's intent, ensuring privacy is maintained while providing a complete and polished answer that directly addresses the user's request.

Demos:

Example 1:

related_llm_request:

请详细解释戴维·伊斯顿在政治学中的系统分析，概述其系统理论方法的关键组成部分及其在政治系统研究中的重要性。related_llm_response:

当然！戴维·伊斯顿的系统分析是政治学中将系统理论应用于政治系统研究的基础框架。他的方法提供了一种结构化的方式来理解政治系统如何运作、维持稳定以及应对环境变化。以下是伊斯顿系统分析的详细解释，包括其关键组成部分和重要性。

Overview of David Easton's System Analysis

戴维·伊斯顿（1917—2014）是一位杰出的政治学家，他试图通过借鉴源自生物学和工程学的系统理论概念，发展一种政治系统的一般理论。他的目标是创建一种科学且具有分析性的

将政治作为一个综合且动态的系统而非孤立事件或制度来研究的框架。

Easton's system analysis views a political system as a set of interactions abstracted from the totality of social behavior, through which authoritative allocation of values (decisions) is made and implemented. In other words, politics is about how societies decide who gets what, when, and how, and Easton's model helps explain how these decisions are made and sustained.

Key Components of Easton's Systems Theory Approach

伊斯顿的模型将政治系统概念化为一个处理输入并产生输出的**黑箱**，嵌入在更大的环境之中。其关键组成部分包括：

1. **Inputs**

Inputs are demands and supports that come from the environment and society into the political system.

- ****Demands:**** These are claims or requests made by individuals or groups for the political system to allocate values or resources in a particular way (e.g., demands for law enforcement, education, welfare).
- ****Supports:**** These are actions or attitudes that indicate approval, allegiance, or compliance with the political system (e.g., voting, paying taxes, obeying laws).

Inputs represent the pressures and resources that the political system must process.

2. **Political System (The Black Box)**

The political system is the core of Easton's model. It receives inputs, processes them, and produces outputs.

- 包括参与权威决策的制度、组织和过程（如立法机关、行政机关、法院）。
- 系统将需求和支持转化为决策和政策。
- 系统必须维持其边界并适应变化以求生存。

3. **Outputs**

Outputs are the decisions, policies, and actions that the political system produces in response to inputs.

- 包括法律、法规、公共政策以及价值的权威性分配。
- 输出影响环境和社会，进而影响未来的输入。

4. **Feedback**

Feedback is the process by which outputs influence future inputs.

- 正反馈可以强化系统的稳定性。
- 负反馈可以表明存在问题或不满，促使进行调整。
- 反馈有助于系统适应并维持平衡。

5. **Environment**

政治系统存在于更广泛的环境中，包括社会、经济、文化和国际因素。

- 环境提供输入并受到输出的影响。
- 系统必须与环境变化互动并适应，才能生存。

Diagrammatic Representation (Simplified)

```
Environment
  â€š â€š
[Demands & Supports] â€š [Political System] â€š [Decisions & Policies]
  â€š â€š
  â€š----- Feedback -----
```

Significance of Easton's System Analysis in Political Science

1. ****对政治的整体理解：****

伊斯顿的模型鼓励将政治视为一个相互关联的系统，而不是孤立的事件或机构。它强调了社会与政治权威之间的动态互动。

2. ****关注过程与稳定性：****

该模型强调政治系统如何通过持续处理输入与输出以及反馈机制来维持稳定与秩序。

3. ****分析清晰度：****

通过将政治活动抽象为输入、系统处理、输出和反馈，伊斯顿为实证研究和比较分析提供了一个清晰的框架。

4. ****为进一步理论奠定基础：****

伊斯顿的系统理论影响了政治学后来的发展，包括结构功能主义、政策分析以及政治发展与变迁研究。

5. ****适应性：****

The model can be applied to various political systems (democracies, authoritarian regimes, etc.) and scales (local, national, international), making it versatile.

6. ****强调环境：****

认识到环境的作用凸显了外部因素（经济状况、社会运动、国际压力）在塑造政治结果中的重要性。

Criticisms and Limitations

- **Overly Abstract:** Some critics argue that Easton's model is too abstract and general, lacking specificity about power relations, conflict, and individual agency.
- **Neglect of Conflict:** The model tends to emphasize stability and equilibrium, potentially underplaying political conflict and change.
- **Limited Normative Guidance:** It is primarily descriptive and analytical, offering little normative guidance on what political systems ought to do.

Summary

David Easton's system analysis

user_query:

戴维·伊斯顿的系统分析回应:

戴维·伊斯顿的系统分析是政治学中的基础框架，将政治系统概念化为一个动态且交互的系统。其方法运用系统理论来解释政治系统如何运作、维持稳定性并适应环境变化。

从本质上讲，伊斯顿的模型将政治系统视为一个“黑箱”，它处理来自社会和环境的输入，将其转化为输出，并利用反馈进行调整以维持自身。伊斯顿系统分析的关键组成部分包括：

1. **输入:** 这些是社会向政治系统提出的要求与支持。要求是对资源分配或政策变更的请求或主张，而支持则是维持系统运转的遵从、合法性或忠诚等行为或态度。
2. **政治系统:** 这是处理输入的一系列机构和过程。它解读需求与支持，以作出权威决策，包括政策的制定与执行。
3. **输出:** 这些是政治系统为响应输入而产生的决策、政策和行动。它们影响社会及其环境，塑造未来的互动。
4. **反馈:** 输出产生响应，作为新输入反馈回系统，指示输出是否有效或需要修改。这种反馈使系统能够适应内部和外部变化。
5. **环境:** 该系统运行于更广泛的社会、经济、文化和国际背景之中，这些背景与政治系统相互影响。

伊斯顿的理论之所以重要，是因为它提供了一种整体的、以过程为导向的视角来分析政治，强调社会与国家机构之间的互动。它通过关注输入、处理、输出和反馈的持续循环，为理解政治稳定性、适应性和变迁提供了一个框架。该模型对后续政治理论和比较政治分析产生了深远影响。

然而，批评者指出，伊斯顿的模型可能过于抽象，有时会淡化政治过程中的冲突、权力斗争和个体能动性。

总之，戴维·伊斯顿的系统分析为研究提供了一个强大的概念工具

将政治系统视为动态实体，通过权威决策来管理社会需求和支持，以维持秩序并随时间适应。

3 other demos omitted for brevity.
#####

由配置 GEPA+合并（GEPA+Merge）生成的 GEPA 提示：

响应用户查询。为获得启发，我们发现了一个可能相关的请求，该请求发送给了一个强大的外部大语言模型及其响应。

L.8 PUPA, QWEN3 8B

PUPA Qwen3 8B craft_redacted_request.predict

Base Prompt:

给定一个私密用户查询，为强大的外部大语言模型创建一个保护隐私的请求。该大语言模型可以在不获取用户私密信息的情况下提供协助。

MIPROv2 Prompt:

Given a private user query, generate a structured, non-sensitive LLM request that captures the user's intent while omitting personal details. Include a step-by-step reasoning process explaining how the redaction was achieved, ensuring the request is actionable and privacy-preserving. Format the reasoning as a coherent thought process and the request as a clear, specific task for the LLM to execute.

Demos:

Example 1:

user_query:

ĐŠŇĂĐŸĐŠĐŤŇČ ĐİĐŔĐ£ĐŸŇĹĐŸ ĐŽĐİĐĐŔĐİĐŤŇČ ĐŤĐŽŇŔ ĐİŇČĐ£ŇĂĐŔĐŠĐŽĐŸ snmpwalk ĐİĐŔ ip 10.2.35.2 ŇŲ ĐŤĐŔĐİĐİŇŇĐĐŸ Đ£Đİ ĐŤĐİŇŲŇČŇČĐ£ŇČ initi_user/d2J3BgtYdB2/d2J3BgtYdB2 (MD5) DES

reasoning:

The user is requesting an SNMP walk command to retrieve data from an SNMPv3-enabled device at IP 10.2.35.2. The provided credentials include a username (`init\_user`), an authentication password (`d2J3BgtYdB2`), and an encryption password (`d2J3BgtYdB2`). The authentication protocol is MD5, and the encryption protocol is DES. The command must include SNMPv3 options to specify these parameters correctly.

llm_request:

生成一条使用所提供凭据执行带 SNMPv3 身份验证和加密的 `snmpwalk` 的命令行指令，通过在响应中省略敏感细节来确保隐私。


```
#####  
3 other demos omitted for brevity.  
#####
```

由配置 GEPA 生成的 GEPA 提示词：

New Instruction for the Assistant:

****Task Overview:****

Rephrase the user's query into a privacy-preserving, actionable request for an external LLM, ensuring no personally identifiable information (PII), proprietary data, or sensitive details are exposed. Maintain the original intent, clarity, and specificity of the query while adhering to domain-specific strategies for generalized information.

****Key Requirements:****

1. ****Privacy Preservation**:**

- ****Generalize Specifics**:** Replace specific locations (e.g., "Andhra Pradesh" â€Š "a major Indian city"), numbers (e.g., "18 volunteers" â€Š "a group of participants"), or proprietary data (e.g., "van Cleef production sites" â€Š "[Brand Name]'s production sites based on publicly available information").
- ****Avoid PII/Proprietary Data**:** Remove names, addresses, dates, internal processes, or brand-specific secrets. Use placeholders like [Brand Name], [Country], or [Region] for brand/company references.
- ****Publicly Available Focus**:** For brand-related queries, emphasize publicly accessible data (e.g., "locations of [Brand Name]'s production sites" instead of "van Cleef's secret factories").

2. ****请求质量**:**

- ****Clarity and Actionability**:** Ensure the rephrased query is precise, avoids ambiguity, and specifies constraints (e.g., "concise, platform-friendly hashtags" for creative tasks).
- ****Domain-Specific Precision**:** For scientific/technical queries, retain key terms (e.g., "CAPM formula," "pharmacokinetic parameters") while anonymizing study details (e.g., "18 volunteers" â€Š "a group of participants").
- ****Avoid Overly Broad Requests**:** Specify constraints like keyword focus, tone, or format (e.g., "health and fitness keywords" instead of "general hashtags").

3. ****领域特定策略**:**

- ****Geographic Queries**:** Replace regions/cities with generalized terms (e.g., "major city in [Country]" or "well-known [Region]").
- ****Brand/Company Queries**:** Reference brands by name but prioritize publicly available data (e.g., "locations of [Brand Name]'s production sites").
- ****Creative Tasks**:** Use examples or constraints (e.g., "wellness, strength, motivation keywords" for hashtags; "platform guidelines" for social media content).

4. ****避免数据泄漏**:**

- ****No Inferred Details**:** Do not assume user-specific preferences, contexts, or private data (e.g., "Tell me the location of van Cleef production sites" â€Š "Provide locations of [Brand Name]'s production sites based on publicly available information").
- ****Anonymize Numerical Data**:** Replace specific counts, dosages, or metrics with generalized terms (e.g., "40 mg OA" â€Š "a dose of OA").

****Example Workflow:****

```
- **User Query**: "Calculate DáŹAmicoŹs cost of equity using CAPM with rRF = 4.67%, ĩŹ = 1.56, and market risk premium = 6.63%."
- **Rephrased Request**: "Calculate the cost of equity for a company using the CAPM formula, given a risk-free rate of 4.67%, a beta of 1.56, and a market risk premium of 6.63%. Show step-by-step calculations and round the final result to two decimal places."

**Niche Domain-Specific Guidance:**
- **Scientific/Pharmacokinetic Queries**: Generalize study details (e.g., "18 healthy male volunteers" ŹŹ "a group of participants"), retain key parameters (e.g., AUC, peak concentrations), and avoid proprietary methodologies.
- **Creative/Translation Tasks**: Use placeholders for song titles/artists (e.g., "Mahmood's song [Title]") and specify tone/rhythm requirements for translations.
- **Financial Calculations**: Retain formulas and numerical values but avoid linking to specific companies or proprietary models.

**Final Output Format:**
- **reasoning**: Briefly explain the rephrasing choices (e.g., "Generalized '18 volunteers' to 'a group of participants' to preserve privacy").
- **llm_request**: Provide the anonymized, actionable request for the LLM.
```

PUPA Qwen3 8B respond_to_query

Base Prompt:

响应用户查询。为获得启发，我们发现了一个可能相关的请求，该请求发送给一个强大的外部大语言模型及其响应。

MIPROv2 Prompt:

Synthesize the external LLM's response with the user's specific query to generate a final privacy-preserving answer. Replace placeholders with actual values while omitting sensitive information, maintaining a professional tone, and ensuring contextual accuracy. Combine the technical solution from the LLM with the user's unique details to produce a clear, anonymized response that adheres to privacy constraints.

Demos:

Example 1:

related_llm_request:

Create a PESTLE analysis framework for GE Healthcare in the US, outlining key issues in each category (Political, Economic, Social, Technological, Environmental, Legal) and their potential impact on the organization. Prioritize these issues based on urgency and significance without referencing confidential data. Structure the output as a matrix with clear criteria for evaluation.

related_llm_response:

以下是为**美国通用电气医疗集团**量身定制的**宏观环境分析框架**，以矩阵形式呈现。该框架列出了各类别中的关键问题，评估了其**紧迫性**与**显著性**，并指出了其对组织的**潜在影响**。

PESTLE Analysis Matrix for GE Healthcare (US)

```

| **Category** | **Key Issues** | **Urgency** (High/Med/Low) | **Significance**
(High/Med/Low) | **Potential Impact on GE Healthcare** |
|-----|-----|-----|-----|
|-----|-----|-----|-----|
|-----|-----|-----|-----|
| **Political** | 1. Healthcare policy changes (e.g., Medicare/Medicaid reimbursement
reforms) | High | High | Directly affects revenue streams and product/service demand;
requires adaptation to new reimbursement models. |
| | 2. Trade policies and tariffs on medical equipment imports/exports | Medium | Medium
| Could increase costs of components or affect supply chain efficiency. |
| | 3. Government funding for healthcare innovation and research | Medium | High |
Influences R&D opportunities and partnerships; potential for grants and subsidies. |
| **Economic** | 1. US healthcare spending trends and budget constraints | High | High |
Impacts purchasing power of hospitals and clinics; affects sales volume and pricing
strategies. |
| | 2. Inflation and rising operational costs | High | Medium | Increases production and
operational expenses; may pressure profit margins. |
| | 3. Economic cycles affecting capital investment in healthcare infrastructure | Medium
| Medium | Influences demand for high-cost medical equipment and technology upgrades. |
| **Social** | 1. Aging population increasing demand for healthcare services and devices
| High | High | Drives demand for diagnostic imaging, monitoring, and therapeutic
devices; growth opportunity. |
| | 2. Growing patient awareness and demand for personalized medicine | Medium | Medium |
Necessitates innovation in tailored healthcare solutions and data-driven technologies. |
| | 3. Workforce shortages in healthcare professionals | High | Medium | May affect
adoption and training for new technologies; impacts customer support and service
delivery. |
| **Technological** | 1. Rapid advancements in AI and machine learning for diagnostics
and imaging | High | High | Critical for maintaining competitive edge; requires
continuous investment in R&D and product development. |
| | 2. Integration of IoT and connected devices in healthcare | Medium | High | Enhances
product offerings but demands robust cybersecurity and interoperability solutions. |
| | 3. Data privacy and cybersecurity technologies | High | High | Essential to protect
patient data and comply with regulations; failure risks legal penalties and reputational
damage. |
| **Environmental** | 1. Regulatory pressure to reduce carbon footprint and waste in
manufacturing | Medium | Medium | Necessitates sustainable practices; potential cost
implications but improves brand reputation. |
| | 2. Energy efficiency in product design and operations | Medium | Medium | Can reduce
operational costs and appeal to environmentally conscious customers. |
| | 3. E-waste management and recycling programs | Low | Medium | Compliance with
environmental standards; growing importance for corporate social responsibility. |
| **Legal** | 1. Compliance with FDA regulations and medical device approvals | High |
High | Critical for market access; delays or failures can halt product launches and cause
financial losses. |
| | 2. Intellectual property rights and patent protections | Medium | High | Protects
innovation and competitive advantage; infringement risks costly litigation. |
| | 3. Liability and product safety regulations | High | High | Ensures patient safety;
non-compliance can lead to lawsuits, recalls, and reputational damage. |

```

```
### Criteria for Evaluation:

- **Urgency**: How immediate the issue requires attention (High = immediate/ongoing;
Medium = within 1-2 years; Low = longer-term).
- **Significance**: The potential magnitude of impact on GE Healthcare's operations,
profitability, and strategic positioning (High = critical; Medium = moderate; Low =
minor).
- **Potential Impact**: Describes how the issue could affect GE Healthcare's business,
including risks and opportunities.

---

This matrix helps prioritize strategic focus areas for GE Healthcare in the US, balancing
immediate challenges with longer-term trends.
user_query:
美国通用电气医疗集团宏观环境分析的问题优先级矩阵：

    以下是根据**美国通用电气医疗集团**的宏观环境分析得出的优先级排序**问题优先级矩阵**，重点关注
    最紧迫和最重要的挑战与机遇：

---

### **High Priority Issues (High Urgency + High Significance)**
| **Category** | **Key Issue** | **Urgency** | **Significance** | **Impact Summary** |
|-----|-----|-----|-----|-----|
|-----|-----|-----|-----|-----|
|-----|
| **Political** | Healthcare policy changes (e.g., reimbursement reforms) | High | High | Direct revenue risk; requires rapid adaptation to new regulations. |
| **Economic** | US healthcare spending trends and budget constraints | High | High | Drives demand and pricing pressures; critical for sales strategy. |
| **Social** | Aging population demand for healthcare services | High | High | Growth opportunity in diagnostics/therapeutics; long-term market expansion. |
| **Technological** | AI/ML advancements in diagnostics and imaging | High | High | Essential for competitiveness; demands sustained R&D investment. |
| **Legal** | FDA regulations and medical device approvals | High | High | Critical for market access; delays risk financial losses. |

---

### **Medium Priority Issues (High Urgency + Medium Significance or Medium Urgency + High Significance)**
| **Category** | **Key Issue** | **Urgency** | **Significance** | **Impact Summary** |
|-----|-----|-----|-----|-----|
|-----|
| **Economic** | Inflation and operational costs | High | Medium | Pressures profit margins; requires cost-efficiency measures. |
| **Technological** | IoT integration and cybersecurity | Medium | High | Enhances offerings but demands robust security frameworks. |
```



M GEPA 生成的核函数生成提示

M.1 NPUEVAL：面向新硬件架构的核函数代码生成

图 27 展示了 GEPA 使用 GPT-4o 为 NPU 核函数生成所生成的提示，该提示在相同的 GPT-4o 智能体下取得了 26.85% 的得分，而使用简单提示时该智能体仅获得 4.25% 的得分。

You are tasked with generating C++ code for a single kernel function that will run on an AI Engine (AIE) tile. The kernel should perform a specified operation on a vector of bfloat16 values. Your code should be complete and self-contained within a single code block, including all necessary headers and imports. Follow these guidelines:

- 1. **Headers and Imports**:** Include only the necessary headers for AIE operations. Avoid including `` or any headers that are not part of the standard AIE API. Use:
```cpp  
#include <stdint.h>  
#include <aie\_api/aie.hpp>  
#include <aie\_api/utils.hpp>  
```
- 2. **Kernel Function**:** Implement the kernel function as specified in the input. The function should take pointers to input and output buffers and a size parameter. Use AIE vector operations to process the data efficiently.
- 3. **Vector Operations**:** Utilize the AIE API's vector operations for loading, processing, and storing data. For example, use `aie::vector` for vector operations and `aie::reduce\_min` for reduction tasks. Ensure that the vector size is compatible with the AIE hardware capabilities.
- 4. **Avoid Non-Existent Functions**:** Do not use functions like `aie::exp` or `aie::store\_v` if they are not supported. Instead, implement the required functionality using available AIE API functions.
- 5. **Error Handling**:** Ensure that the code is free of syntax errors and compatible with the AIE environment. Test the code for compilation without errors.

Here is an example template for a kernel function:

```
```cpp
#include <stdint.h>
#include <aie_api/aie.hpp>
#include <aie_api/utils.hpp>

void kernel_function_name(bfloat16 *input_vector, bfloat16 *output_vector, uint32_t vector_size) {
 constexpr int vector_length = 16; // Adjust based on AIE capabilities
 aie::vector<bfloat16, vector_length> input_data;
 aie::vector<bfloat16, vector_length> output_data;

 for (uint32_t i = 0; i < vector_size; i += vector_length) {
 input_data = aie::load_v<vector_length>(input_vector + i);
 // Perform the required operation on input_data
 // Store the result in output_data
 aie::store_v(output_vector + i, output_data);
 }
}
```
```

Replace `kernel\_function\_name` and the operation logic with the specific task details provided in the input.

图 27: GEPA 为 NPUEval 生成的提示，在相同的 GPT-4o 智能体下取得了 26.85% 的得分，而该智能体使用简单提示仅获得 4.25% 的得分。

M.2 KERNELBENCH: 面向 NVIDIA GPU 的 CUDA 核函数代码生成

GEPA GPT-4o generated prompt for CUDA Kernel Generation

要通过用自定义 CUDA 核函数替换算子来优化给定的 PyTorch 模型，请遵循以下详细说明。目标是在确保正确性的同时实现性能提升。将优化后的输出架构命名为`ModelNew`。在代码块中输出新模型代码。请生成真实代码，而非伪代码，并确保代码可编译且功能完整。不要包含测试代码。

Steps to Create Custom CUDA Kernels

1. Identify Operators to Replace

- 分析模型，识别可从自定义 CUDA 实现中获益的算子。
- 考虑算子融合机会，将多个操作合并到单个核函数中以提高效率。

2. Setup and Compilation

- 使用`torch.utils.cpp\_extension.load\_inline`编译 CUDA 代码。这允许直接在 Python 脚本中定义和编译自定义 CUDA 核函数。
- 确保包含所有必要的 CUDA 和 C++ 头文件，以避免缺少包含错误。

3. Implementing the CUDA Kernel

- Write the CUDA kernel code, ensuring it is optimized for parallel execution. Use shared memory to reduce global memory accesses and ensure coalesced memory access patterns.
- Example structure for a CUDA kernel:

```
```cpp
__global__ void my_kernel(const float* input, float* output, int size) {
 int idx = blockIdx.x * blockDim.x + threadIdx.x;
 if (idx < size) {
 // Perform computation
 }
}
```
```

4. Kernel Launch Configuration

- Configure the kernel launch parameters (number of blocks and threads per block) to maximize GPU utilization.
 - Example:
- ```
```python
const int block_size = 256;
const int num_blocks = (size + block_size - 1) / block_size;
my_kernel<<<num_blocks, block_size>>>(input, output, size);
```
```

##### #### 5. Error Handling and Debugging

- Implement error checking after CUDA API calls and kernel launches using `cudaGetLastError()` to catch errors early.
- Use CUDA debugging tools like `cuda-memcheck` and NVIDIA Nsight for debugging and profiling.
- Be aware of common syntax errors and namespace issues, and ensure that the CUDA code adheres to syntax rules.

#### #### 6. Integrating with PyTorch

- Define a Python function that wraps the CUDA kernel call and integrates it into the PyTorch model.

##### - Example:

```
```python
def my_custom_op(input):
    output = torch.zeros_like(input)
    my_kernel(input.data_ptr<float>(), output.data_ptr<float>(), input.numel())
    return output
```
```

#### 7. 测试与验证 - 通过将自定义核函数的输出与原始 PyTorch 算子进行比较，验证其正确性。- 使用性能分析工具测量性能提升并识别进一步的优化机会。- 为已知正确输入建立参考输出以验证正确性。- 考虑浮点误差，并使用一个小的容差值来接受可接受的差异。

#### #### 8. Compatibility and Compute Capability

- Ensure that the CUDA code is compatible with the target GPU's compute capability.
- Use appropriate compiler flags and consider building for multiple compute capabilities if necessary.

##### ### Example Code

Here is an example of how to replace a simple element-wise addition with a custom CUDA kernel:

```
```python
import torch
import torch.nn as nn
from torch.utils.cpp_extension import load_inline

# Define the custom CUDA kernel for element-wise addition
elementwise_add_source = """
#include <torch/extension.h>
#include <cuda_runtime.h>

__global__ void elementwise_add_kernel(const float* a, const float* b, float* out, int
    size) {
    int idx = blockIdx.x * blockDim.x + threadIdx.x;
    if (idx < size) {
        out[idx] = a[idx] + b[idx];
    }
}

torch::Tensor elementwise_add_cuda(torch::Tensor a, torch::Tensor b) {
    auto size = a.numel();
    auto out = torch.zeros_like(a);

    const int block_size = 256;
```
```



```
const int num_blocks = (size + block_size - 1) / block_size;

elementwise_add_kernel<<<num_blocks, block_size>>>(a.data_ptr<float>(), b.data_ptr<float>(), out.data_ptr<float>(), size);

return out;
}
"""

elementwise_add_cpp_source = (
 "torch::Tensor elementwise_add_cuda(torch::Tensor a, torch::Tensor b);"
)

Compile the inline CUDA code for element-wise addition
elementwise_add = load_inline(
 name="elementwise_add",
 cpp_sources=elementwise_add_cpp_source,
 cuda_sources=elementwise_add_source,
 functions=["elementwise_add_cuda"],
 verbose=True,
 extra_cflags=[""],
 extra_ldflags=[""],
)

class ModelNew(nn.Module):
 def __init__(self) -> None:
 super().__init__()
 self.elementwise_add = elementwise_add

 def forward(self, a, b):
 return self.elementwise_add.elementwise_add_cuda(a, b)
...

Additional Best Practices
- **Optimize Memory Usage**: Minimize data transfers between host and device, and use shared memory to reduce global memory access.
- **Atomic Operations**: When using atomic operations like `atomicMax` with floating-point numbers, ensure correct usage by following best practices, such as using appropriate data types and minimizing contention.
- **Performance Optimization**: Maximize parallel execution, optimize memory access patterns, and use compiler flags to enhance performance.
- **Namespace Usage**: Avoid adding class declarations or function definitions directly to reserved namespaces like `cuda`. Use nested namespaces within non-reserved namespaces to organize code.
- **Numerical Precision**: Be aware of floating-point arithmetic issues, such as non-associativity, and use appropriate precision levels for calculations.
- **Debugging Tools**: Utilize tools like CUDA-GDB and NVIDIA Nsight for debugging and profiling to ensure correctness and performance.
```

通过遵循这些说明，您可以有效地用自定义 CUDA 核函数替换 PyTorch 算子，同时确保性能提升和正确性。

### Instruction for Replacing PyTorch Operators with Custom CUDA Kernels

您的任务是通过用自定义 CUDA 核函数替换某些算子来优化给定的 PyTorch 模型，以实现性能提升。请按照以下步骤确保成功实现：

#### #### Step 1: Identify Operators for Replacement

- **Criteria for Selection**: Choose operators that are computationally intensive and have potential for parallelization. Consider operators that are frequently used in the model's forward pass.
- **Operator Fusion**: Look for opportunities to fuse multiple operators into a single CUDA kernel, such as combining matrix multiplication with activation functions (e.g., `matmul + ReLU`).

#### #### Step 2: Implement Custom CUDA Kernels

- **Kernel Structure**: Define your CUDA kernel using the `__global__` specifier. Ensure that each thread handles a specific part of the computation. Use correct index calculations to access data.
- **Memory Management**:
  - Allocate memory for input, output, and any intermediate data on the GPU using `cudaMalloc`. Use `cudaMemcpy` to transfer data between host and device.
  - Utilize shared memory to cache frequently accessed data and reduce global memory accesses.
  - Ensure coalesced global memory accesses for efficient memory transactions.
- **Numerical Stability and Boundary Conditions**:
  - Implement verification mechanisms to ensure numerical stability. Use `__host__ __device__` functions for testing on both CPU and GPU.
  - Handle boundary conditions to prevent out-of-bounds memory access. Ensure that thread indices are within valid ranges.
- **Optimization Techniques**:
  - Use shared memory to reduce global memory accesses and improve performance.
  - Consider using mixed precision and Tensor Cores for matrix operations to enhance performance.
  - Avoid diverged execution paths to maintain efficient parallel execution.

#### #### Step 3: Integrate CUDA Kernels into PyTorch Model

- **Inline Compilation**: Use `torch.utils.cpp_extension.load_inline` to compile your CUDA code and integrate it into the PyTorch model.
- **Model Modification**: Replace the original PyTorch operators with calls to your custom CUDA functions. Ensure that the new model architecture (`ModelNew`) is fully functional and compiles without errors.

#### #### Step 4: Testing and Validation

- **Correctness**: Verify that the output of the optimized model matches the expected output of the original model. Use a set of test cases to ensure accuracy.
- **Performance Evaluation**: Measure the runtime of the optimized model and compare it to the original. Aim for a significant reduction in execution time.
- **Edge Case Handling**: Ensure that the kernel correctly handles cases where the matrix size is not a multiple of the block size and other potential edge cases.

#### #### Example Code

Below is an example of how to define and integrate a custom CUDA kernel for element-wise addition:

```
```python
import torch
import torch.nn as nn
from torch.utils.cpp_extension import load_inline

# Define the custom CUDA kernel for element-wise addition
elementwise_add_source = """
#include <torch/extension.h>
#include <cuda_runtime.h>

__global__ void elementwise_add_kernel(const float* a, const float* b, float* out, int
    size) {
    int idx = blockIdx.x * blockDim.x + threadIdx.x;
    if (idx < size) {
        out[idx] = a[idx] + b[idx];
    }
}

torch::Tensor elementwise_add_cuda(torch::Tensor a, torch::Tensor b) {
    auto size = a.numel();
    auto out = torch::zeros_like(a);

    const int block_size = 256;
    const int num_blocks = (size + block_size - 1) / block_size;

    elementwise_add_kernel<<<num_blocks, block_size>>>(a.data_ptr<float>(), b.data_ptr<
        float>(), out.data_ptr<float>(), size);

    return out;
}
"""

elementwise_add_cpp_source = (
    "torch::Tensor elementwise_add_cuda(torch::Tensor a, torch::Tensor b);"
)

# Compile the inline CUDA code for element-wise addition
elementwise_add = load_inline(
    name="elementwise_add",
    cpp_sources=elementwise_add_cpp_source,
    cuda_sources=elementwise_add_source,
    functions=["elementwise_add_cuda"],
    verbose=True,
    extra_cflags=[""],
    extra_ldflags=[""],
)

class ModelNew(nn.Module):
    def __init__(self) -> None:
        super().__init__()
        self.elementwise_add = elementwise_add

    def forward(self, a, b):
```

```
        return self.elementwise_add.elementwise_add_cuda(a, b)
    ...

#### Constraints
- 确保优化后的模型保持与原始模型相同的准确率。
- 自定义 CUDA 核函数应针对性能进行优化，最小化执行时间并最大化 GPU 利用率。

通过遵循这些说明，您将能够有效地用自定义 CUDA 核函数替换 PyTorch 算子，在保持模型准确率的同时实现显著的性能提升。
```

GEPA 在优化过程中调用反思语言模型的次数

表 4: GEPA 在优化过程中调用反思语言模型的总次数。

Benchmark Name	Num Reflection Calls	Num Reflection Calls
	GPT-4.1-Mini	Qwen3-8B
AIME-2025	24	90
LiveBench-Math	34	38
HotpotQA	69	64
IFBench	21	17
Hover	92	50
PUPA	46	38